NICE: neural-initialised static condensation with equilibrium correction, with application to the thickness design of cut TPMS lattices

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Abstract

Thickness optimisation of lattices trimmed to a part's shape needs fine-scale accuracy and fast analysis at every design iteration. Fine-scale finite element models are too costly. Homogenisation assumes periodic boundary conditions, which fail without scale separation or when cells are cut. Static condensation eliminates each cell's interior degrees of freedom, reducing the global system without loss of accuracy, but must refactorise every interior stiffness matrix at every design iteration. This paper proposes NICE to avoid these factorisations. It retains all degrees of freedom on cell faces and in elements cut by the trimming plane; a network predicts the interior displacements from the cell geometry, a two-grid correction reduces the interior force imbalance, and the condensed stiffness follows from the strain energy of the corrected field. The resulting condensed stiffness bounds the exact one from above, with an error of second order in the interior displacement error, independent of the network accuracy. Thickness sensitivities have an error of first order in the interior displacement error, so accurate compliance does not ensure accurate sensitivities; the network is trained and checked on both. For cut Schwarz-P cells, the mean energy error over 80 validation geometries is 0.074%, an eight-cell analysis with sensitivities is about ten times faster than a direct solve of the full lattice, and thickness optimisation reproduces the exact-condensation design of an eight-cell lattice. For a 24-cell layer clamped at its cut, where homogenisation underestimates compliance by 27%, NICE agrees with exact condensation to 0.03% in compliance and 0.3% in gradient.

Keywords: superelement, static condensation, neural network, multigrid, cut finite element method, TPMS lattice, thickness optimisation

1. Introduction

The stiffness of a lattice can be tailored by grading the thickness of its cells Yu et al. (2019). In the thickness design of a graded lattice, the thickness parameters of the cells are the design variables, and their distribution is determined by iterative optimisation. The design depends on local responses, such as the stiffness of the cells next to supports and loads and the sensitivity of the compliance to the thickness of each cell, so the response of

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every cell must be computed accurately. At the same time, the whole structure is reanalysed at every design iteration, and the cost of each analysis, which grows with the size of the model, accumulates over the optimisation. Accuracy thus calls for a model that resolves the deformation at the scale of the cells, whereas efficiency calls for a model that is as small as possible.

Fine-scale finite element models discretise every cell and meet the accuracy requirement; with large-scale parallel solvers, such models have been used in topology optimisation at very large scale Aage et al. (2017). Because every design iteration changes the geometry of every cell, however, a fine-scale model must be reassembled and solved again at every iteration (Section 5.9). Homogenisation replaces the cells by an equivalent material; it is inexpensive and is the usual basis for the design of graded lattices Li et al. (2018), Wu et al. (2021). It assumes periodic boundary conditions on a unit cell, an assumption that fails without scale separation or when cells are cut: near supports, loads and boundaries, the displacement field varies on the scale of a cell, and when a lattice is trimmed to the shape of a part, the cut cells at its boundary are no longer complete periodic cells. For the plate of Section 5.10, a single layer of cells clamped at its cut (Figure 1), the homogenised model underestimates the compliance of the uniform starting design by 27%.

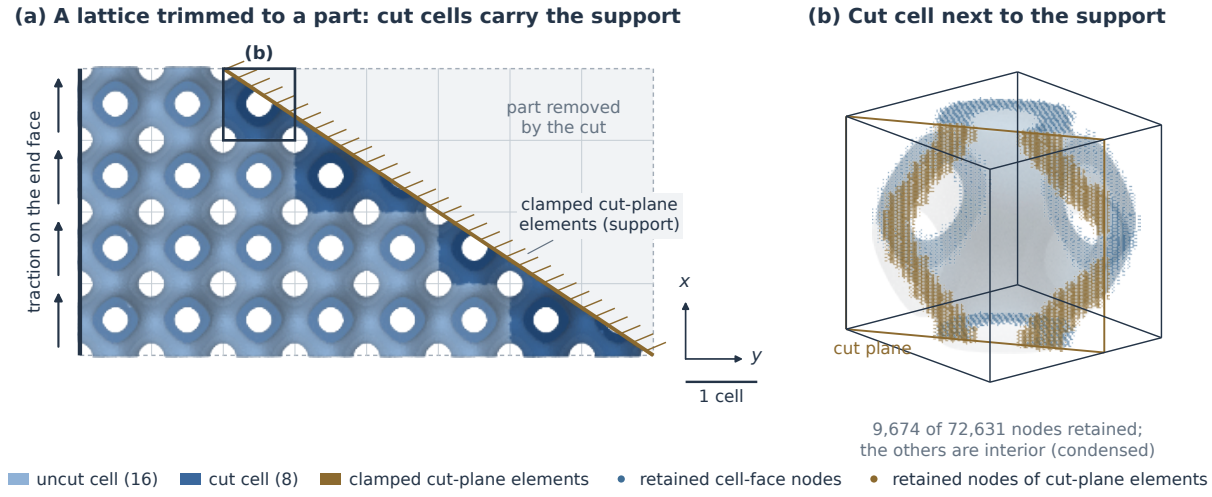


Figure 1: **A lattice trimmed to the shape of a part, whose cut cells carry the support.** (a) Plan view of a single layer of 8×4 Schwarz-P cells (Eq. (1), uniform $\tau = 0.40$) trimmed by a plane. The layer has 16 uncut cells (light) and eight cut cells (dark); the part removed by the cut is drawn pale with a dashed outline. The plate is clamped at its cut: every degree of freedom of the cut-plane elements of every cut cell is fixed (hatched line). The end face carries an in-plane traction load of unit resultant (arrows). (b) The cut cell outlined in (a), with a translucent material surface and the cut plane. Of the 72,631 nodes of its Q_2 model, the 9,674 nodes on the cell faces (blue) and in the cut-plane elements (sand) are retained. All other nodes are interior and are eliminated by static condensation. The surfaces are reconstructed from Eq. (1) for visualisation only.

Substructuring lies between these two approaches. Each cell is treated as a substructure, or superelement, whose interior degrees of freedom are eliminated by static condensation, so that only the retained (master) degrees of freedom through which the cell connects to its neighbours, supports and loads remain Guyan (1965), Irons (1965). The global system is

assembled and solved on the retained degrees of freedom only, and the interior displacements of each cell are recovered afterwards to evaluate local quantities such as the sensitivities. For cut thin-walled cells, the retained degrees of freedom are those of all nodes on the cell faces and of all nodes of the elements intersected by the trimming plane, hereafter referred to as cut-plane elements; for the 80 validation geometries of this paper, a cell has 2,679 to 45,900 retained degrees of freedom, with a median of about 2.4×10^4, and several times as many interior ones (Figure 1b). Static condensation introduces no approximation, but it requires a sparse Cholesky factorisation of the interior stiffness matrix of every cell, from which a dense condensed stiffness matrix is formed. For a cell with the median number of retained degrees of freedom, this matrix occupies 4.2 GiB in double precision, and for the deployed cells of Section 5.9 the Cholesky factor of a single cell occupies 0.60–9.4 GiB. Because every design iteration changes the geometry of every cell, these factorisations must be repeated for all cells at every iteration; on the lattices of Section 5.9, one analysis by conventional exact static condensation takes longer than a direct solve of the whole lattice.

Learned component models obtain the stiffness relation of an element or a substructure from a neural network, so that no interior equilibrium problem of the component has to be solved during the analysis. For example, Capuano & Rimoli (2019) map the nodal displacements of an element directly to its nodal forces, and learned substructures Huang et al. (2023), Guo et al. (2026a) predict numerical shape functions with a neural network, compute the substructure stiffness matrix as $N^T K N$ and assemble the substructures in the manner of finite elements. To keep the network output small, learned substructures describe the boundary displacement of each substructure by a few degrees of freedom, for example through linear or cubic Bézier interpolation, and the network outputs one shape function for each boundary degree of freedom; port reduction Eftang & Patera (2013), Smetana & Patera (2016), Ballani et al. (2018) likewise reduces the degrees of freedom shared between components. Reducing the boundary degrees of freedom restricts the displacements that neighbouring substructures can exchange, and the resulting model error cannot be removed by a more accurate interior approximation. For cut cells, the supports and loads on the cut surface act on all nodes of the cut-plane elements, whose degrees of freedom must therefore all be retained, so a description of the boundary displacement by a few degrees of freedom is unsuitable. Since the retained degrees of freedom are numerous and vary with the cell geometry, one network output per retained degree of freedom is also impractical.

This paper therefore proposes neural-initialised static condensation with equilibrium correction (NICE). NICE retains all degrees of freedom on the cell faces and in the cut-plane elements and approximates only the interior displacements: a neural network that takes the cell geometry as input provides an initial interior displacement field, and one two-grid cycle of Chebyshev smoothing and coarse-grid correction Xu (1992), applied with the retained displacements held fixed, reduces the interior force imbalance. The network is nonlinear in the geometry but exactly linear in the retained displacements. Its parameters act on local node positions, feature channels and grid levels rather than on individual degrees of freedom, so that one network of about 6×10^5 parameters serves cells with different numbers of retained degrees of freedom. The smoothing mainly reduces the high-frequency part of the interior error, and the coarse-grid correction reduces its smooth part. The network and the correction together define a linear map from the retained displacements to the displacements of the whole cell, and the condensed stiffness is computed from the strain energy of the resulting

displacement field. The assembled system is solved by preconditioned conjugate gradients, which need only products of the condensed stiffness with vectors; each product is obtained by applying the network, the correction, the cell stiffness matrix and the transposes of the correction and the network in turn, so neither the map nor the condensed stiffness has to be formed.

If the interior displacement approximation is linear in the retained displacements and reproduces them and rigid-body motions, the condensed stiffness computed from its strain energy is symmetric positive semidefinite, with rigid-body motions as its only zero-energy modes. By the principle of minimum potential energy, it is then an upper bound on the exact condensed stiffness, with an error of second order in the interior displacement error, and the two coincide when the interior displacements are exact. The network of NICE satisfies these conditions by construction, so these properties do not depend on its training accuracy. Since the two-grid correction is a fixed linear operation, the conditions still hold after the correction; when the smoothing interval and the coarse solve satisfy the conditions of Proposition 2, the correction does not increase the error Golub & Varga (1961), Xu & Zikatanov (2002). The network is trained with the correction applied, and the two complement each other: under the same correction, a graph-harmonic initial field, which has no trainable parameters, leaves 5.4 to 265 times the mean energy error of NICE (Section 5.5).

The interior displacement error affects the compliance and the sensitivities differently. The compliance error is a sum of cell errors weighted by the fractions of the strain energy that the cells carry, and is of second order in the interior displacement error. The thickness sensitivities, by contrast, contain a term of first order in this error, introduced through the derivative of the stiffness matrix with respect to the thickness, which the energy error bounds only at the order of its square root. A cell that carries a small fraction of the energy therefore has little effect on the compliance but may still have inaccurate sensitivities. This is not specific to static condensation: whenever the compliance and the sensitivities are computed from an approximate displacement field, an accurate compliance does not ensure accurate sensitivities, and similar observations have been made for approximate reanalysis and inexact solves in topology optimisation Amir et al. (2009), Amir et al. (2010), Gogu (2015). Section 4 derives these relations for any interior approximation that meets the conditions above (Propositions 3 and 4); accordingly, NICE is trained and checked on both the compliance and the sensitivities (Section 5.6).

The main contributions are as follows.

- All degrees of freedom on the faces of a cut cell and in its cut-plane elements are retained, and only the interior displacements are approximated. Although these degrees of freedom change with the geometry, supports and loads on the cut surface act on the original degrees of freedom and can be applied without retraining (Sections 2, 5.7 and 5.10).
- A neural network for the interior displacements is constructed whose output is, by construction, exactly linear in the retained displacements and reproduces them and rigid-body motions; its size does not depend on the number of retained degrees of freedom, and it forms no matrix. The condensed stiffness computed from its strain energy is symmetric positive semidefinite and an upper bound on the exact condensed stiffness, with an error of second order in the interior displacement error (Sections 3.1,

3.2 and 3.6).

- A fixed two-grid correction is introduced into the condensation and reduces the error of a trained network without retraining. It preserves these properties for any discrete model with a symmetric positive semidefinite stiffness matrix and a positive definite interior block, and under the conditions of Proposition 2 it does not increase the energy error (Sections 3.3 and 3.4); in every example of Supplementary Table ST07, a larger number of smoothing steps reduced the energy error further (Section 5.5).
- The different routes by which the interior displacement error enters the compliance and the sensitivities are derived. Both quantities are checked accordingly, and the thickness designs are verified against exact static condensation and compared with homogenisation (Sections 4, 5.6, 5.8 and 5.10).

Section 1.1 reviews related work. Section 2 describes the finite element model of the cut cells, static condensation, the design problem and the assumptions of the analysis. Section 3 presents NICE: the computation of the condensed stiffness, the network that predicts the interior displacements, the two-grid correction, the training and the application of the condensed stiffness without forming it. Section 4 analyses the effect of the interior displacement error on the compliance and the sensitivities; the proofs are given in the appendices. Section 5 presents numerical examples for cut thin-walled Schwarz-P cells in a stabilised cut finite element model: the accuracy on 80 validation geometries and on two-cell assemblies, an ablation that restricts the cell-face displacements, eight-cell lattices of cells used neither in training nor in checkpoint selection, the computational cost compared with a direct solve of the whole lattice and with conventional exact static condensation, and thickness optimisation checked against exact static condensation and compared with a homogenised model. Section 6 discusses the scope and limitations of the method, and Section 7 concludes.

1.1. Related work

Among learned substructures, problem-independent machine learning (PIML) has been studied most systematically. PIML predicts numerical shape functions of a substructure for a few boundary degrees of freedom and imposes constraints for rigid-body motion; its self-consistent form computes the substructure stiffness matrix as $N^T K N$ Huang et al. (2022), Huang et al. (2023). Subsequent work trained the networks without precomputed shape-function data by minimising the potential energy Huang et al. (2024), imposed symmetry equivariance Jiang, C. et al. (2026), treated isoparametric substructures Zhang, L. et al. (2024), transferred learned substructures to complex three-dimensional design domains Zhang et al. (2026) and applied them to lattice optimisation Xu et al. (2025). Another study joins oversampled numerical basis functions by an overlapping partition of unity and corrects the reconstructed fine-scale field by a few Jacobi-preconditioned conjugate gradient iterations on the global equilibrium equations Guo et al. (2026b). Since each substructure has only a few boundary degrees of freedom, the assembled model is small enough for very large structures, and the resulting boundary model error is controlled by refining the partition, raising the order of the boundary interpolation or oversampling. NICE lies at the other end of this trade-off: it keeps all retained degrees of freedom of a cell, which avoids the boundary model error but enlarges the assembled model. NICE learns the interior displacements of each geometry in a family whose discrete space changes with the cut, instead

of numerical shape functions for a fixed set of boundary degrees of freedom. Another PIML variant replaces the linear interpolation of the boundary displacement by cubic Bézier interpolation and was compared, in the same study, with a variant that predicts numerical shape functions for every boundary node (the full-node variant). For voxel substructures, the extra cost of the full-node variant outweighed its gain in accuracy, and on the larger substructures it was less accurate, because its network had to predict many more outputs than the Bézier variant Guo et al. (2026a). In NICE, the interior of a thin-walled cell is several times larger than its retained part, so condensation still removes most unknowns; the cell of Figure 1b retains only 9,674 of its 72,631 nodes. The network size of NICE does not grow with the number of retained degrees of freedom, and the residual error of the network prediction is further reduced by the two-grid correction (Section 5.5).

Closest to NICE are component models that keep a mechanical structure. If a condensed stiffness is computed from the strain energy of an approximate displacement field that is linear in the retained displacements and reproduces them and rigid-body motions, then, by the principle of minimum potential energy, it is an upper bound on the exact condensed stiffness, with an error of second order in the interior displacement error (Section 3.1). The same property underlies static condensation itself Toselli & Widlund (2005), multiscale finite elements Hou & Wu (1997), component reduced-basis methods Huynh et al. (2013) and learned shape functions evaluated as $N^T K N$ Huang et al. (2023). NICE applies it to network-predicted interior displacements on a set of retained degrees of freedom that changes with the cut. Learned and reduced component models differ mainly in three respects: the degrees of freedom kept on the interface, the representation of the interior and the stage at which the approximation is improved. The learned substructures of Huang et al. (2023) and Guo et al. (2026a) reduce the interface degrees of freedom and learn shape functions for the reduced boundary. NICE keeps all interface degrees of freedom and learns only the response of the interior, which it represents by a network that takes the cell geometry as input, and it improves the approximation within the assembled condensed stiffness.

As to the interface degrees of freedom, component reduced-basis methods keep the complete interface in the condensation but represent the interior on a discrete space shared by the parameter family and construct reduced bases for it offline Huynh et al. (2013). For lattices, such component-wise models have been reused in design optimisation, with error bounds on the displacement, the compliance and its sensitivity McBane & Choi (2021); for unfitted discretisations, solutions on varying active sets have been extended to a common background space before reduction Chasapi et al. (2023). NICE also keeps the complete set of retained degrees of freedom, but this set changes with the cut, so its network takes the discrete geometry of each cell as input instead of the parameters of a shared discretisation.

As to the representation of the interior, lattice methods and learned elements differ. One lattice method condenses the cells into superelements and interpolates their stiffness in a density parameter by a reduced-order surrogate, which supplies explicit stiffness derivatives for optimisation without scale separation Wu et al. (2019). For lattices built by spline composition, reduced-order cell operators have replaced the cell-wise local solves of inexact finite element tearing and interconnecting, dual–primal (FETI-DP) methods Hirschler et al. (2024). For unfitted two-dimensional lattices, balancing domain decomposition by constraints (BDDC) has been combined with full cell stiffness matrices, whose integrals over trimmed cells are approximated by matrix discrete empirical interpolation Bonilla Moreno

et al. (2027). Each of these lattice methods keeps one discretisation across cells: a common lattice pattern, a composed parametric cell and one high-order unfitted element per cell, respectively. For unfitted coarse elements, the conditioning of numerical shape functions under small cuts has also been improved Chen & Li (2026). Learned elements condense an embedded inner domain into a symmetric positive semidefinite interface stiffness Parish et al. (2024), and convex neural energy elements learn boundary energies from targets computed with the exact condensed stiffness and recover the interior fields by exact lifting Jiang, H. et al. (2026). NICE instead learns the displacement recovery itself: a single corrected recovery operator F, built without an interior factorisation and applied without forming a matrix, defines the assembled condensed stiffness, the recovered displacements and the thickness sensitivities.

As to the stage at which the approximation is improved, inexact substructuring methods Dohrmann (2007), Li & Widlund (2007), Klawonn & Rheinbach (2007) improve approximate subdomain and coarse solves inside a preconditioner, and their iteration converges to the exact discrete solution (Section 6.2). In a nonlinear example, convex neural energy elements refine the exact lifting by Newton steps and differentiate through these steps Jiang, H. et al. (2026). Component reduced-basis methods bound the error of the component or assembled solution a posteriori Huynh et al. (2013). NICE provides a computable interior residual (Eq. (7)) but no certified error bound; its correction acts on the interior displacements of each cell at fixed retained displacements, inside the strain energy that defines the condensed stiffness, and therefore changes the assembled condensed stiffness itself.

Learning has also been combined with multilevel and domain decomposition solvers, for example in learned multigrid prolongations Greenfeld et al. (2019), Luz et al. (2020), machine-learning-assisted adaptive FETI-DP coarse spaces Heinlein et al. (2021) and networks trained through the solver Um et al. (2020). Hybrid neural solvers combine learned predictions with relaxation, learned inverse actions or coarse spaces Zhang, E. et al. (2024), Kopaničáková & Karniadakis (2025), Xing et al. (2026). In optimisation without scale separation, spectral multiscale coarse bases Efendiev et al. (2013) have served as the coarse space of a two-level preconditioner for the fully resolved problem Alexandersen & Lazarov (2015). Geometry-aware neural operators and graph networks Li et al. (2023a), Li et al. (2023b), Pfaff et al. (2021), multigrid-structured operators He et al. (2024), learned Green's functions Boullé & Townsend (2023) and training by energy minimisation E & Yu (2018) learn the solution map of the whole problem. Most neural operators learn maps that are nonlinear in their input and do not obey superposition, so the strain energy of the predicted field is not a quadratic form in the retained displacements, whereas the properties of the condensed stiffness rest on this quadratic form. These methods accelerate a global solve or approximate the solution of the whole problem; the map learned in NICE recovers the displacements of a single cell from its retained displacements, and the strain energy of these displacements defines a condensed stiffness that can be assembled, differentiated and improved.

Static condensation is itself a two-level method. The exact recovery E maps the retained displacements to the equilibrium displacements of the whole cell; in algebraic multigrid terms, E is the ideal interpolation from the retained to all degrees of freedom, and $S = E^T K E$ is the corresponding coarse-grid matrix Falgout & Vassilevski (2004). NICE replaces E by a recovery operator F from the retained to the active degrees of freedom and condenses with the coarse-grid matrix $\widehat{S} = F^T K F$ of this interpolation. This operator is predicted by a

network and then smoothed by the two-grid cycle, in the same spirit as smoothed-aggregation multigrid, which improves a tentative prolongation by smoothing Vaněk et al. (1996). The operator F is nonlinear in the cell geometry and linear in the retained displacements. Learned multigrid prolongations Greenfeld et al. (2019), Luz et al. (2020) are likewise nonlinear in the operator and linear in the vector they act on, but they are part of a solver. The operator F instead defines a component model that serves sets of retained degrees of freedom changing with the cut, and its coarse-grid matrix is assembled into the lattice and differentiated in design.

2. Cut cells, static condensation and the design problem

The cells are discretised by a stabilised cut finite element method (CutFEM) Burman et al. (2015) on a fixed Cartesian background mesh. The material boundaries (the TPMS surfaces and the cut plane) are not meshed; they only determine which background elements are active and enter the stiffness through integrals over the material part of each element. A change of thickness or of the cut therefore requires no remeshing, and the thickness sensitivity follows from the derivatives of these integrals (Appendix H). All cells lie on the same background mesh, on which the network of Section 3.2 defines its fixed grid hierarchy; the active elements, however, change with the geometry, and so does the set of retained degrees of freedom, which Sections 3 and 4 must account for. Their relations use only the symmetry and positive semidefiniteness of the resulting stiffness matrix and the positive definiteness of its interior block.

2.1. Geometry and finite element model

This paper considers three-dimensional thin-walled cells based on triply periodic minimal surfaces (TPMS). Each cell occupies the reference box $\mathcal{B} = [0,1]^3$, whose six faces are called the cell faces. The material domain of a cell is defined by a Schwarz-P-type trigonometric level-set function, a spatially varying band parameter and an optional planar cut:

$$\begin{aligned} \phi(x) &= \sum_{a=1}^{3} \cos(2\pi x_a), \qquad \tau(x) = \sum_{c\in\{0,1\}^3} N_c^{Q_1}(x)\tau_c, \\ \Omega(\eta) &= \{x \in \mathcal{B} : |\phi(x)| \le \tau(x),\ \boldsymbol{n}\cdot x \le b_{\mathrm{cut}}\}. \end{aligned} \tag{1}$$

The eight parameters τ_c are interpolated by the trilinear functions $N_c^{Q_1}$; they set the width of the implicit band, and thereby the thickness and the material distribution of the cell. They are called corner thickness parameters, and the derivatives with respect to them are called thickness sensitivities. When the eight corner values are collected in a vector, such as $\boldsymbol{\tau}$ or the sensitivity vector $\boldsymbol{s}$, the corners are numbered $c = 1, \dots, 8$. The cut plane $\boldsymbol{n}\cdot x = b_{\mathrm{cut}}$ has the offset b_{cut} and the unit normal $\boldsymbol{n}$, which points away from the material that remains after the cut; for an uncut cell, the half-space constraint is omitted. The symbol η collects the geometry, material and discretisation parameters.

The displacements are approximated by continuous tensor-product Q_2 functions on a Cartesian background mesh. A background element is active if its intersection with $\Omega(\eta)$ has positive measure, which is certified by interval enclosures of the level-set and band functions over the element (Appendix A.1); the displacement degrees of freedom (DOFs) of the nodes

of the active elements are called the active DOFs. Integration over the material domain gives the elastic stiffness of the cell, and ghost-penalty stabilisation Burman (2010) adds terms that couple neighbouring active elements to control the effect of small cuts, active elements that contain very little material Burman et al. (2015). The symmetric matrix K defines the stabilised discrete energy $\frac{1}{2}u^T K u$. The reference solution in this work is the equilibrium of this discrete system, and all errors are measured against this discrete solution, not against the solution of the continuum problem. Figure 1 shows a cut lattice and one of its cut cells, Table 1 (Section 5.1) lists the settings of the examples, and Appendix A gives the details of integration and assembly.

2.2. Retained degrees of freedom and static condensation

For each cell, the retained DOFs form the set P, which has two parts (Appendix A.1): the DOFs of the cell-face nodes of every material patch of positive area on the cell faces, and all displacement DOFs of the active elements that carry a cut-surface patch of positive area. The latter elements form the layer of active elements intersected by the cut plane and are called the cut-plane elements; some of their nodes lie away from the cut plane, but the basis functions of these nodes contribute to the displacement and the virtual work on that plane. The complete set of retained DOFs is used in both the network prediction and the assembly; it includes the DOFs coupled to neighbouring cells and the DOFs on the exterior boundary.

The remaining active DOFs are the interior DOFs and form the set I; the two sets have $p = |P|$ and $i = |I|$ DOFs, and $n_a = p + i$. The selection matrices J_P and J_I extract the two sets from the vector u of active displacements, and $q = J_P u$ is the vector of retained displacements. Boundary loads act through q, and interior body loads are zero. With the DOFs ordered as (P, I), static condensation gives the exact displacement recovery E and the exact condensed stiffness S (the Schur complement):

$$
\begin{aligned}
&K = \begin{bmatrix} K_{PP} & K_{PI} \\ K_{IP} & A \end{bmatrix}, \qquad A = K_{II}, \qquad E = \begin{bmatrix} I_p \\ -A^{-1}K_{IP} \end{bmatrix}, \\
&S = K_{PP} - K_{PI}A^{-1}K_{IP} = E^T K E.
\end{aligned}
\tag{2}
$$

The stiffness matrix K is symmetric positive semidefinite; its null space is spanned by the six rigid-body modes, which form the columns of $R \in \mathbb{R}^{n_a \times 6}$, and their restriction $R_P = J_P R$ to the retained DOFs has rank six. Two features of the discrete model ensure these properties (Appendix A.1): the geometry generator accepts only cells whose active elements form a single face-connected set, and material that is connected only through partially filled elements is coupled through shared basis functions and the ghost penalty. An interior field with zero energy would be a rigid-body motion that vanishes on the retained DOFs, so the interior block A is positive definite, and prescribing q removes every interior motion with zero energy. The exact displacement recovery $u = Eq$ is the unique minimiser of the discrete energy subject to $J_P u = q$, and Sq is the vector of the corresponding nodal forces on the retained DOFs.

2.3. Assembly, compliance and the design problem

For substructure m, the Boolean assembly map B_m extracts $q_m = B_m U$ from the vector U of free global retained displacements. Coincident cell-face DOFs, identified by their position

on the background mesh, are shared between neighbouring cells, whereas retained DOFs of the cut-plane elements that do not lie on the cell faces belong only to their own substructure. The thickness field on a shared face depends only on the four corner values of that face, and ϕ is periodic, so the material patches of two neighbouring cells coincide on every face that is not intersected by a cut. Where a cut removes material from one side of a face only, the unmatched face DOFs remain DOFs of the cell that carries them. Fixed homogeneous supports are incorporated into B_m, and the reference and the approximate assembled systems use the same maps B_m:

$$\mathbb{K} = \sum_m B_m^T S_m B_m, \qquad \widehat{\mathbb{K}} = \sum_m B_m^T \widehat{S}_m B_m, \qquad \mathbb{K}U = f_g, \quad \widehat{\mathbb{K}}\widehat{U} = f_g. \tag{3}$$

Here $\widehat{S}_m$ is the approximate condensed stiffness of Section 3, and f_g is the prescribed force on the free retained DOFs. Because the interior DOF sets of different cells are disjoint, condensing each cell before assembly gives the same matrix as condensing the assembled fine-scale system. The supported reference matrix $\mathbb{K}$ is assumed to be positive definite.

The compliance is $C = f_g^T U$ and its approximation $\widehat{C} = f_g^T \widehat{U}$, with the relative error $e_C = |\widehat{C}/C - 1|$; the local accuracy of the condensed stiffness is measured by the relative energy error of Section 3.1. The thickness sensitivities of a cell form the eight-component vector $\boldsymbol{s}$ of Section 4.2, and with $\widetilde{\boldsymbol{s}}$ computed from the approximate displacements, their relative error in the Euclidean norm is $e_s = \|\widetilde{\boldsymbol{s}} - \boldsymbol{s}\|_2 / \|\boldsymbol{s}\|_2$.

In design, the corner thickness parameters at the lattice vertices are the design variables; they are collected in $\boldsymbol{\tau}_g$ and mapped to the corners of each cell by $\boldsymbol{\tau}_m = \boldsymbol{\tau}_m(\boldsymbol{\tau}_g)$. The design problem is

$$\min_{\boldsymbol{\tau}_g} \; C(\boldsymbol{\tau}_g) \quad \text{subject to} \quad V(\boldsymbol{\tau}_g) \le V^*, \qquad \tau_{\min} \le \tau \le \tau_{\max},$$

where V is the material volume of the discrete model. In addition, the corner span and the gradient norm of each cell are bounded to keep every cell within the range of geometries used in training (Section 5.10). The compliance gradient collects the cell sensitivities at each vertex, $\partial C/\partial \boldsymbol{\tau}_g = \sum_m (\partial \boldsymbol{\tau}_m/\partial \boldsymbol{\tau}_g)^T \boldsymbol{s}_m$, where $\boldsymbol{s}_m$ is the eight-component sensitivity vector of cell m (Section 4.2).

An approximate condensed stiffness $\widehat{S}_m$ that replaces S_m in this design problem must meet three requirements. First, $\widehat{S}_m$ must be symmetric positive semidefinite, with a null space consisting only of the rigid-body modes of the retained DOFs, the columns of R_P. Second, the supported assembled matrix $\widehat{\mathbb{K}}$ of Eq. (3) must be positive definite, which follows from $\widehat{S}_m \succeq S_m$ and $\mathbb{K} \succ 0$ (Appendix B.2). Third, the error of $\widehat{S}_m$ must be traceable to the compliance error e_C and to the sensitivity error e_s, which weight this error differently. Section 3 constructs a condensed stiffness that meets these requirements: for a new geometry it needs no interior factorisation, even when the cut changes the discrete space, and it can be improved at deployment without retraining. Sections 3.1 and 4 derive what any recovery operator that satisfies the assumptions of Section 2.4 provides towards these requirements.

2.4. Assumptions

Sections 3 and 4 consider an approximate displacement recovery operator F, which maps the retained displacements q of a cell to the displacements Fq of all its active DOFs. Propo-

sitions 1, 3 and 4 are stated under the following assumptions, given in full in Appendix B.1, and Proposition 2 also uses (A1) and (A2).

- (A1) The cell stiffness matrix K is symmetric positive semidefinite, its null space consists of the rigid-body modes, $A = K_{II} \succ 0$, and interior body loads vanish (Section 2.2).
- (A2) The recovery operator F is linear, reproduces the retained displacements, $J_P F = I_p$, and reproduces rigid-body motions, $F R_P = R$.
- (A3) Cells share only retained DOFs, and the supported assembled stiffness matrix $\mathbb{K}$ is positive definite (Section 2.3).
- (D) On a design interval, the discrete choices listed in Appendix B.1 are fixed: the active elements, the ghost-penalty faces, the retained and interior DOF sets, the assembly maps with the supports, and the load.

3. The NICE method

NICE is defined by the corrected recovery operator F, which maps the retained displacements of a cell to the displacements of the whole cell; the condensed stiffness $\widehat{S} = F^T K F$ follows from the strain energy of the recovered displacement field, and $F = E$ gives exact static condensation. Section 3.1 computes the condensed stiffness from the strain energy and applies it with the transpose of F; for any recovery operator that reproduces the retained displacements and rigid-body motions, the condensed stiffness is then symmetric positive semidefinite and never softer than the exact one. The interior displacements are predicted by a network that takes the cell geometry as input, and the resulting recovery is exactly linear in the retained displacements and reproduces them by construction (Section 3.2). A fixed two-grid correction $\mathcal{W}$ is applied to the predicted displacements before the strain energy is computed and reduces the interior force imbalance left by the network (Sections 3.3 and 3.4); the network is trained with this correction applied (Section 3.5). Section 3.6 describes how the condensed stiffness is applied, and Figure 2 shows how these parts define both the action of the condensed stiffness and the displacements recovered after assembly.

3.1. Condensed stiffness from the strain energy

The condensed stiffness is defined through the strain energy, under the original stiffness K, of a recovered displacement field. Let F be a linear recovery operator that maps the retained displacements q to the displacements Fq of the whole cell and reproduces the retained displacements, $J_P F = I_p$, and the rigid-body motions, $F R_P = R$, where R contains the rigid-body modes of Section 2.2; together with linearity, these conditions form assumption (A2) of Section 2.4. In NICE, F consists of the network recovery operator $\widehat{E}$ followed by the two-grid correction $\mathcal{W}$, both described at the end of this section. The recovery operator and its condensed stiffness are

$$F = \mathcal{W}\widehat{E}, \qquad \widehat{S} = F^T K F, \qquad \widehat{\mathcal{U}}(q) = \tfrac{1}{2} q^T \widehat{S} q, \qquad \widehat{f}_P = \widehat{S} q. \tag{4}$$

Here $\widehat{\mathcal{U}}(q)$ is the strain energy of the recovered field, and $\widehat{f}_P$ are the corresponding nodal forces on the retained DOFs. The definitions of $\widehat{S}, \widehat{\mathcal{U}}$ and $\widehat{f}_P$ in Eq. (4) apply to any recovery operator F; only the relation $F = \mathcal{W}\widehat{E}$ is specific to NICE. The form $\widehat{S} = F^T K F$ coincides

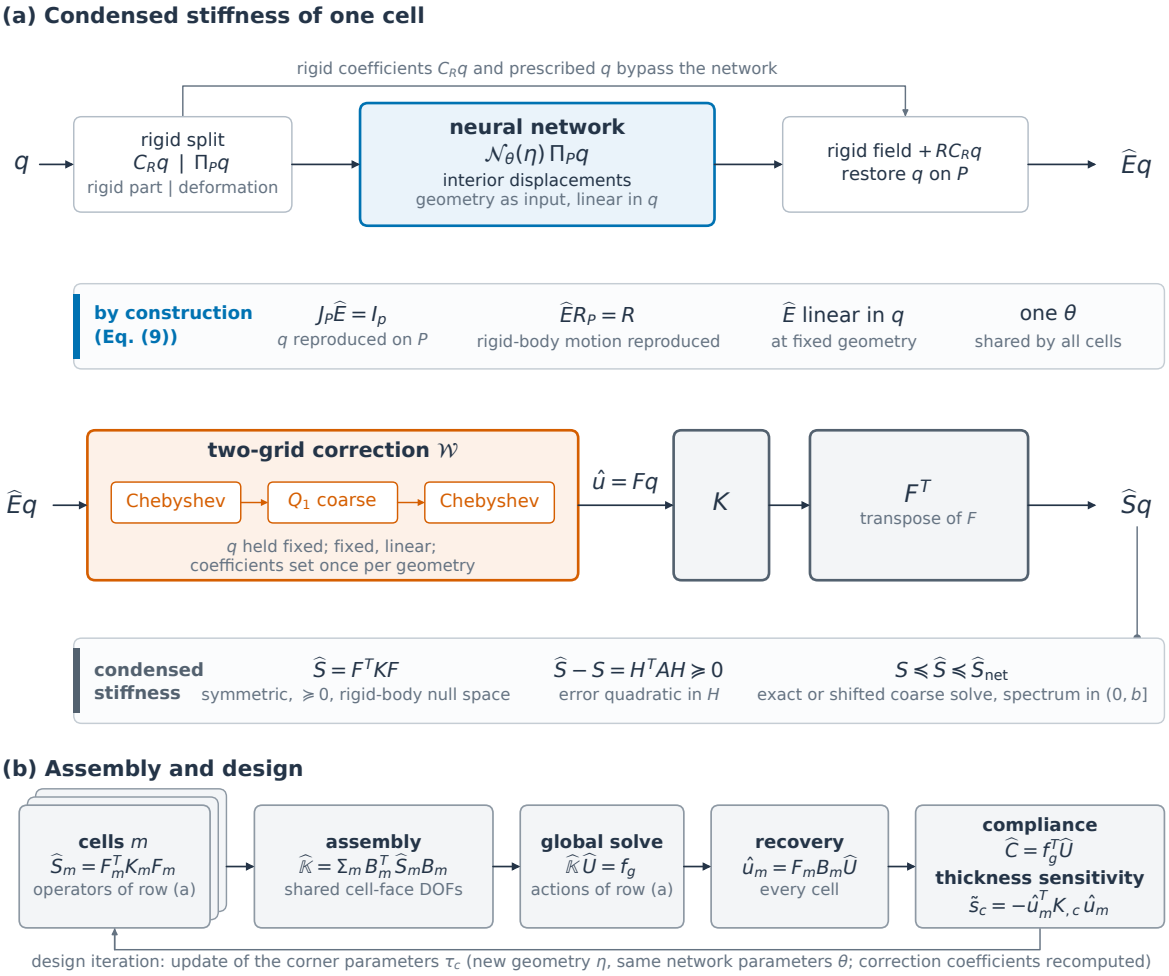


Figure 2: **Overview of NICE: condensed stiffness of one cell, assembly and design.** (a) The rigid-body part of the retained displacements is separated before the network predicts the interior displacements, so that the network acts only on the deformation; the rigid-body field is then added back and the prescribed retained displacements are restored, which gives $\widehat{E}q$. The two-grid correction $\mathcal{W}$ reduces the interior force imbalance with the retained displacements held fixed, which gives $\hat{u} = Fq$, and applying the cell stiffness K and then the transpose F^T gives the corresponding nodal forces on the retained DOFs, $\widehat{S}q$. The ordering $S \preceq \widehat{S} \preceq \widehat{S}_{\text{net}}$ in (a), where $\widehat{S}_{\text{net}}$ is the condensed stiffness without correction, is that of Proposition 2; it concerns the energy error and does not apply to the sensitivity error. (b) The condensed stiffness matrices of the cells are assembled on the shared cell-face DOFs and the assembled system is solved; the displacements of every cell are recovered from the assembled solution, and the compliance and the thickness sensitivity are computed. A design iteration updates the corner thickness parameters and therefore changes the geometry of every cell; the network parameters stay the same, and the coefficients of $\mathcal{W}$ are recomputed.

with that of substructure methods in which the condensed stiffness matrix is computed from a matrix of numerical shape functions N as $N^T K N$ Huang et al. (2023); NICE, however, does not form F. For a virtual retained displacement δq, the virtual displacement of the whole cell is $F\delta q$, and the variation of the strain energy is $\delta\widehat{\mathcal{U}} = (F\delta q)^T KFq = \delta q^T F^T KFq$, so applying the condensed stiffness requires the transpose F^T after the stiffness action. In NICE, this transpose includes every step of F: the rigid-body reconstruction, the restoration of the retained displacements and the correction. With $F_I = J_I F$, the action of the condensed stiffness has the block form

$$\widehat{S}q = (KFq)_P + F_I^T (KFq)_I. \tag{5}$$

The second term transfers the work of the interior residual $(KFq)_I$, the interior force imbalance of the recovered field, to the retained DOFs; it vanishes when the recovered field is in interior equilibrium and is otherwise needed for the returned forces to be the derivative of the strain energy $\widehat{\mathcal{U}}$. The symmetry and positive semidefiniteness of $\widehat{S}$ follow from $K = K^T \succeq 0$ and do not require the gather, scatter and grid-transfer maps of the network to be symmetric (Section 3.2). Since the null space of K consists of the rigid-body modes (Section 2.2) and $FR_P = R$, the only zero-energy modes of $\widehat{S}$ are the retained rigid-body modes (Appendix B). The transpose F^T is derived in Appendix E, and Appendix B.2 shows that, with this transpose, the error of the condensed stiffness is quadratic in the interior error.

Under assumption (A1), the exact field Eq has the smallest strain energy among all displacement fields with retained displacements q, by the principle of minimum potential energy. Any recovery operator F with $J_P F = J_P E = I_p$ differs from E only in the interior DOFs; let $H = J_I(F - E)$ denote this interior error. Because the exact field is in interior equilibrium, $J_I KE = 0$, the cross terms between E and H vanish from $\widehat{S} = F^T KF$ (Appendix B.2).

Proposition 1 (Ritz identity). Under (A1) and (A2), the condensed stiffness of F exceeds the exact condensed stiffness S by the energy of the interior error:

$$\boxed{\widehat{S} - S = H^T A H \succeq 0.} \tag{6}$$

An approximate recovery operator therefore adds stiffness equal to the energy of its interior error; the error has no first-order term, because the exact field is in interior equilibrium, and exact static condensation is the case $F = E$, $H = 0$. For a given retained displacement q, called a test displacement, the accuracy of $\widehat{S}$ is measured by the relative energy error $\varepsilon(q) = q^T(\widehat{S} - S)q/(q^T Sq)$, which is nonnegative by Eq. (6). Let $u = Eq$, $\widehat{u} = Fq$ and $d_I = Hq$; the interior residual $r_I = (K\widehat{u})_I$ satisfies $r_I = Ad_I$, so that the energy error is

$$\varepsilon(q) = \frac{d_I^T A d_I}{q^T S q} = \frac{r_I^T A^{-1} r_I}{q^T S q}. \tag{7}$$

The residual can be computed without the exact recovery E, but evaluating its A^{-1}-norm requires one interior solve per test displacement, and a bound on the energy error over all retained displacements follows only from how well the sampled test displacements cover them (Appendices B and G.4). Proposition 1 and the residual form of Eq. (7) are classical results Fraeijs de Veubeke (1965), Toselli & Widlund (2005), Becker & Rannacher (2001).

In NICE, $F = \mathcal{W}\widehat{E}$. Section 3.2 constructs the network recovery operator $\widehat{E}$, which reproduces the retained displacements, $J_P\widehat{E} = I_p$, and the rigid-body motions, $\widehat{E}R_P = R$, by construction (Eq. (9)). The correction $\mathcal{W}$ is one symmetric two-grid cycle applied with the retained displacements held fixed (Sections 3.3 and 3.4); it is linear and leaves the retained displacements unchanged, and taking $\mathcal{W}$ as the identity gives the uncorrected recovery $F = \widehat{E}$. For each geometry, the steps of the cycle and their coefficients are fixed independently of the retained displacements, and a correction driven by the interior residual, such as $\mathcal{W}$, leaves a rigid-body field unchanged (Appendix B.2). The corrected operator F thus also reproduces the retained displacements and the rigid-body motions, and Proposition 1 applies to it. By Eq. (6), the error of the condensed stiffness is the energy of the interior error of F, which Sections 3.3 and 3.4 reduce. After the assembled system is solved, $F_m B_m \widehat{U}$ gives the displacements of cell m for evaluating the response and the sensitivities.

3.2. Network for the interior displacements

The network approximates the interior part of the exact recovery, the map $q \mapsto E_I q = -A^{-1}K_{IP}q$; this approximation yields a displacement field and, through its strain energy, the condensed stiffness on all retained DOFs (Section 3.1). Since a cell has 2,679 to 45,900 retained DOFs and this set changes with the cut, two direct approaches are unsuitable. Forming the recovery matrix for every geometry requires one interior solve per retained DOF, and a network with one output per retained DOF would need an output dimension that changes with the cut.

The relations of Sections 3.1 and 4 rest on three properties, which the architecture enforces exactly by construction: (i) at fixed geometry the recovery is linear in q, since all nonlinearity is confined to a geometry branch and every operation on the displacement features is linear; (ii) the recovery reproduces the retained displacements and the rigid-body motions, since the rigid-body motion is separated before the network and reconstructed after it, and the retained displacements are overwritten at the output (Eq. (9)); (iii) the transpose of the recovery, which the nodal forces of Eq. (5) require, is obtained by transposing the linear displacement path operation by operation (Appendix G.1). Section 5.2 verifies the three properties numerically.

The remaining design choices determine how well the network approximates the map, and they were not tested individually by ablation. For each active element and each element face on which the ghost penalty acts, the network uses a fixed set of nodes, called a stencil. The weights of the network act on the local node positions of the stencils, on the feature channels and on the grid levels, and the coefficients of each stencil are computed from the geometry, so one network serves the sets of retained DOFs of all geometries within the range covered by training. Because the material boundaries intersect the background elements, the element moments, the integrals of local monomials over the material part of each background element (Appendix A.1), enter as geometry input. The map from the retained displacements to the interior displacements couples distant parts of the cell, and this long-range coupling is carried by a U-shaped grid hierarchy, which passes the features to successively coarser grids and back. Additional local interactions act on the stencils that contain weakly connected nodes, which are coupled to the rest of the cell by few and small stiffness entries.

The network therefore has two branches (Figure 3): a nonlinear geometry branch computes, once per geometry, the coefficients of a linear operator, and the displacement branch

is this linear operator.

The rigid-body motion is separated before the network is applied. With the columns of R taken as three translations and three rotations about a common centre, the matrix $C_R = (R_P^T R_P)^{-1} R_P^T$ extracts the rigid-body coefficients and the projector $\Pi_P = I_p - R_P C_R$ removes the rigid-body part, which splits the input into the rigid-body coefficients $C_R q$ and the retained deformation $q_d = \Pi_P q$. The network acts only on q_d, so the reconstruction of the rigid-body motion does not depend on how well the network is trained.

In the geometry branch, the element moments describe the material distribution within each background element, and the node features indicate whether a node carries retained DOFs, whether it belongs to the cut-plane elements and whether it is weakly connected, and also contain the local stiffness magnitude and the node position. A node is weakly connected when the norm of its diagonal stiffness block is below 1% of the median over the active nodes (Appendix G). Encoders map these inputs to element and node feature vectors, which exchange information in two rounds, and coefficient heads, small output networks, then compute the geometry-dependent coefficients: the weights of the fixed element–node and ghost-face–node connections, of the transfers between the grids of the hierarchy and of the coarse-grid convolutions. For a given geometry, these coefficients do not depend on the retained displacements.

The displacement branch lifts the three components of q_d to 32 feature channels on the retained nodes, the nodes that carry retained DOFs, and sets the features of the interior nodes to zero, which defines the initial features $X^0(q)$. Local interactions propagate these values through the neighbourhoods of the elements and of the ghost-penalty faces. Each interaction gathers the features at the 27 local node positions of a stencil with four separate sets of geometry-dependent weights, called heads, mixes the channels within each head and scatters the result back to the nodes of the stencil. With M_I and M_P the masks of the interior and retained nodes, acting on nodal rows, one interaction updates the features by

$$X^{\ell+1} = \mathsf{M}_I \left[X^\ell + \sum_h \mathcal{S}_{\ell h}(\eta) \Big(\mathcal{G}_{\ell h}(\eta) X^\ell W_{\ell h} \Big) \right] + \mathsf{M}_P X^0(q). \tag{8}$$

Here $\mathcal{G}_{\ell h}$ and $\mathcal{S}_{\ell h}$ are the gather and scatter maps with geometry-dependent weights, $W_{\ell h}$ mixes the channels, and the last term restores the retained features. The U-shaped grid hierarchy, which provides the long-range coupling, passes through grids with 65, 33, 17 and 9 positions per axis, the finest of which is the Q_2 node grid of the cell. Pairs of local interactions before and after the hierarchy, and further pairs on the stencils that contain weakly connected nodes, complete the branch, and a linear map then converts the features to nodal displacements (Figure 3b,c). Because the weights do not act on individual DOFs, the same network of about 6×10^5 parameters serves every set of retained DOFs (Supplementary Table ST15). For a given channel width and stencil size, the work of one network application is proportional to the number of element and ghost-face stencils and to the size of the fixed grids, independent of how many of the nodes carry retained DOFs; one application processes a batch of retained displacement vectors at once. Appendix G gives the complete order of operations and the definitions of the coefficients.

Every operation on the displacement features is linear, and the nonlinear encoders act only on the geometry, so the raw map $\mathcal{N}_\theta(\eta)$, with network parameters θ, obeys superpo-

sition at fixed η. Adding back the rigid-body field and restoring the prescribed retained displacements gives the network recovery operator

$$\widehat{E}q = J_P^T q + J_I^T J_I \left[RC_R q + \mathcal{N}_\theta(\eta)\Pi_P q \right]. \tag{9}$$

It follows that $J_P \widehat{E} = I_p$, from the final overwrite of the retained displacements, and $\widehat{E} R_P = R$, from the separation of the rigid-body motion; both relations hold by construction. The transpose $\mathcal{N}_\theta(\eta)^T$ required by Eq. (5) is the composition of the transposed operations in reverse order (Eq. (E.1)), so that, in exact arithmetic, the returned nodal forces are the derivative of the strain energy $\widehat{\mathcal{U}}$.

The data flow of the displacement branch resembles that of an iterative solver for the interior problem, in which the retained displacements are prescribed (Dirichlet) boundary values. Each local interaction of Eq. (8) updates the interior features from their stencil neighbours with geometry-dependent coefficients and, at the same time, resets the retained features to their lifted prescribed values, and the grid hierarchy carries information between distant parts of the cell, as the coarse levels do in multigrid. The resemblance is limited to the data flow, however: the branch never applies the stiffness matrix and does not compute residuals, the equilibrium solution of the interior problem is not a fixed point of its layers, and the branch is a fixed sequence of learned linear maps. The two-grid correction of Sections 3.3 and 3.4, in contrast, acts on the residual of the same interior problem with prescribed coefficients. Because the network is trained with this correction applied (Section 3.5), it provides the part of the interior displacement field that a fixed number of correction steps does not recover.

3.3. Chebyshev smoothing

Equation (7) expresses the energy error of a cell through its interior residual, and Eq. (14) expresses the compliance error as the energy of the error of the recovered field. Both suggest reducing the interior force imbalance of the network prediction with the retained displacements held fixed. Polynomial smoothing Adams et al. (2003) and energy minimisation on a coarse grid reduce different components of this imbalance.

Let $D = \mathrm{diag}(A)$ be the diagonal of the interior stiffness matrix A, and let the generalised eigenvectors v_j, ordered by increasing eigenvalue, satisfy $Av_j = \lambda_j D v_j$ and $v_j^T D v_k = \delta_{jk}$. With the coefficients $c_j = v_j^T D d_I$ of the interior error d_I of Section 3.1, the error energy and its fraction in the first ℓ Jacobi-scaled interior modes are

$$d_I^T A d_I = \sum_j \lambda_j c_j^2, \qquad \chi_\ell(q) = \frac{\sum_{j=1}^{\ell} \lambda_j c_j^2}{d_I^T A d_I}. \tag{10}$$

The corresponding fraction for the exact interior field u_I is defined in the same way, with $u_I^T A u_I$ as its total energy.

The network prediction serves as an initial approximation to the solution of $Au_I = -K_{IP}q$, which Chebyshev semi-iteration with Jacobi preconditioning Golub & Varga (1961) improves with q held fixed. The number of iterations is prescribed and the coefficients depend on the geometry but not on q, so the corrected recovery operator remains linear in q. Smoothing a learned initial approximation resembles both the smoothing of a tentative prolongation in smoothed aggregation Vaněk et al. (1996) and hybrid solvers in which

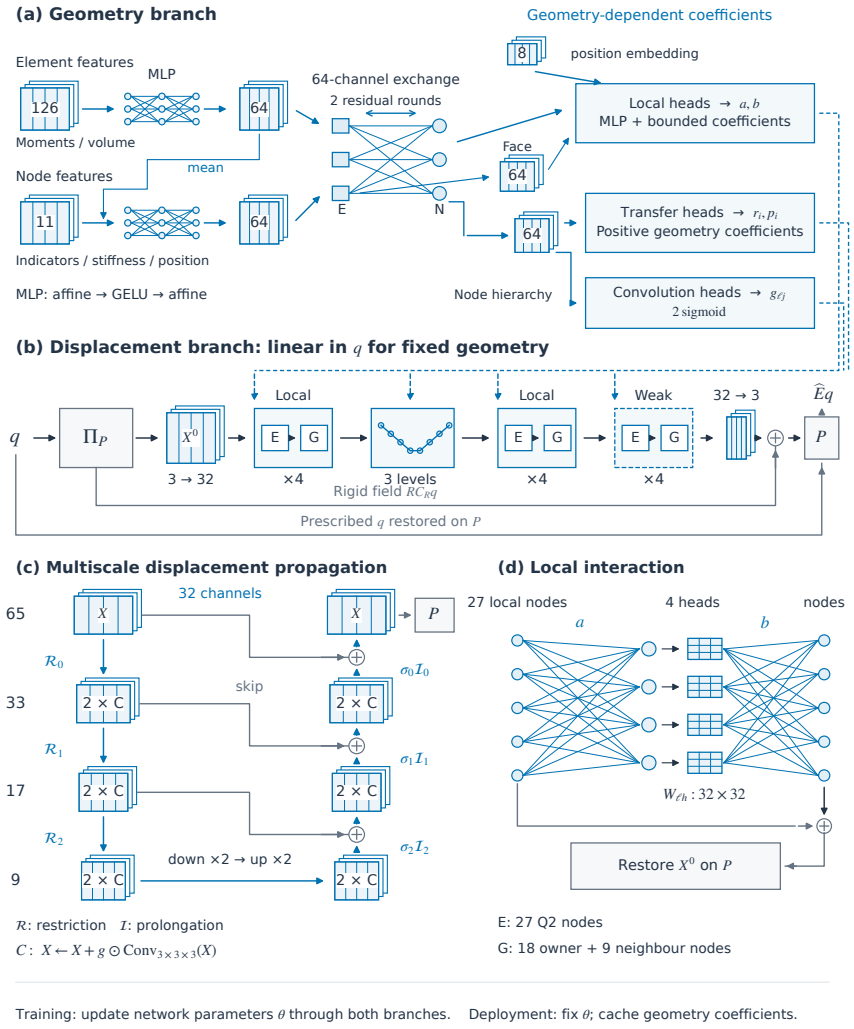


Figure 3: **Architecture of the network for the interior displacements.** (a) Geometry branch. Element and node encoders produce geometry feature vectors with 64 channels, which then exchange information in two residual rounds. Coefficient heads compute from these vectors the coefficients of the local interactions, the grid transfers and the convolutions. The heads of the local interactions also receive a position embedding, which identifies the local node position within a stencil (Appendix G.1). (b) Displacement branch. The retained displacements, with their rigid-body part removed, pass through four pairs of local interactions, the grid hierarchy, four further pairs and four pairs on stencils that contain weakly connected nodes (labelled 'Weak'). Linear maps convert the three displacement components to 32 feature channels at the input and back at the output. Bypass paths without learned parameters reconstruct the rigid-body motion and restore the prescribed retained displacements. (c) Multiscale displacement propagation. Restriction $\mathcal{R}$ and prolongation $\mathcal{I}$ connect the grids of the hierarchy, which have 65, 33, 17 and 9 background positions per axis. Each coarse grid applies two residual convolutions on the downward pass and two on the upward pass. (d) Local interaction. The features at the 27 local nodes of a stencil are gathered with geometry-dependent weights into four heads, and the channels are mixed within each head. The result is scattered back to the nodes and added to the incoming features, and the retained features are then restored. E and G denote element and ghost-face interactions. Blue dashed arrows carry geometry-dependent coefficients, and solid arrows carry features or displacements. For fixed geometry, the complete displacement path is linear.

relaxation removes the high-frequency error left by a learned prediction Zhang, E. et al. (2024).

After k smoothing steps, the interior error d_I becomes $\Phi_k d_I$, where $\Phi_k = p_k(D^{-1}A)$ and p_k is the error polynomial of degree k; each coefficient c_j in Eq. (10) is thus multiplied by $p_k(\lambda_j)$. The Chebyshev polynomial is constructed for a target interval $[a, b]$ with $0 < a < b$. If the actual positive spectrum of $D^{-1/2}AD^{-1/2}$ lies in $(0, b]$, the polynomial does not increase the error in the A-energy norm, but modes with eigenvalues below a can decay slowly and require the complementary coarse-grid correction of Section 3.4. All correction sequences in this paper use $a = b/30$, where b is a power-iteration estimate of the largest eigenvalue of $D^{-1}A$ increased by 5%; Section 5.2 verifies that b bounds the spectrum. Appendix D gives the polynomial, its recurrence, the estimate of the interval and the contraction bound.

3.4. Coarse-grid correction and the two-grid cycle

Let $\widehat{u}_I$ be the current approximation of the interior displacements, either the network prediction or a smoothed iterate, and let $r_I = (K\widehat{u})_I$ be its interior residual. The coarse basis $V \in \mathbb{R}^{i \times n_c}$ is assumed to have full column rank; Appendix F.1 treats a basis without it. With the coarse-grid matrix $A_c = V^T A V$ and $b_r = V^T r_I$, minimising the energy over $\widehat{u}_I + \operatorname{range} V$ gives

$$\widehat{u}_I^{\mathrm{c}} = \widehat{u}_I - V A_c^{-1} b_r, \qquad d_I^{\mathrm{c}} = (I_i - V A_c^{-1} V^T A) d_I, \qquad \|d_I\|_A^2 - \|d_I^{\mathrm{c}}\|_A^2 = b_r^T A_c^{-1} b_r. \tag{11}$$

The coarse-grid correction enforces interior equilibrium for every virtual displacement in the coarse space, and the last relation in Eq. (11) gives the energy removed by this projection Xu (1992). The principal coarse basis, denoted $Q_1(17)$, consists of trilinear vector functions on a grid of 17^3 vertices, restricted to the interior DOFs (Appendix F.1). All coarse spaces act on the interior DOFs only, so the retained displacements stay fixed.

The correction $\mathcal{W}$ of Section 3.1 is the two-grid cycle, which places the coarse-grid correction between two smoothing stages of k steps each. Let $C_V = I_i - V A_c^{-1} V^T A$ be the error operator of the coarse-grid correction, and let $H_{\mathrm{net}} = J_I(\widehat{E} - E)$ be the interior error of the network recovery operator. The two-grid error operator $\Phi_k C_V \Phi_k$ Hackbusch (1985), Trottenberg et al. (2001), Xu & Zikatanov (2002), Falgout et al. (2005) maps H_{net} to the interior error of F, and by Eq. (6) the error of the condensed stiffness is

$$H = \Phi_k C_V \Phi_k H_{\mathrm{net}}, \qquad \widehat{S} - S = H^T A H. \tag{12}$$

Proposition 2 (correction ordering). Let the network recovery operator $\widehat{E}$ satisfy (A2), let the coarse solve be exact or use the coarse inverse with the nonnegative shift of Appendix F.1 (Eq. (D.6)), and let $\|\Phi_k\|_A \le 1$. Then, under (A1), $S \preceq \widehat{S} \preceq \widehat{S}_{\mathrm{net}}$, where $\widehat{S}_{\mathrm{net}} = \widehat{E}^T K \widehat{E}$ is the condensed stiffness without correction (Appendix D.1). The same ordering holds for any linear correction that is fixed for a given geometry and whose interior error map Θ satisfies $\Theta^T A \Theta \preceq A$.

By Proposition 2, the correction cannot increase the error in the A-energy norm. The assembled stiffness matrices inherit this ordering and their inverses are ordered in reverse (Appendix C), so the correction moves the assembled compliance towards the reference value

(Appendix D.1). Proposition 2 guarantees the ordering but not the size of the reduction (Appendix D), which Section 5.5 measures.

Remark 1. The guarantee of Proposition 2 rests on the two conditions on the smoothing and on the coarse solve, and it covers the energy only. If the spectrum of $D^{-1}A$ extends above $a+b$, the Chebyshev polynomial amplifies the corresponding modes and the cycle can increase the energy error (Appendix D). Since the polynomial changes with the number k of smoothing steps, the ordering holds for each k separately, and the reduction need not be monotone in k. For an approximate coarse inverse G, the correction does not increase the energy error if $2G - GA_cG \succeq 0$ (Eq. (D.6)); the nonnegative shift satisfies this condition, whereas an arbitrary approximation need not. The ordering does not extend to the sensitivity: in the example of Appendix D.1, a projection halves the energy of the interior error but changes the sensitivity difference $\widetilde{s}_c - s_c$ of Eq. (17) from zero to $2t - t^2$, whereas the reference sensitivity is -1.

3.5. Training

The network is trained on sets of test displacements. Prescribed polynomial and multi-scale displacements cover a broad range of spatial variation, and responses to equilibrated nodal point loads, traction loads and spring supports, together with displacements imposed by a neighbouring cell, provide test displacements that arise from mechanical loading. The rigid-body components are removed, and each test displacement is normalised with the exact condensed stiffness to $q_j^T S q_j = 1$. For a batch of B test displacements, the loss function is

$$\mathcal{L}(\theta) = \frac{1}{B}\sum_{j=1}^{B} \log(q_j^T \widehat{S} q_j) + w_s \frac{1}{|\mathcal{J}_s|} \sum_{j \in \mathcal{J}_s} \frac{\|\widetilde{\boldsymbol{s}}_j - \boldsymbol{s}_j\|_2^2}{\|\boldsymbol{s}_j\|_2^2}. \tag{13}$$

Because $q_j^T S q_j = 1$, the first term is the mean logarithm of the energy ratio $q_j^T \widehat{S} q_j / (q_j^T S q_j)$. By Proposition 1, for a recovery operator that reproduces the retained displacements, this term is smallest when the recovered field equals the exact field for every sampled test displacement. The second term compares the thickness sensitivities $\widetilde{\boldsymbol{s}}_j$ computed from the recovered field (Section 4.2) with the reference sensitivities $\boldsymbol{s}_j$, both eight-component vectors, on the subset $\mathcal{J}_s$ of test displacements that have reference sensitivities. All trained networks reported in this paper use $w_s = 1$. The gradients of the loss with respect to the network parameters θ pass through the recovered field to the geometry encoders, the coefficient heads and the linear displacement maps. Test displacements with large energy ratios are added during training; they are found by a block search on the complement of the rigid-body modes, which iterates eight candidate test displacements at once (Appendix G.3). The geometries are augmented with the 48 symmetries of the cube (Appendix G). Table 2 lists the variants, and Supplementary Table ST01 gives the settings for training and model selection.

NICE is trained with the operator used at deployment. The loss of Eq. (13) is evaluated with the corrected recovery operator $F = \mathcal{W}\widehat{E}$, where $\mathcal{W}$ is the two-grid cycle of Table 1, written 8 / $Q_1(17)$ / 8 for 8 pre-smoothing steps, the coarse-grid correction with $Q_1(17)$ and 8 post-smoothing steps. Because $\mathcal{W}$ is linear and fixed for a given geometry, the gradient passes through the smoothing recurrence and the coarse-grid correction to the network parameters, as in solver-in-the-loop training Um et al. (2020). The smoothing interval and

the factorisation of the coarse-grid matrix depend only on K and are recomputed for each geometry, so the correction adds no trainable parameters.

3.6. Applying the condensed stiffness

The element moments, the geometry encoding of the network with its coefficients and maps, the smoothing interval and the factorisation of the coarse-grid matrix depend on the geometry but not on the retained displacements. They are computed once for each geometry and reused for all retained displacements, and Supplementary Table ST13 gives the cost of this preparation. Each product of $\widehat{S}$ with a vector of retained displacements then requires one forward pass of the network, one two-grid cycle, one product with K and the transposes of the cycle and of the network (Appendices E–F). Neither $\widehat{E}$ nor $\widehat{S}$ is formed as a matrix, and each cell stores its geometry-dependent coefficients and the factorisation of its coarse-grid matrix. By comparison, a dense $\widehat{S}$ for a cell with the median number of retained DOFs, 23,604, would take 4.2 GiB in double precision.

Algorithm 1. Action of the corrected condensed stiffness.

1. Compute $\widehat{E}q$ with the network, and apply the two-grid cycle $\mathcal{W}$ with the retained displacements held fixed to obtain $\widehat{u} = Fq$.
2. Compute $y = K\widehat{u}$ and return $F^T y$ by applying $\mathcal{W}^T$ and then the transpose of the network recovery operator.
3. Assemble these cell actions through Eq. (3). After the global solve, recover the cell displacements $F_m B_m \widehat{U}$ to evaluate the displacement field and the sensitivities.

4. Effect of interior displacement errors on compliance and sensitivities

Because all retained DOFs are kept, a cell is approximated only through its interior displacements. For any recovery operator that satisfies assumption (A2) of Section 2.4, this section traces the error of the operator through the condensed stiffness to the two quantities used in design, the compliance and the sensitivities. By Proposition 1 (Section 3.1), the condensed stiffness exceeds the exact one by the energy of the interior error. It is therefore an upper bound on the exact condensed stiffness, with an error of second order in the interior displacement error that can be computed from the interior residual (Eq. (7)). This error reaches the assembled compliance weighted by the fraction of the strain energy that each cell carries (Section 4.1), and it reaches the thickness sensitivity through the derivative of the stiffness matrix, with a term of first order in the interior displacement error (Section 4.2). An accurate compliance therefore does not imply an accurate local sensitivity.

The relations below are stated for a single substructure and apply to every cell of an assembly. Proposition 1, the residual form of Eq. (7) (Section 3.1) and the self-adjoint compliance sensitivity Haftka & Gürdal (1992), Bendsøe & Sigmund (2004) are classical. The bound weighted by the energy fractions (Proposition 3) and the sensitivity relations (Proposition 4) are derived in this paper for approximate recovery operators under the assumptions of Section 2.4.

4.1. Compliance

Consider the exact solutions of the reference and the approximate assembled systems of Eq. (3), which use the same cell stiffness matrices, assembly maps and homogeneous supports and the same nonzero load on the retained DOFs. Let $u_m = E_m B_m U$ and $\widehat{u}_m = F_m B_m \widehat{U}$ be the displacements of cell m recovered from the two solutions, and let $a(v, v) = \sum_m v_m^T K_m v_m$ be the energy form of the assembly. Global equilibrium and the energy orthogonality within each cell then give (Appendix C)

$$C - \widehat{C} = a(\widehat{u} - u, \widehat{u} - u) = \|\widehat{U} - U\|_{\mathbb{K}}^2 + \sum_m \|H_m B_m \widehat{U}\|_{A_m}^2. \tag{14}$$

The compliance is therefore underestimated by the total energy of the error of the recovered displacement field. This energy consists of two orthogonal parts: the first comes from the change of the assembled retained solution, and the second from the departure of the interior displacements from equilibrium at that solution. How much of the error of a cell reaches the compliance depends on the strain energy that the cell carries. At the exact assembled retained displacements $q_m = B_m U$, the energy fraction of cell m is $w_m = q_m^T S_m q_m / C$, and these fractions sum to one. With ε_m the energy error of cell m defined in Eq. (7), the weighted energy error is $\beta = \sum_m w_m \varepsilon_m(q_m)$; a cell whose response is a rigid-body motion has zero energy, and its contribution to β is taken as zero.

Proposition 3 (energy-weighted compliance error). Under (A1)–(A3), the relative compliance error is bounded by the weighted energy error β (Appendix C.1):

$$\boxed{0 \le \frac{C - \widehat{C}}{C} \le \frac{\beta}{1 + \beta} \le \beta.} \tag{15}$$

A cell with a small energy fraction can therefore leave the compliance accurate even when its own interior displacements are inaccurate.

Equation (14) also separates the error of the condensed stiffness from the error of an incomplete assembled solve, and the numerical checks of Section 5.8 use this separation. Let $\bar{U}$ be an approximate solution of the assembled system, with the recomputed residual $\rho = f_g - \widehat{\mathbb{K}}\bar{U}$ and the recovered cell displacements $\bar{u}_m = F_m B_m \bar{U}$. Then

$$C - f_g^T \bar{U} = a(\bar{u} - u, \bar{u} - u) + \bar{U}^T \rho, \tag{16}$$

and the signed residual work $\bar{U}^T \rho$ measures the effect of stopping the solve early (Appendix C.2).

4.2. Sensitivities

Consider a differentiable design interval on which the active and retained DOFs, the assembly maps and the homogeneous supports are fixed and the load does not depend on the design. On such an interval, the exact compliance sensitivity is $s_c = -u^T K_{,c} u$, where $K_{,c} = \partial K / \partial \tau_c$, and for a parameter that affects several substructures their contributions are summed. The design derivative of the exact recovery E does not appear in s_c, because the exact field is in interior equilibrium Giles & Pierce (2000), Haftka & Gürdal (1992). The sensitivity computed from the recovered field, $\widetilde{s}_c = -\widehat{u}^T K_{,c} \widehat{u}$, uses the same stiffness

derivative and ignores the dependence of the recovery operator F, and hence of the network, on the design. Over the eight corners, s_c and $\widetilde{s}_c$ form the vectors $\boldsymbol{s}$ and $\widetilde{\boldsymbol{s}}$.

Proposition 4 (sensitivity error). Under (A1), (A2) and (D), with both sensitivities evaluated at the same retained displacements:

(a) the sensitivity $\widetilde{s}_c$ differs from the exact sensitivity by a term linear and a term quadratic in the interior displacement error,

$$\boxed{\widetilde{s}_c - s_c = -2d^T K_{,c} u - d^T K_{,c} d, \qquad u = Eq, \quad d = Fq - Eq;} \tag{17}$$

(b) with $\mathcal{E} = d^T K d$ the energy of the interior error, $\|\widetilde{\boldsymbol{s}} - \boldsymbol{s}\|_2 \le 2L\sqrt{\mathcal{E}} + Q\mathcal{E}$, where the constants L and Q of Eq. (H.4) depend on the cell, the test displacement and the stiffness derivatives. The energy error therefore controls the relative sensitivity error only at the order $\sqrt{\varepsilon}$ (Appendix H.1).

Interior equilibrium gives $(Ku)_I = 0$, but in general $(K_{,c}u)_I \neq 0$, so the linear cross term remains and a decrease of the energy error need not decrease the sensitivity error monotonically. Thickening a corner enlarges the material domain, so that $K_{,c} \succeq 0$ under exact integration (Eq. (H.7)); the quadratic term of Eq. (17) is then nonpositive, whereas the linear term can have either sign (Appendix H.2).

The sensitivity $\widetilde{s}_c$ is not the derivative of the approximate compliance $\widehat{C}$, because $\widehat{C}$ also depends on the design through F. For a parameter that affects one substructure, differentiation with the retained DOFs fixed gives

$$\widehat{C}_{,c} = -\widehat{u}^T K_{,c} \widehat{u} - 2\widehat{q}^T F_{,c}^T K \widehat{u} = \widetilde{s}_c - 2(F_{I,c}\widehat{q})^T r_I, \qquad \widehat{u} = F\widehat{q}, \tag{18}$$

where $\widehat{q}$ is the assembled retained solution obtained with $\widehat{S}$, $r_I = (KF\widehat{q})_I$ and $F_{I,c} = J_I \partial F/\partial \tau_c$. The second term vanishes when the recovered field is in interior equilibrium, but energy accuracy alone does not control it. Whether Eq. (18) or $\widetilde{s}_c$ gives the more accurate gradient depends on whether $\|H_{,c}q\|_A$ is small compared with $\|E_{I,c}q\|_A$ (Eq. (H.6)), where $H_{,c}$ and $E_{I,c} = J_I E_{,c}$ are the design derivatives of the interior error of F and of the exact interior recovery (Appendix H.2). All sensitivities reported in this paper are $\widetilde{s}_c = -\widehat{u}^T K_{,c} \widehat{u}$, which estimate the exact sensitivity and ignore the design dependence of the network. The derivatives in this section hold on intervals on which the discrete choices of Appendix B.1 are fixed. Across a switch of the discrete model the reference compliance itself can jump, but since $0 \le C - \widehat{C} \le \beta C$ holds at every design by Eq. (15), any jump that the approximate compliance adds is bounded by its own error level (Supplementary Note S4.3). Appendix H specifies the numerical stiffness derivatives.

These relations imply that a condensed stiffness based on network-predicted interior displacements must be checked separately for the compliance and for the sensitivities. By Proposition 3, the relative compliance error is bounded by the energy errors of the cells at the exact assembled retained displacements, weighted by their energy fractions. By Proposition 4, the energy error of a cell controls its relative sensitivity error only at the order $\sqrt{\varepsilon}$. Section 5.6 checks both quantities in two-cell assemblies, and Section 5.10 checks them in thickness designs.

5. Numerical examples

The numerical examples cover single cells, assemblies of cells and thickness design; Sections 5.1 and 5.2 give the settings and verify the reference model. On single cells (Sections 5.3–5.5), the mean energy error of NICE over the 80 validation geometries is 0.074%. Applying the correction to the base network without retraining lowers the mean energy error under traction loads from 6.89% to 0.0965%, and under the same correction a graph-harmonic initial field without trainable parameters leaves 5.4 to 265 times the error of NICE. In assemblies (Sections 5.6–5.8), the compliance error follows the relation of Proposition 3, in which the energy error of each cell is weighted by its energy fraction, whereas the local sensitivity has to be checked separately. In design, an analysis of an eight-cell lattice with sensitivities is about ten times faster than a direct solution of the whole lattice (Section 5.9). For the optimised designs, the compliance and the gradient computed with NICE agree with those of exact static condensation (Section 5.10). Table 3 summarises the accuracy of NICE against exact static condensation in all assembled examples.

5.1. Geometries, variants and loads

The 80 validation geometries form four cut-severity groups of 20 cells each: uncut, lightly cut, moderately cut and heavily cut. Their thickness fields are uniform, affine or mixed trilinear, and their corner parameters range from 0.1762 to 0.6983 (Supplementary Table ST02). Each cell has at most one planar cut, whose normal is $(\cos\vartheta, \sin\vartheta, 0)$ with $0 < \vartheta < \pi/4$; a heavy cut retains less than one third of the volume of the cell box, a moderate cut between one and two thirds, and a light cut more than two thirds. The cells used for detailed comparisons are U1, U2, L1, M1, M2 and H1–H3 (Supplementary Section R1), where U, L, M and H denote uncut, lightly, moderately and heavily cut cells; four of them are shown in Figure 4. Table 1 lists the discretisation and correction settings.

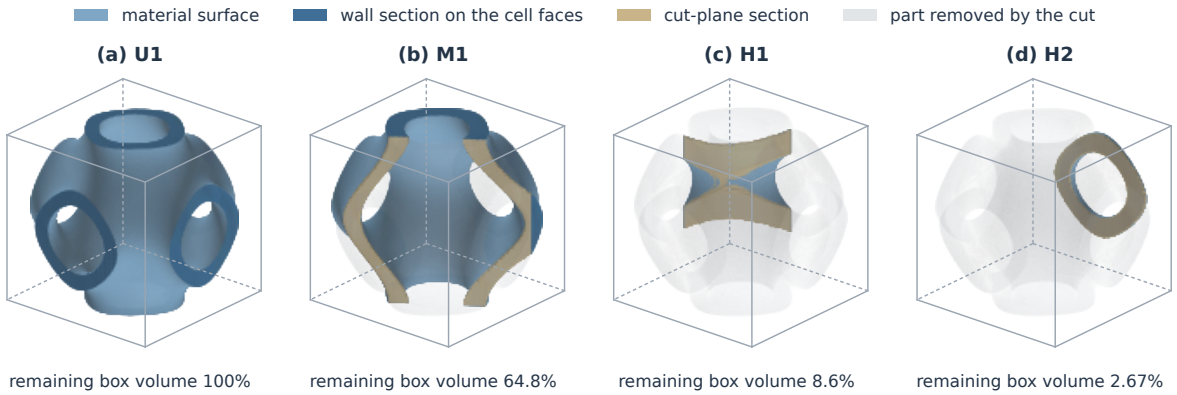


Figure 4: **Representative validation geometries.** (a) Uncut cell U1; (b) moderately cut cell M1; (c,d) heavily cut cells H1 and H2. All panels use the same unit-box scale and viewing direction and show the cells from the side of the cut. Light blue shows the material surface, dark blue the material section on the cell faces, sand the material section on the cut plane and translucent grey the part removed by the cut. The percentages give the box volume that remains after the cut, before the intersection with the thin-walled material, relative to the unit box. The surfaces are reconstructed from Eq. (1) on a grid of 129 points per axis and serve only for visualisation.

Each validation geometry is evaluated with the test displacements of the nine load classes of Supplementary Table ST03, which are defined in Appendix G.2. The classes comprise imposed polynomial and multiscale displacements, the responses to equilibrated nodal point loads and to traction loads applied on all cell faces or on a single face, the responses with spring supports, and displacements imposed by a neighbouring cell. Traction loads, which represent surface loads and the tractions exerted by neighbouring cells, form the principal load class, with 64 test displacements per geometry. Equal nodal point loads also act on weakly connected nodes and are a severe test of the stabilised problem. In the five cells of Supplementary Table ST04, the ghost penalty carries on average less than 0.05% of the energy of the exact field under traction loads, but 49–81% under equal nodal point loads.

Table 1: Discretisation and correction settings.

Quantity	Setting
Reference geometry	Unit box; Schwarz-P-type implicit band with trilinear corner parameters and an optional planar cut
Displacement approximation	Continuous tensor-product Q_2 solid elements on a Cartesian background mesh
Background resolution, validation geometries	$n = 32$ elements per axis; 65 Q2 nodes per axis
Isotropic material, validation geometries	Normalised Young's modulus $E_Y = 1$, Poisson's ratio $\nu = 0.3$
Ghost-penalty coefficient, validation geometries	$\gamma = 10^{-4}$
Retained DOFs	DOFs of the cell-face nodes of material patches of positive area on the cell faces; all DOFs of the active elements that carry a cut-surface patch of positive area (cut-plane elements)
Standard volume integration	4^3 initial subcells; one local refinement of partially filled subcells; clipped Kuhn tetrahedra with quadrature-rule parameter 4
Smoothing interval	$[b/30, b]$, with b equal to 1.05 times a 40-step power-iteration estimate of $\lambda_{\max}(D^{-1}A)$ (Section 3.3)
Principal coarse space	Trilinear vector functions on a 17^3-vertex grid, restricted to the interior DOFs
Finite-difference step of the thickness derivative	$h_c = 10^{-5}\tau_c$, with the active DOFs and the ghost-penalty contribution held fixed
Two-grid correction of NICE	8 Chebyshev steps, coarse-grid correction with $Q_1(17)$, 8 Chebyshev steps; same sequence in training and evaluation
Floating-point precision	Network in single precision; stiffness matrix–vector products and energies in double precision; correction in double precision in the accuracy studies, and in single precision in the timed analyses of Section 5.9 and in the optimisations of Section 5.10 (Appendix F.2)

The mesh, material and stabilisation settings apply to all 80 validation geometries; lengths are relative to the unit box, and the Young's modulus is normalised. For other correction sequences, the coarse-space dimensions and the numbers of smoothing steps are given with the comparisons in which they appear.

The main text compares three variants, two of which are the trained networks of Table 2. The base network was trained without correction, and NICE continues the training of the base network for 15,000 training steps with the complete correction $\mathcal{W}$ of Sections 3.3 and 3.4 inside the training loop (Section 3.5).

Table 2: Variants compared in the numerical examples.

Variant	Network parameters	Training set (geometries)	Correction in training	Correction at evaluation
NICE	Base network continued for 15,000 training steps	591	8 / $Q_1(17)$ / 8	8 / $Q_1(17)$ / 8
Base network	40,000 training steps; parameters of step 30,000 selected	305	None	None

The third variant, denoted 'Base network + correction', is the base network with the correction $\mathcal{W}$ applied at deployment, without retraining, and is evaluated in Sections 5.3, 5.5 and 5.6.

All variants use the network architecture of Section 3, with about 6×10^5 trainable parameters (Supplementary Table ST15). NICE draws its training geometries from a set of 591 geometries that contains 304 of the 305 training geometries of the base network, and in its 15,000 training steps it sees at most 153 of them (Supplementary Table ST01). Supplementary Note S8 reports two further continuations of the base network, which draw from the same set and see the same geometries in the same order as NICE.

Twenty of the 80 validation geometries, 6 uncut and 14 cut, were used for checkpoint selection (Supplementary Table ST01), so statistics are also given for the 60 geometries outside checkpoint selection. The test cells of the two-cell assemblies of Section 5.6 belong to these 20 selection geometries, and nine further cells outside checkpoint selection are assembled in that section as an additional check. NICE was designated the principal variant after all variants, including the two continuations of Supplementary Note S8, had been compared on the 80 validation geometries and the two-cell configurations (Supplementary Table ST01); the sixteen cells of the lattices of Section 5.8 entered no selection step. Population statistics first average the energy error over the test displacements of each geometry and load class and then weight the geometries equally (Appendix A.1), so that the population maximum is the largest mean of a single geometry.

Table 3: Accuracy of NICE against exact static condensation in the assembled examples. Compliance error: $|\widehat{C}/C - 1|$. Sensitivity error: largest relative error e_s of the eight-component thickness-sensitivity vector of a cell. Gradient error: relative error of the lattice gradient with respect to the shared vertex parameters. Entries are maxima or ranges over the listed loads and designs. In the two-cell assemblies only the test cell is learned; in all other examples every cell is learned.

Example	Section	Cells (cut)	Loads or designs	Compliance error (%)	Sensitivity or gradient error (%)
Two-cell assemblies, seven selection cells	5.6	2, 14 configurations	six face loads each	≤ 0.056	sensitivity ≤ 0.67
Two-cell assemblies, cut-surface traction loads	5.6	2, 7 configurations	three cut-surface traction loads each	≤ 0.10	sensitivity ≤ 0.16
Two-cell assemblies, nine held-out cells	5.6	2, 18 configurations	six face loads each	≤ 0.27	sensitivity ≤ 1.49
Lattices with every cell learned, face loads	5.8	8 (4), 8 (3)	three face loads	≤ 0.015	sensitivity ≤ 0.144; gradient 0.04–0.09
Lattices with every cell learned, random loads	5.8	8 (4), 8 (3)	three random loads	≤ 0.069	sensitivity ≤ 0.454; gradient 0.22–0.28
Case A, thickness optimisation	5.10	8 (4)	designs at iterations 0, 12, 23	0.011–0.028	gradient 0.069–0.33
Plate supported on its cut	5.10	24 (8)	start, final, homogenisation and two uniform designs	0.011–0.032	gradient 0.029–0.32
Plates of the scale study	5.10	24 (8), 51 (12)	first design	0.013, 0.011	gradient 0.045, 0.047

5.2. Verification of the reference model and of the deployed operator

The reference model with $n = 32$ was verified by mesh refinement on seven cells: U1 was refined to $n = 40$, M1 and M2 to $n = 48$ (Figure S01), and H1, H2 and two further validation cells to $n = 64$ (Supplementary Table ST10). Except in the heavily cut cell H2, refinement changes the compliance by at most 0.28% and the thickness sensitivities by at most 0.44%; in H2, the changes reach about 1% in the compliance and 3.6% in the sensitivity. On uncut cells and assemblies of uncut cells, the compliance of the discrete model agrees with that of an independent body-fitted discretisation with quadratic tetrahedra to 0.05–0.25% (Supplementary Note S1). Varying the ghost-penalty coefficient between 10^{-5} and 10^{-3} or refining the volume integration changes the compliance and the sensitivities by at most 0.12%. Supplementary Note S1 also verifies the finite-difference stiffness derivative and its step size (Figure S01(c)). All subsequent comparisons are made against this discrete reference model (Section 2.1).

On five cells, U2, M1, M2, H1 and H2, the deployed operator is symmetric and satisfies the work–energy identity, both to relative deviations of at most 3×10^{-8}. The identity states that the work of the returned nodal forces on the retained displacements equals the energy of the recovered field. The deployed operator gives the rigid-body modes zero energy up to round-off, and its recovered field agrees with the field computed by the training implementation to within 1.1×10^{-7} (Supplementary Table ST04).

The Chebyshev smoothing of Section 3.3 is guaranteed not to increase the energy error when $b \geq \lambda_{\max}(D^{-1}A)$. On all 80 validation geometries, the endpoint b obtained from the power iteration exceeds the converged largest eigenvalue by 2.1–5.0% (Appendix D), which supports this spectral condition numerically. The Gershgorin bound is a guaranteed upper bound, but it is 4.7 to 19.4 times larger than b, with a median of 18.4, and, used as b, it would place the smoothing interval far above the actual spectrum. The cells of the lattices and optimisations of Sections 5.8–5.10 use the same power-iteration estimate, but the condition was not checked for them; the symmetry and positive semidefiniteness of $\widehat{S}$ and the ordering $\widehat{S} \succeq S$ do not depend on it.

5.3. Accuracy on single cells

In every cut-severity group, the mean energy error of NICE is about 0.1% or less (Figure 5), and under traction loads its group means are lower than those of the base network by factors of 71 to 117. From uncut to heavily cut cells, the group means of NICE rise from 0.016% to 0.109%, about sevenfold, whereas those of the base network rise from 1.14% to 12.7% (Supplementary Table ST03b). Over all 80 geometries, the mean energy errors of NICE and the base network are 0.074% and 6.89%, and their largest single-geometry means are 0.65% and 48.6%, respectively. Over the 5,120 sampled test displacements under traction loads, 99% of the energy errors of NICE lie below 0.69%, and the largest is 1.24% (Supplementary Table ST03).

Under nodal point loads, the mean energy error of the base network rises from 0.908% for uncut to 11.3% for heavily cut cells, with a largest single-geometry mean of 70.1%, whereas the group means of NICE stay between 0.013% and 0.090%. Under the other load classes, the means of NICE are of the same order (Supplementary Table ST03), and Figure S02 shows the distributions over the geometries. On the 60 geometries outside checkpoint selection, the class means of NICE differ from those over all 80 geometries by at most 0.005 percentage points. Under traction loads, the mean of NICE for the heavily cut group is higher on these 60 geometries, 0.126% against 0.109%, and the five largest single-geometry means of NICE all belong to these 60 geometries (Supplementary Table ST03).

Most of the improvement comes from the correction itself. Applying the correction to the base network at deployment, without retraining, lowers the mean under traction loads from 6.89% to 0.0965%, with a largest single-geometry mean of 0.91%, whereas continuing the training of the base network without the correction leaves its mean nearly unchanged (Supplementary Note S8). NICE continues the training with the correction inside the training loop and draws its training geometries from a larger set; this lowers the mean by a further factor of 1.31, with a 95% bootstrap interval over the geometries of 1.20–1.40 (Supplementary Table ST03). Under the same correction, a deterministic graph-harmonic initial field leaves 5.4 to 265 times the error of NICE (Section 5.5).

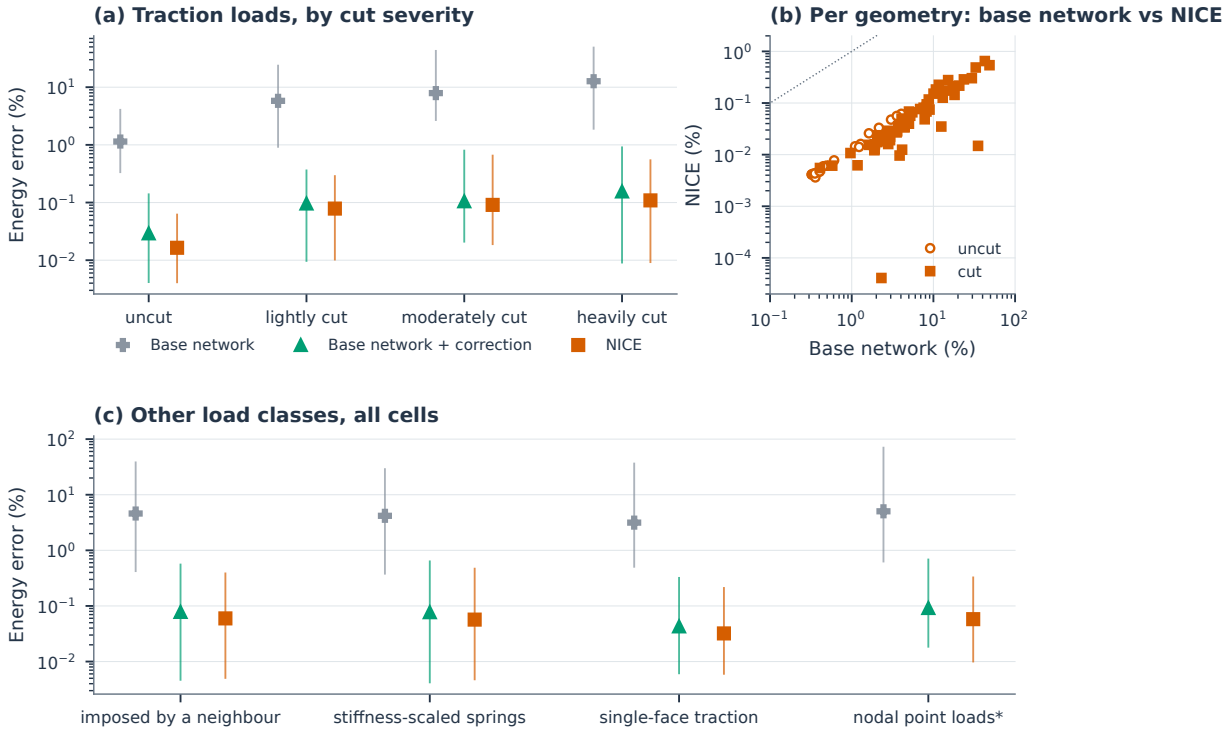


Figure 5: **Energy error of the base network, Base network + correction and NICE on the 80 validation geometries.** Each observation is the mean energy error $q^T(\widehat{S} - S)q/(q^T S q)$ of one geometry over the validation test displacements of one load class. (a) Traction loads, 20 geometries per cut-severity group. Markers give the mean over the geometries, and bars give the range from the 10th percentile to the maximum. (b) Single-geometry means of the base network and of NICE under traction loads. Open markers denote uncut geometries and filled markers cut geometries; the dotted line denotes equality. (c) Displacements imposed by a neighbouring cell, stiffness-scaled spring supports, traction loads on a single face and equal nodal point loads. The first two classes use 75 geometries and the other two use 80. The asterisk marks the nodal point loads, which serve as a severe test of the stabilised problem. Twenty geometries (6 uncut, 14 cut) were used for checkpoint selection. Variants as in Table 2; 'Base network + correction' is the base network with the correction applied at deployment, without retraining.

5.4. Where the network error lies

Part of the error left by the network lies in modes that smoothing damps slowly, and part lies next to the cut. Relative to its amplitude, the error also carries far more strain energy than the exact solution and is in this sense stiffer. This section examines the error before the correction: its distribution over the interior spectrum and in space, the reason why an interior displacement error of about 1% produces an energy error of several percent, and the split of its effect on the sensitivity between the linear and the quadratic term of Eq. (17).

The error of the base network occupies different parts of the Jacobi-scaled interior spectrum in different cells (Figure 6). In M1 under traction loads, the lowest 200 modes contain 24.5% of the error energy but only 4.87% of the energy of the exact field (Supplementary Table ST05b), and all 200 lie below the lower endpoint $a = 0.173$ of the smoothing interval, with $a = b/30$ (Table 1). In H2, only 66 of the 200 modes lie below $a = 0.138$. Smoothing damps these modes slowly and leaves them to the coarse-grid correction (Section 5.5).

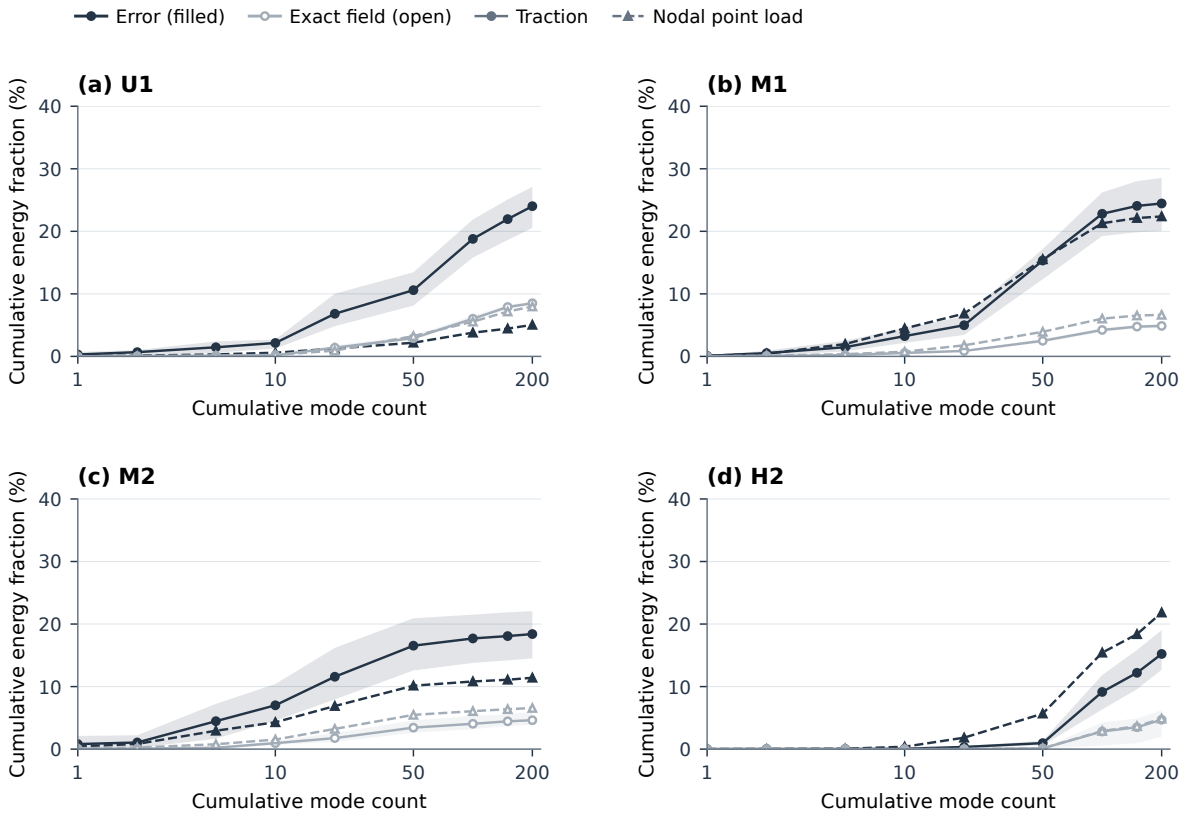


Figure 6: **Spectral distribution of the interior displacement error of the base network.** Panels show U1, M1, M2 and H2. The modes solve $Av = \lambda Dv$, with $D = \mathrm{diag}(A)$, and are ordered by increasing eigenvalue. Filled dark markers denote the error and open light-grey markers the exact interior field. Solid lines with circles denote traction loads and dashed lines with triangles nodal point loads. Curves are means over the test displacements. Bands give the 10th to 90th percentiles over the test displacements under traction loads. Each cumulative fraction is normalised by the total interior energy of its own field or error.

Part of the error of the base network lies next to the cut plane (Figure 7). In M1, the elements within two element widths of the cut plane make up 13% of the elements. For

the traction-load test displacement plotted, the exact field places 8% of its energy in these elements, whereas over the four test displacements evaluated, 39–43% of the error energy of the base network lies in this layer. The layer adjoins the retained DOFs of the cut-plane elements, whose values the network must propagate into the interior. Relative to the base network, NICE reduces the total error energy by about two orders of magnitude, by factors of 113 to 138, and the element error energies by factors of roughly 25 to 300 (10th to 90th percentiles). NICE also halves the fraction of the error energy in this layer, to 19–22%.

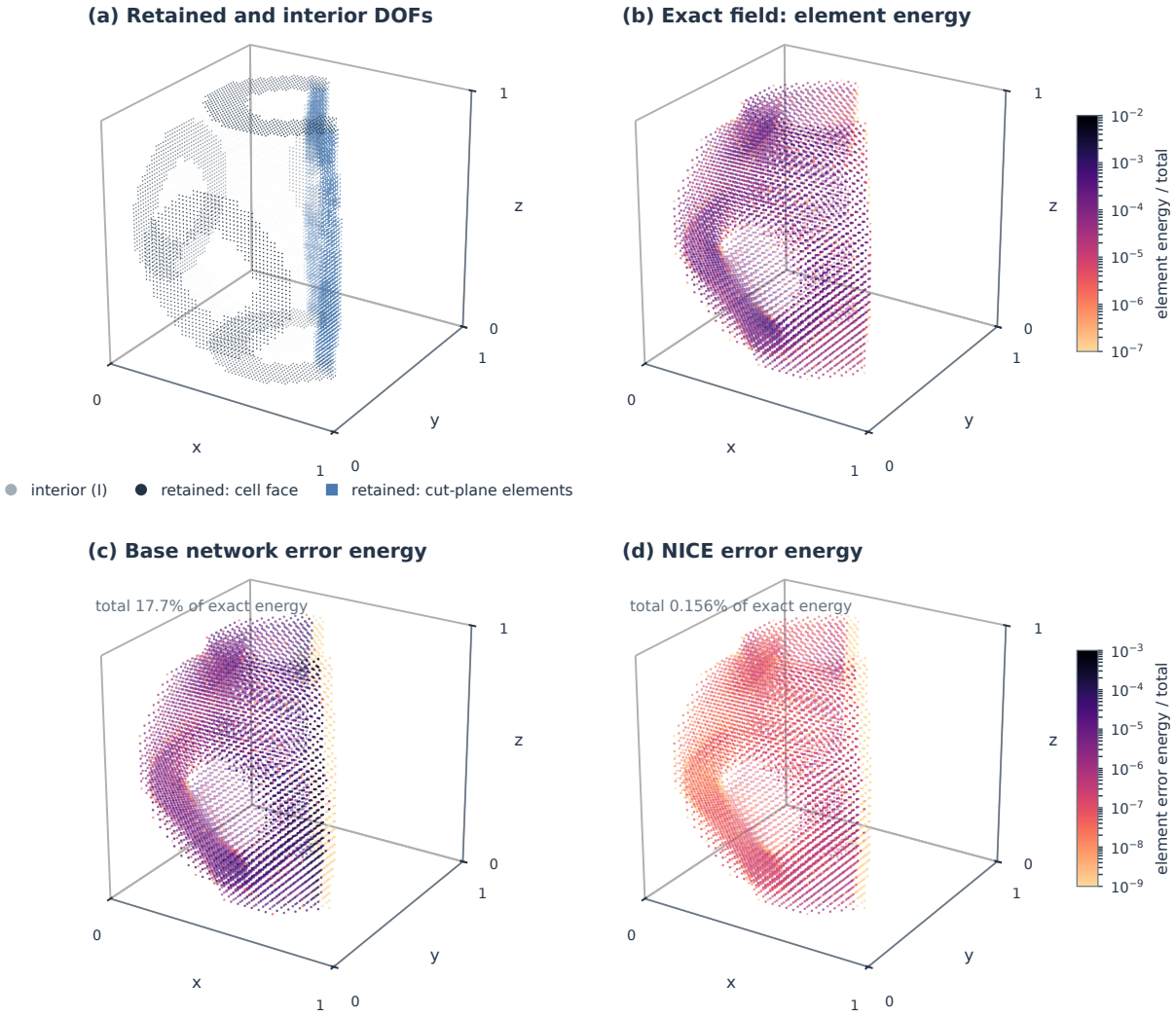


Figure 7: **Retained DOFs and spatial distribution of the interior displacement error in M1.** (a) Retained DOFs on the cell faces, retained DOFs of the cut-plane elements, and interior DOFs. (b) Bulk element energies of the exact field for one traction-load test displacement, relative to its total energy. (c,d) Bulk element energies of the error of the base network and of NICE for the same retained displacements, relative to the total energy of the exact field in (b), on a common colour scale. The cut-plane elements carry no error because their DOFs are retained.

A small displacement error produces a large energy error when the error is stiff. To measure this amplification, let d_I be the interior displacement error, u_I the exact interior displacements and $\delta^2 = d_I^T D d_I / u_I^T D u_I$ the Jacobi-weighted relative displacement error. With

$\kappa = (d_I^T A d_I / d_I^T D d_I)/(u^T K u / u_I^T D u_I)$, the ratio of the Rayleigh quotients of the interior error and of the exact field, $\varepsilon = \delta^2 \kappa$ holds identically (Appendix B.3). Under traction loads, the base network has mean δ of 0.7–1.3% in M1, M2, H1 and H2, but mean κ of 124 to 7089 (Supplementary Table ST04). The displacement error is therefore small, but relative to its amplitude it is far stiffer than the exact field, and the resulting energy errors are 1.8–35%. In the uncut cell U2, a similar δ of 0.74% combines with $\kappa = 85$ and gives an energy error of 0.40%. For NICE, δ is 0.05–0.19% and κ is 33 to 582 in the cut cells, and the two values are 0.11% and 47 in U2.

Where the errors of the base network are large, the quadratic term of Eq. (17) dominates the sensitivity error. In M1 and H2 under traction loads, the sensitivity errors at fixed retained displacements are 14.1% and 75.1%, and the linear term contributes only 7.2% and 0.79%, respectively, of the sum of the norms of the linear and quadratic terms. Under traction loads, the linear term contributes 28–43% in the three cells whose energy errors are below 2%, U1, U2 and H1, and under nodal point loads it contributes 24–72% in all six cells. Because the linear term is first order in the displacement error, it dominates as this error tends to zero (Supplementary Table ST05 and Figure S03).

5.5. Correction of a fixed network

With the base network held fixed, the effect of the correction can be examined in isolation (Figure 8). Eight Chebyshev smoothing steps at fixed retained displacements reduce the mean energy error in every cell, but unevenly (Supplementary Table ST05a). Under traction loads, the error of H2 falls from 35.0% to 0.205%, whereas that of M1 falls only from 13.5% to 4.69%, and to 2.95% after 32 steps, because much of it lies in modes below the smoothing interval (Section 5.4). The sensitivity error of M1 falls from 14.1% to 2.76%. In U1, the energy error decreases at every step, whereas the sensitivity error varies non-monotonically with the number of steps (Figure 8a,b).

A trilinear coarse-grid correction applied before the same eight steps reduces the error of M1 to 0.230%, and eight additional pre-smoothing steps reduce it further to 0.186%; the coarse space of M1 has 5,601 DOFs for 165,927 interior DOFs. With two, four and eight steps per smoothing stage, the two-grid cycle gives 1.02%, 0.422% and 0.186% on M1, and the eight-step cycle gives about 0.027% on M2 and U1 (Supplementary Table ST07). Supplementary Table ST06 compares other coarse spaces.

Table 4 applies the same correction, at the same retained displacements, to four initial fields: zero interior displacements, a graph-harmonic initial field, the field of the base network and the field of NICE. The graph-harmonic field is computed with a volume-weighted graph Laplacian on the element connectivity, after the same exact separation of the rigid-body part as for the network. After the correction, the field of the base network has energy errors of 0.004–0.19%; the graph-harmonic field has 0.08–17%, higher by factors of 7 to 290, and zero interior displacements have 28–1100%, higher by factors of 2,400 to 12,000. Even with 64 steps per stage, eight times the smoothing work, the graph-harmonic field remains 2.6 to 50 times above the eight-step result of the base network under traction loads in five of the six cells of Supplementary Table ST07, which are the cells of Table 4 and U1. Only in H2 does it reach the result of the base network.

Table 4: Mean energy error (%) after the same 8 / $Q_1(17)$ / 8 correction applied to different initial fields. Traction loads, 32 test displacements.

Cell	Base network, no correction	Zero interior displacements	Graph-harmonic	Base network + correction	NICE
M1	13.5	1097	17.2	0.186	0.131
M2	3.64	324	6.72	0.0273	0.0358
H1	1.81	39.9	1.28	0.0131	0.0115
H2	35.0	28.3	0.0806	0.0117	0.0148
U2	0.402	47.2	1.23	0.00424	0.00465

The first column gives the error of the base network without correction; Supplementary Table ST07 gives the results with 16 to 64 smoothing steps per stage.

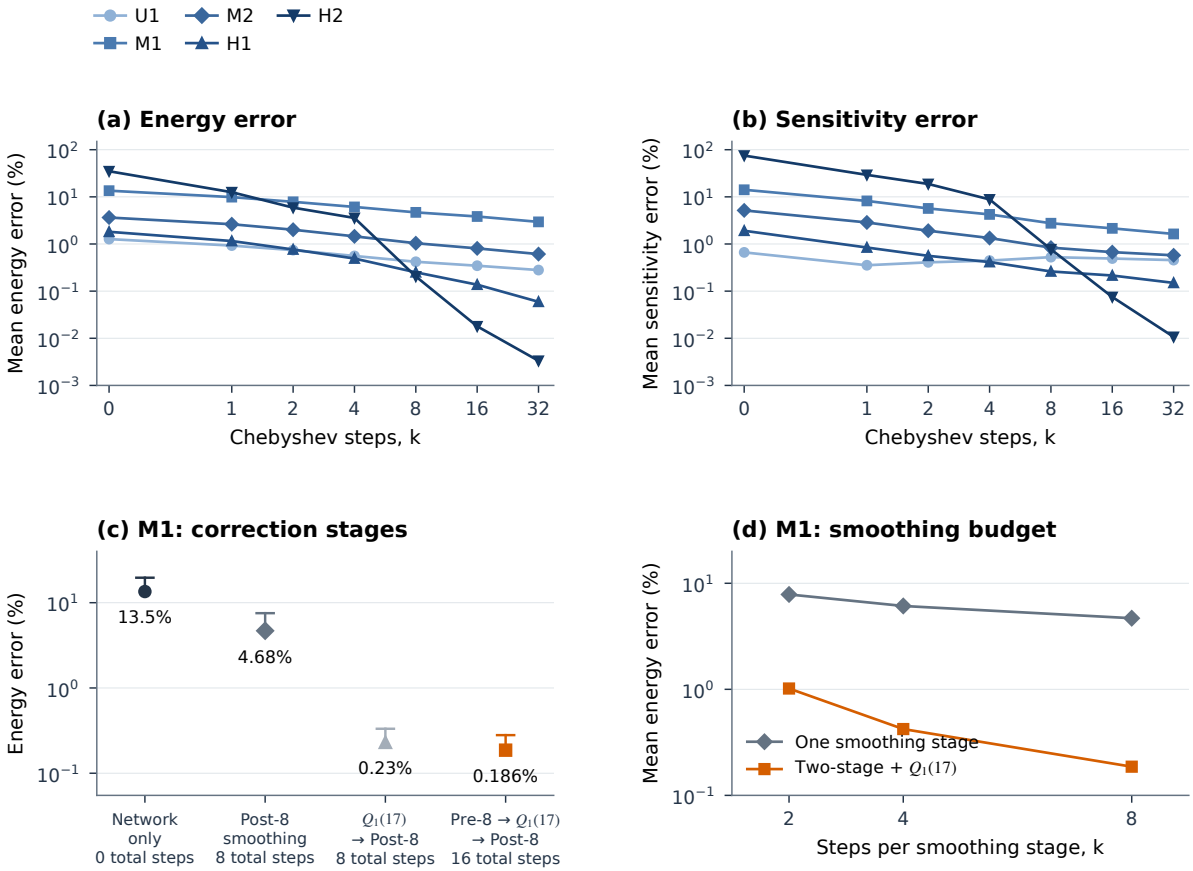


Figure 8: **Correction of the base network with fixed network parameters.** (a,b) Mean energy error and sensitivity error during Chebyshev smoothing on five cells. (c) M1 without correction, with eight smoothing steps, with a coarse-grid correction followed by eight steps, and with eight steps on each side of the coarse-grid correction; dots show means and caps the 90th percentile. (d) M1: mean energy error against the number k of steps per smoothing stage, for a single smoothing stage and for the cycle of k pre-smoothing steps, the $Q_1(17)$ coarse-grid correction and k post-smoothing steps. All panels use traction loads, fixed retained displacements and $a = b/30$. In (a,b) the step axis is linear from zero to one and logarithmic thereafter.

5.6. Two-cell assemblies: compliance and sensitivity

This section checks whether the small single-cell energy errors of Sections 5.3–5.5 give an accurate compliance and accurate local sensitivities after assembly. Each two-cell assembly joins a learned test cell to an exact, uncut neighbouring cell, and the thickness field is continuous across their interface (Figure 9). The seven test cells, U1, U2, L1, M1, M2, H1 and H3, belong to the geometries used for checkpoint selection (Section 5.1); each is assembled in two configurations, x and y, and 'M1/x' denotes test cell M1 in configuration x. Each configuration has six face loads: traction loads in the three Cartesian directions, applied separately on the loaded face of the test cell and on that of the neighbouring cell. Three traction loads on the cut surface of a test cell are considered separately. For each load, the compliance and the eight-component thickness-sensitivity vector of each cell are compared with their exact values; the nodal load is held at its value for the base design, and Supplementary Note S4.2 gives the load normalisation and the solver used for the pairs. A common reference tolerance of 3% is used for the compliance and for the sensitivity vector of each cell and is drawn as the 3% line in the figures. The reported errors are maxima over the six loads and, for the sensitivity, also over both cells.

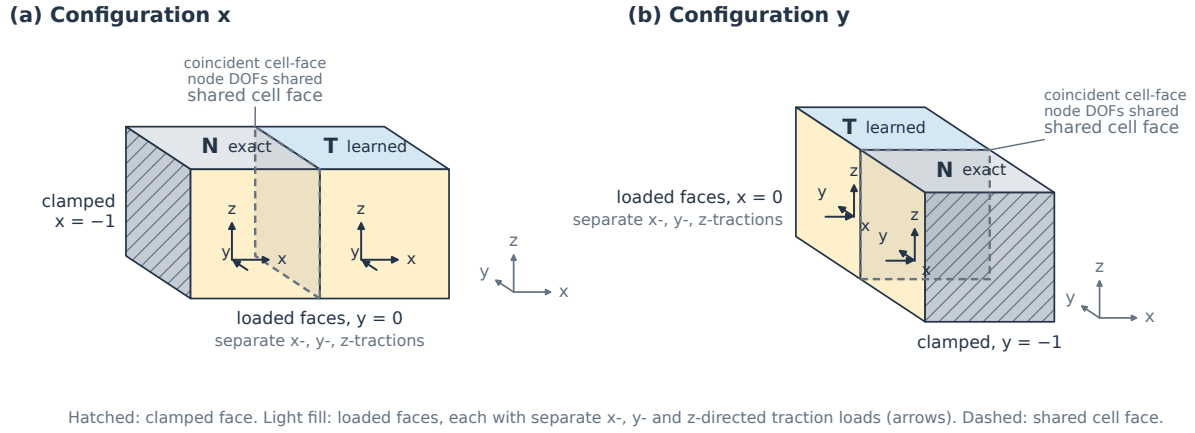


Figure 9: **Supports and loads of the two-cell assemblies.** (a) Configuration x: the neighbouring cell is translated by $(-1, 0, 0)$, the face $x = -1$ is clamped, and the traction loads act on the faces at $y = 0$. (b) Configuration y: the translation is $(0, -1, 0)$, the face $y = -1$ is clamped, and the traction loads act on the faces at $x = 0$. The loaded faces of the test cell (T) and of the neighbouring cell (N) each carry separate traction loads in the x, y and z directions. The coincident DOFs of the cell-face nodes are shared across the interface, and the DOFs of cut-plane elements that do not lie on the cell faces remain local to their cell. Cut test cells also carry three traction loads on the cut surface, which are analysed separately.

Figure 10 compares Base network + correction and NICE in these configurations. NICE stays below both 3% lines in all fourteen configurations of the seven test cells: its largest compliance error over the six face loads is 0.056%, on M1/x, and its largest sensitivity error over both cells is 0.67%, on U1/y. The directions of the thickness-sensitivity vectors are reproduced more closely than their magnitudes; in these fourteen configurations, the cosine between each sensitivity vector of NICE and the exact vector is at least 0.999998 (Supplementary Note S4.3).

Nine further validation cells, which were not used for checkpoint selection, were assembled in the same way: the five geometries with the largest single-cell errors of NICE, and four

cells chosen at random, one from each of the uncut, lightly cut, moderately cut and heavily cut groups. All eighteen configurations stay below both 3% lines, with largest errors of 0.27% in the compliance and 1.49% in the sensitivity, both on the geometry with the largest single-cell error (Supplementary Table ST08b). All compliance errors above 0.02% occur on the five geometries with the largest single-cell errors; on the four randomly chosen cells, the compliance errors stay below 0.011% and the sensitivity errors below 0.22%.

Base network + correction also stays below both 3% lines, with a largest sensitivity error of 0.95% on U1/x. Without the correction, the base network exceeds the 3% sensitivity line on U1 and M1, with sensitivity errors up to 12.2%, and on M1/x its compliance error of 4.38% also exceeds the 3% compliance line (Supplementary Table ST08). Continuing the training of the base network without the correction, or with smoothing alone, also leaves sensitivity errors above the 3% line on U1 and M1 (Supplementary Note S8). The correction therefore brings the assembled responses below the 3% lines.

Under the traction loads on the cut surface of the test cell, the errors of NICE are at most 0.10% in the compliance and 0.16% in the sensitivity (Supplementary Table ST09).

An accurate compliance does not guarantee an accurate local sensitivity, so the two quantities have to be checked separately. For NICE, the sensitivity error exceeds the compliance error, but both lie far below the 3% line. Under the x traction load on the test-cell face of H1/x, the sensitivity error is about twice the compliance error (Supplementary Table ST25). Under the z traction load on the neighbour face of U1/x, where the test cell carries a small energy fraction, the two errors differ by four orders of magnitude. For the variants without the complete correction, the sensitivity error under this load exceeds the 3% line, while the compliance error stays far below it (Supplementary Table ST08 and Note S8).

The energy fraction of the test cell explains why the compliance error can be orders of magnitude smaller than the sensitivity error (Supplementary Table ST25). Under the z traction load on the neighbour face of U1/x, for example, the test cell of NICE carries 0.127% of the exact assembled energy and has a local energy error of 0.0575%; their product $\beta = w\varepsilon$ bounds the relative compliance error by 7.33×10^{-5}%, close to the observed 7.31×10^{-5}%. The sensitivity error of the test cell is 0.598%, about 8,200 times the compliance error, a ratio made up of two factors. First, the compliance error is relative to the compliance of the whole assembly, so the local energy error enters it weighted by w, which gives a factor $1/w \approx 780$, whereas the sensitivity error is relative to the sensitivity of the test cell itself. This sensitivity scales with the energy of the test cell, as its error does, so that w cancels. Second, at fixed retained displacements the energy error controls the sensitivity error only at the order $\sqrt{\varepsilon}$ (Proposition 4(b)), and here the sensitivity error is about ten times ε. Figure 11 shows this weighting for NICE over all loads, and the base network, Base network + correction and the two variants of Supplementary Note S8 follow the same relations (Figure S04).

5.7. Ablation: restricting the cell-face displacements

Keeping all retained DOFs matters more for the local sensitivity than for the compliance. To isolate this effect, pairs in configuration x are solved with the exact condensed stiffness of each cell, while the cell-face displacements of each cell are restricted to tensor-product Bernstein polynomials of degree r; the DOFs of the cut-plane elements that do not lie on the cell faces remain unrestricted. The ablation restricts only the representation of the

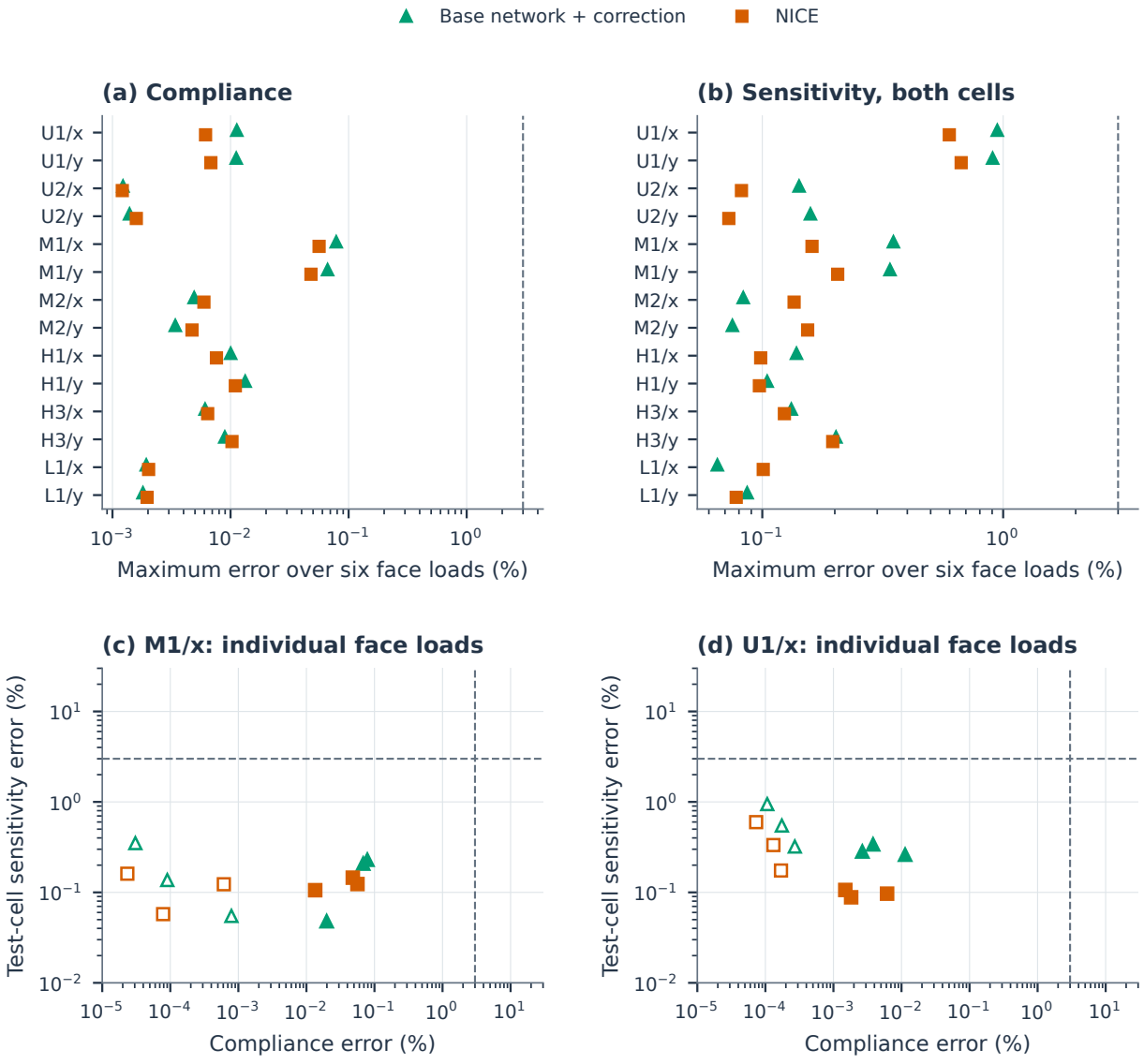


Figure 10: **Compliance and thickness sensitivity in two-cell assemblies.** The test cell is represented by a learned substructure and the neighbouring cell by exact static condensation. The figure shows NICE and Base network + correction, which is the base network with the correction applied at deployment, without retraining. (a,b) Maximum errors over the six face loads in each configuration; the sensitivity error is also maximised over both cells. (c,d) Compliance error and test-cell sensitivity error under each face load on M1/x and U1/x. Filled markers denote loads on the face of the test cell, and open markers loads on the face of the neighbouring cell. Dashed lines mark 3%. Supplementary Table ST08 gives all results. It also includes the base network and the two variants of Supplementary Note S8. None of these three variants was evaluated in every configuration.

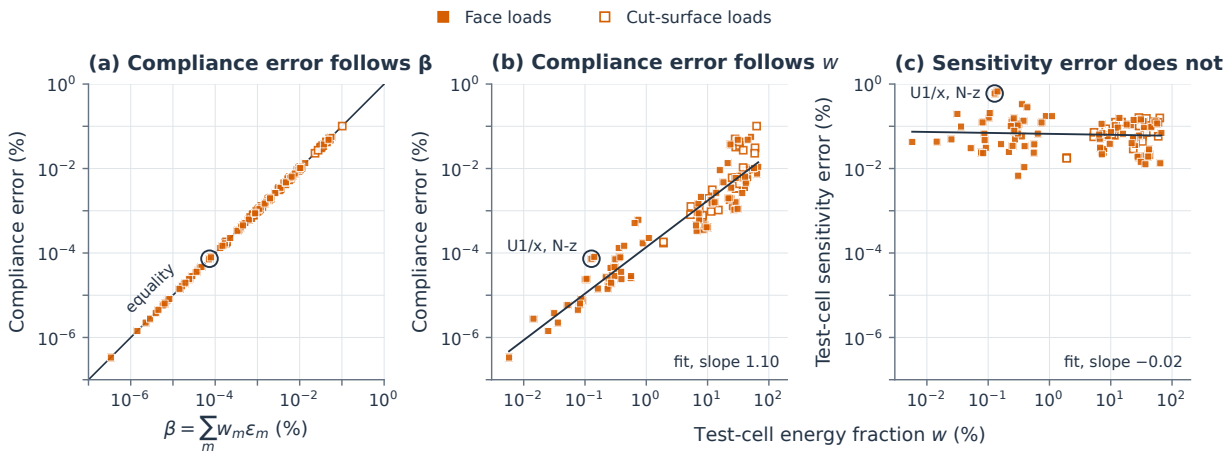


Figure 11: **The energy fraction of the test cell scales down the compliance error but not the sensitivity error.** Results of NICE in the fourteen configurations of the seven test cells, one point per load: 84 face loads (filled markers) and 30 cut-surface loads (open markers). Supplementary Tables ST08 and ST09 give the maxima over the loads. (a) Compliance error against $\beta = \sum_m w_m \varepsilon_m$, with the energy fractions $w_m = q_m^T S_m q_m / C$ and the energy errors $\varepsilon_m = q_m^T (\widehat{S}_m - S_m) q_m / (q_m^T S_m q_m)$ evaluated at the exact assembled retained displacements. Only the learned test cell contributes to β. The line denotes equality. For every load, the compliance error lies between 0.963 and 1.004 times β. The five ratios above 1 correspond to an excess of at most 7.8×10^{-10} of the compliance, whereas the residual work of the pair solves reaches 1.8×10^{-8} of the compliance (Supplementary Note S4.3). (b) Compliance error and (c) test-cell sensitivity error against the exact energy fraction w of the test cell; solid lines are least-squares fits in logarithmic coordinates. The circled point is U1/x under the z traction load on the neighbour face. Figure S04 shows the corresponding results of the other variants.

retained displacements of the present cells, at a fixed cell size. Learned substructures with a reduced boundary control the resulting error by refining the partition, enriching the boundary interpolation or oversampling Huang et al. (2024), Guo et al. (2026a), Guo et al. (2026b); these remedies are not reproduced here (Supplementary Note S2). With every cell face restricted, degree one gives compliance errors of 78–85% on U1, M1, M2 and H1, which decrease to about 4% at $r = 5$ and to 0.49–0.74% at $r = 8$. At $r = 8$, the H1 pair has 14,001 retained DOFs instead of 32,991 (Figure 12).

The local sensitivity converges more slowly: at $r = 8$, the test-cell sensitivity error is 1.3–4.5% under the loads on the test-cell face and 24–64% when the loads on the neighbour face are included. Restricting only the shared interface removes most of the compliance error, which at $r = 3$ is then 0.17–0.44% on U1, M1 and M2; under the neighbour-face loads, however, the sensitivity error remains 8–21% at $r = 3$ and 2.2–3.4% at $r = 5$. A restricted interface is therefore adequate for the compliance at low degree, but not for the local sensitivity under neighbour-face loads. Keeping all retained DOFs removes this part of the error, at the cost of a larger assembled model (Supplementary Table ST11).

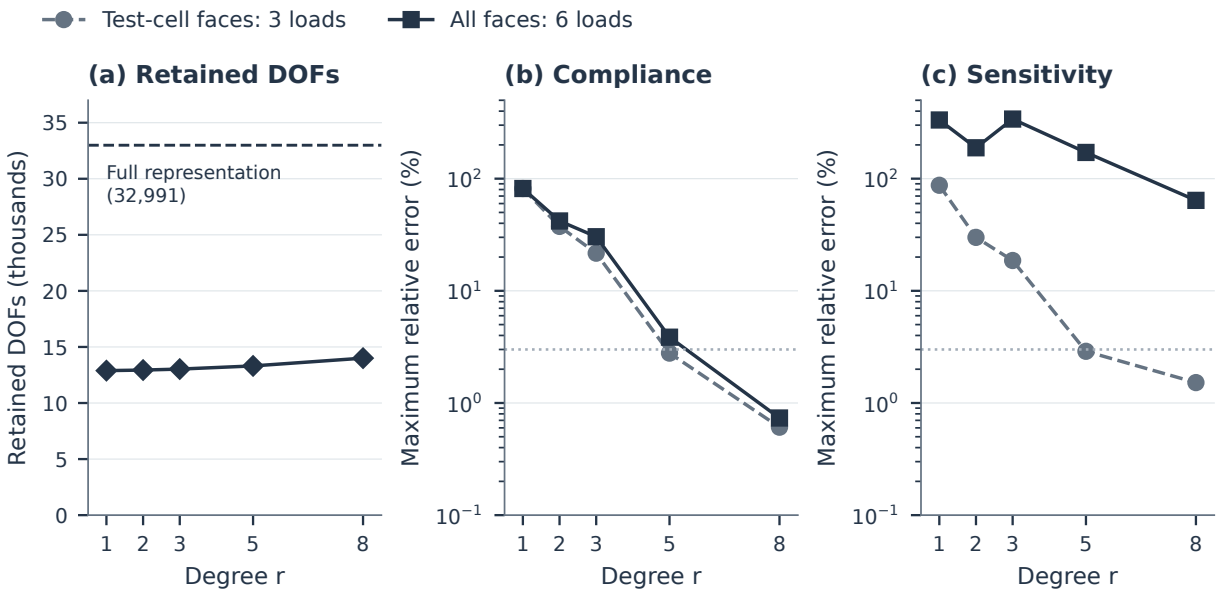


Figure 12: **Response errors caused by restricting the cell-face displacements.** Both cells of H1/x use their exact condensed stiffness and Bernstein polynomials of degree r on every cell face. The DOFs of the cut-plane elements that do not lie on the cell faces remain unrestricted. (a) Number of retained DOFs; the dashed line marks the 32,991 DOFs of the full representation. (b,c) Maximum compliance error and maximum test-cell sensitivity error, over the three loads on the test-cell face or over all six loads on the test-cell and neighbour faces. Traction loads on the cut surface are excluded. Errors are relative to the solution with all retained DOFs, and the horizontal lines mark 3%.

5.8. Lattices of learned cells

In a design, every cell is a learned cell represented by NICE, so the errors of neighbouring cells enter the same assembled solution. Two lattices with a continuously graded thickness field and a planar boundary cut test whether these errors accumulate. The first is a $2 \times 2 \times 2$

block of eight distinct cells, four of which are cut, with remaining volume fractions of 62% and 25%; the second is a $3 \times 3 \times 1$ layer of eight cells, three of which are cut, with one corner position left empty. Neighbouring cells share the corner parameters of their common faces, and the corner parameters of the block range from 0.25 to 0.54. The sixteen cells were generated for these examples and were used neither in training nor in checkpoint selection. Each lattice is clamped on one face and loaded by unit traction loads on the opposite face in the three Cartesian directions, called the face loads, and by three random load vectors; for the sensitivities, the nodal load is held at its value for the evaluated design (Appendix H). The reference solution assembles the dense exact condensed stiffness matrices and is solved to a recomputed relative residual of at most 1.3×10^{-10}. The lattices of learned cells are solved by preconditioned conjugate gradients on the free retained DOFs, with a balanced two-level preconditioner (Supplementary Note S4.1). Its fine-level action is the inverse of the assembled stiffness block of the retained DOFs, $\mathbb{K}_{PP} = \sum_m B_m^T K_{PP,m} B_m$, which is factorised once per design iteration, and its coarse space consists of the trilinear lattice-vertex functions multiplied by the six rigid-body modes.

The errors of the cells do not accumulate in these lattices. Under the face loads, the compliance error stays below 0.015% in both lattices, and the sensitivity error of every cell is at most 0.144% (Table 3). The sensitivities of the cells, summed at the shared corner parameters, give the lattice gradient, which under the face loads agrees with the exact one to within 0.09%, with cosines above 0.999999. Over all six loads, the assembled retained solutions differ from the exact ones by 0.13% and 0.12% in the relative Euclidean norm. These lattice errors are smaller than the largest errors of the two-cell assemblies of Section 5.6 because Eq. (15) bounds the compliance error of the lattice by the weighted energy error β. This error weights the energy error of each cell at the exact retained displacements by the energy fraction of the cell. Under the face loads, these cell energy errors range from 0.006% to 0.026% in the block and from 0.005% to 0.052% in the layer; under the random loads, they range from 0.02% to 0.12%, and they are largest in the cut cells. Under the face loads, β exceeds the compliance error of the lattice by less than 0.4% of β, and under the random loads by about 4–5%. The excess arises because the assembled solution adapts to the stiffer learned cells, whereas β evaluates the cell errors at the exact retained displacements.

The recomputed residual of these solves stagnates, but the errors above are errors of the condensed stiffness and not of the incomplete solve. The solves reach the prescribed tolerance on the recursively updated residual, whereas the recomputed residual $\|f_g - \widehat{\mathbb{K}}\bar{U}\|/\|f_g\|$ stagnates at 3.6×10^{-4} in the block and at 3.2×10^{-3} in the layer. This level is consistent with rounding in the single-precision network and does not come from the correction: running the correction in single instead of double precision changes this level by less than 1% in the block. At the final iterate, the residual work $\bar{U}^T \rho$ of Eq. (16) is at most 2.5×10^{-8} of the compliance. Supplementary Note S4.3 also records a dual-norm bound on the residual work and the action–energy inconsistency, the difference between the energy computed with the applied condensed stiffness and the energy of the recovered displacements (Supplementary Table ST14). Together with the residual work, these records show that the errors above do not come from the incomplete solve.

5.9. Computational cost

A lattice analysis with sensitivities by NICE needs less time and memory than the direct solution of the whole lattice and than conventional exact static condensation, and less time than conjugate gradients on the whole lattice preconditioned by algebraic multigrid. There are two reasons. First, NICE solves the assembled problem on the retained DOFs only: for the two eight-cell lattices, these number 139,002 and 143,685, whereas the whole-lattice model has 1.83 and 2.11 million DOFs. Second, NICE prepares every cell by a network evaluation and a fixed correction instead of a factorisation of the interior stiffness matrix. At the scale of Section 5.10, the cut finite element model of a 110-cell plate has 32.7 million DOFs and its assembled retained system 1.6 million, and a design iteration takes 19 min, about 11 s per cell. The cost comparison below concerns one lattice analysis with sensitivities, from cell setup to the solution, on the two eight-cell lattices of Section 5.8 and on two four-cell lattices, which are the $2 \times 2 \times 1$ layers of the block. Each layer has two uncut cells and two cut cells with remaining volume fractions of 62% and 25%, and keeps the retained DOFs, the clamp and the loads of the block. Table 5 compares three routes.

Route (a) is the direct solution of the whole lattice: the full cut finite element model of the lattice, which contains the retained and interior DOFs of every cell, is assembled and factorised once by the sparse Cholesky factorisation of MKL PARDISO, with 16 threads on one AMD EPYC 9654 CPU. Its recomputed relative residuals are below 3×10^{-9}, so route (a) also serves as the exact reference for the four-cell lattices.

Route (b) is conventional exact static condensation on the same CPU with 16 threads. Each cell is condensed by a Cholesky factorisation of its interior stiffness matrix with the Schur-complement option of PARDISO, and the assembled condensed system is then solved by a block Cholesky factorisation that eliminates the retained DOFs belonging to a single cell before the DOFs of the shared interface. The compliance of route (b) agrees with that of route (a) to within 7×10^{-10}.

Route (c) is NICE on one GPU. It prepares the learned substructure of every cell, solves the assembled system by the preconditioned conjugate gradients of Section 5.8 to a relative residual of 10^{-6} and then computes the sensitivities; the correction runs in single precision.

All routes include cell setup and stiffness assembly, and none includes the one-off cost of data generation and training. The two exact routes solve six loads but do not evaluate sensitivities, which would add to their time; in route (a), the solution of the six loads takes 3–4% of the total time (Supplementary Table ST12b). Since the routes run on different processors, one GPU against 16 CPU threads, the comparison concerns the routes as each is deployed and does not measure hardware-equivalent amounts of work.

Table 5: Cost of one lattice analysis with sensitivities. (a) Direct solution of the whole lattice by sparse Cholesky factorisation with MKL PARDISO and 16 threads on the CPU. Time from cell setup to the solution of six loads (three traction loads and three random loads), and peak process memory. (b) Condensation of every cell with the Schur-complement option of PARDISO, followed by the solution of the condensed lattice system by block Cholesky factorisation, 16 threads: total time and peak process memory. (c) Cell preparation, preconditioner setup, conjugate gradients for the three traction loads to 10^{-6}, and sensitivities: time, peak GPU memory and CPU memory after cell preparation. The compliance error of (c) is computed against (a) for the four-cell lattices and against the exact static condensation of Section 5.8 for the eight-cell lattices. Routes (a) and (b) solve six loads and do not evaluate sensitivities; route (c) solves three loads and evaluates the sensitivities. Memory in GiB. Supplementary Note S3 and Supplementary Table ST12 give all phases.

Lattice (cut / cells)	DOFs: total / free retained	(a) Whole-lattice direct, 16 threads: time (s) / memory (GiB)	(b) Conventional exact condensation, 16 threads: time (s) / memory (GiB)	(c) NICE (one GPU): time (s) / iterations	(c) NICE memory (GiB): GPU / CPU	(c) NICE compliance error (%)
$2 \times 2 \times 1$, $z = 0$ (2 / 4)	957,888 / 77,310	484 / 35.1	589 / 27.9	39.4 / 114	6.3 / 2.4	0.015, 0.012, 0.010
$2 \times 2 \times 1$, $z = 1$ (2 / 4)	884,940 / 71,046	349 / 32.0	632 / 24.7	37.3 / 119	6.1 / 2.4	0.022, 0.016, 0.014
$2 \times 2 \times 2$ (4 / 8)	1,833,474 / 139,002	868 / 71.1	950 / 40.1	79.2 / 129	8.5 / 3.9	0.014, 0.011, 0.0097
$3 \times 3 \times 1$ (3 / 8)	2,113,611 / 143,685	1,042 / 85.9	1,218 / 42.2	105.5 / 164	9.2 / 4.7	0.015, 0.014, 0.015

For the two eight-cell lattices, an analysis with NICE takes 79 and 105 s, whereas the whole-lattice direct solution takes 868 and 1,042 s, about ten times as long, and conventional exact static condensation takes longer still (Table 5). On the four-cell lattices, the time ratios are similar, and the compliance computed with NICE agrees with that of the direct solution to within 0.022%. More threads reduce the time of the direct solution only slightly, because cell setup and assembly do not become faster; Supplementary Table ST12b gives the times with 32 threads and with one thread. In route (a), cell setup and stiffness assembly on the CPU take 41 to 53% of the time. In route (c), the cell preparation on the GPU contains setup and assembly as well as the network encoding and the setup of the correction; with these preparation phases excluded from both routes, NICE remains five to eight times faster (Supplementary Tables ST12b and ST12d). For the eight-cell lattices, the timed compliance errors include the algebraic error of the solve to 10^{-6} (Eq. (16)) and exceed the errors of the condensed stiffness reported in Section 5.8 by up to 0.005 percentage points.

The memory requirements follow the same pattern. For the eight-cell lattices, the peak process memory of the direct solution is 71 and 86 GiB, set by the Cholesky factor, which grows slightly faster than the number of DOFs. Conventional exact static condensation needs 40 and 42 GiB, most of it for the condensed cell matrices, which are held until their

elimination (Supplementary Table ST12). NICE needs at most 9.2 GiB of GPU memory and holds at most 4.7 GiB of CPU memory after cell preparation. Per cell, the stored state of NICE takes 0.25 to 1.37 GiB, whereas the interior Cholesky factor of conventional exact static condensation takes 0.60 to 9.4 GiB (Supplementary Table ST13).

Iterative solvers for the whole lattice were compared on the two eight-cell lattices; they solve the systems of route (a) with PETSc Balay et al. (2026) on the same CPU with 16 MPI processes (Supplementary Table ST12e). Conjugate gradients preconditioned by smoothed-aggregation algebraic multigrid Vaněk et al. (1996) reach a relative residual of 10^{-4} in 52 to 84 s per load, after a setup of 35 and 39 s, and at this residual the compliance differs from that of the converged solution by less than 5×10^{-10}. Including the setup and assembly of route (a), this solver takes 637 and 752 s for the three loads of route (c), eight and seven times as long as NICE. Classical algebraic multigrid (BoomerAMG Falgout & Yang (2002)) gave an indefinite preconditioner for one lattice and needed 250 to 260 iterations per load for the other. For BDDC Dohrmann (2003) with exact local solvers and one subdomain per cell, the factorisation of the Dirichlet and complete Neumann matrices of every cell alone takes 449 and 523 s (Supplementary Note S3). Every fine-scale route also repeats the cell setup and the assembly of the whole-lattice system, which in route (a) take 416 and 474 s (Supplementary Table ST12b), 5.3 and 4.5 times the time of the NICE analysis.

In a design iteration, the geometry of every cell changes, so the preparation of every cell is repeated, as in the analyses of Table 5. For the conventional routes, this preparation is the factorisation of the interior stiffness matrix; for NICE, it is the network encoding and the setup of the correction. The assembled solve can start from the previous solution, and the conventional routes can reuse their symbolic factorisation as long as the pattern of active elements does not change.

5.10. Thickness design

Two cases are optimised with NICE: case A is the $2 \times 2 \times 2$ block of Section 5.8, and the second case is a cut plate supported on its cut, described below and shown in Figure 13. In both cases, the corner thickness parameters solve the design problem of Section 2.3 under one unit traction load on a face, with $V^* = 0.8\,V(\boldsymbol{\tau}^0)$. The parameters at the vertices in the plane of the loaded face are fixed, 9 of 27 parameters in case A and 10 of 74 in the plate, which keeps the nodal load independent of the design and removes the load-derivative term $2f_{g,c}^T\widehat{U}$ of Appendix H. The optimiser receives the vertex gradient of Section 2.3, with the sensitivities $\widetilde{\boldsymbol{s}}_m$ of Section 4.2 in place of $\boldsymbol{s}_m$. The bounds are $0.18 \le \tau \le 0.69$, and in each cell the corner span, the largest difference between two of its corner parameters, and the norm of the gradient of τ are limited to 0.45. These bounds and limits keep every cell inside the parameter domain of the training data, in which the corner parameters lie in [0.175, 0.699] and the corner span and the gradient norm are at most 0.47 (Section 6.3). Each design iteration regenerates the geometry and the learned substructure of every cell and solves the lattice from the previous solution by the preconditioned conjugate gradients of Section 5.8. It then takes one step of the method of moving asymptotes (MMA) Svanberg (1987), in the variant of Svanberg (2007), with a move limit of 5% of the parameter range and no line search (Supplementary Table ST16).

In case A, the NICE design agrees with the design obtained with exact static condensation. The block, loaded by a normal traction on the face opposite the clamp, is optimised

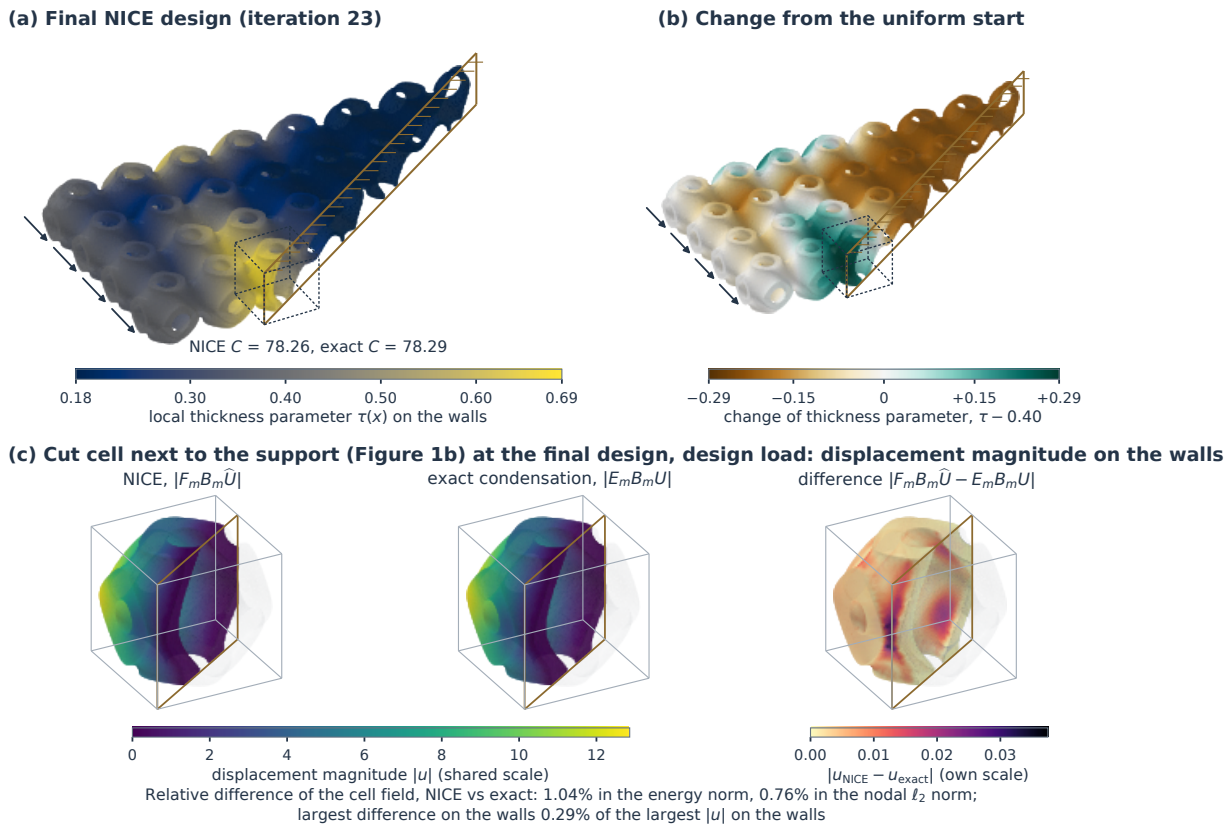


Figure 13: **Thickness design of the plate supported on its cut.** (a) Final NICE design, with the material surface coloured by the local thickness parameter, and the NICE and exact compliances (Supplementary Table ST21). (b) Change from the uniform start, $\tau - 0.40$. Material moves into the cells next to the support near the loaded end (up to $+0.29$), and the cells far from the load are thinned (down to -0.22). (c) The cut cell of Figure 1b at the final design under the design load. The panel shows the displacement magnitude on the material surface from the NICE recovered field $F_m B_m \widehat{U}$ and from exact static condensation $E_m B_m U$ on a common scale, and their difference on a separate scale. The two fields differ by 1.04% in the energy norm of the cell and by 0.76% in the Euclidean norm of the nodal displacements. Applying the same recovery operator to the exact retained displacements also gives 1.04%, so the difference comes from the displacement recovery within the cell. Over the plate, the difference in the energy norm is 1.72%; its square equals the relative compliance difference of 2.96×10^{-4} (Eq. (14)).

from its graded design with NICE and, for comparison, with exact static condensation, called the twin run (Figure 14a). Both runs stop when the relative change of the objective has stayed below 10^{-4} in three consecutive iterations (Supplementary Table ST16), NICE after 24 design iterations and the twin run after 23 (Table 6). No Karush–Kuhn–Tucker (KKT) residual is used for stopping, because the discrete model, and with it the objective, changes between design iterations (Section 4.2); Supplementary Note S6.1 explains this choice. Both runs lower the compliance by 2.0% with 20% less material; their final corner parameters differ by at most 0.0051, and the exact compliances of their final designs differ by 1.25×10^{-5} of their value. At the three checked designs, the NICE compliance lies at most 0.028% below the exact one, with the sign given by Eq. (15), and the lattice gradient has errors of at most 0.33%, with every component of the exact sign (Table 3). Both errors grow as the design approaches the bounds (Supplementary Table ST17).

The plate supported on its cut is a single layer of 8×4 cells, from which a planar cut removes eight cells and leaves 16 uncut and eight cut cells (Supplementary Note S6.3). The plate is clamped at its cut: every DOF of the cut-plane elements of every cut cell is fixed, a support that lies within one element, of size $h = 1/32$, of the cut surface. Because the DOFs of the cut-plane elements are retained, NICE represents this support without retraining. The end face of the plate is four cells wide and carries a traction load of unit resultant that acts in the plane of the plate and normal to its long side, so the plate acts as an in-plane cantilever supported along the oblique cut (Figure 15a). The cut normal, with $\vartheta = 33.7°$, lies in the range $0 < \vartheta < \pi/4$ of orientations in which NICE was mainly evaluated (Sections 5.1 and 6.3).

Starting from the uniform design $\tau = 0.40$, NICE stops after 24 design iterations by the rule on the change of the objective (Figure 14b). The compliance rises from 75.59 to 110.49 while the volume is reduced to V^*, and then falls to 78.26. The final design reaches both bounds as well as the corner-span and gradient limits: it is thickest near the end of the support closest to the loaded face and along the opposite free edge, and the cells far from the load are thinned to the lower bound (Figure 15a). Checked with exact static condensation, the NICE compliance lies 0.011% below the exact one at the uniform start and 0.030% below it at the final design; at these two designs, the lattice gradient has errors of 0.029% and 0.30%, and all 64 components have the exact sign (Supplementary Table ST21). Two uniform designs of the same volume serve as references: in the first, the ten parameters of the loaded face are kept at 0.40 and the other 64 take the common value that meets V^*; in the second, all 74 parameters take the same value. Against the first uniform design, the NICE design lowers the exact compliance by 32.1%, from 115.36 to 78.29, and against the second by 30.9%; the homogenisation design described below lowers it by 30.5% against the first. For both uniform designs, the NICE compliance agrees with the exact one to within 0.019%, and the gradient to within 0.064% (Supplementary Table ST21).

For comparison, the plate is also optimised with a homogenised model, as in homogenisation-based graded design Li et al. (2018). The elasticity tensor of this model is that of the cell with uniform thickness, obtained by periodic homogenisation on the same discrete model (Supplementary Table ST18b). The model is clamped on the plane of the cut and optimised with the same variables, bounds, limits, volume bound and optimiser as the NICE run. Its volume is computed from the homogenised material volume fraction $V^H(\tau)$ and, at the uniform start, equals the material volume of the discrete model to within

1.2×10^{-7}. The homogenised model underestimates the exact compliance of the uniform start by 27.0%, which is consistent with the absence of scale separation through the thickness of a single layer; with 12 instead of 6 elements per cell and axis, the underestimate is 25.8%. The clamp of the homogenised model is imposed by a penalty, and a hundredfold larger penalty factor changes the macroscale compliance by less than 0.001% (Supplementary Note S6.3). Nor is the underestimate caused by the support: the fine-scale model clamps all cut-plane elements, a support at least as stiff as one acting on the cut surface alone, which lowers the exact compliance and therefore narrows the difference. Along its own optimisation path, the macroscale compliance lies 27–33% below the NICE compliance; up to iteration 3, the two runs analyse the same designs (Figure 14b). Analysed on the cut geometry, the homogenisation design uses 0.18% less material than the NICE design, and its compliance is 2.44% higher. This difference is the same whether both designs are evaluated with NICE or with exact static condensation, which gives exact compliances of 80.198 and 78.287 (Supplementary Tables ST18c and ST21). The corner parameters of the homogenisation design (Figure 15b) differ from those of the NICE design by up to 0.26, with a correlation of 0.95 (Supplementary Table ST18c). The homogenised model thus underestimates the compliance of the cut layer by about a quarter on both macroscale meshes and leads to a design that is less stiff on the cut geometry.

The discrete model of a cell can change from one design to the next (Section 4.2). In both cases, the discrete model changed between every pair of consecutive design iterations, in at least 3 of the 8 cells of case A and at least 11 of the 24 cells of the plate (Supplementary Table ST19b). In these runs, the residual work of Eq. (16) stayed at most 5.4×10^{-8} of the compliance. In four design iterations over all runs, the geometry generator required a perturbation of at most 5.8×10^{-4} in a corner parameter (Supplementary Table ST19a). A NICE design iteration took 74 s on average for case A and 200 s for the plate (Table 6); for the plate, single iterations took 155–270 s, and the 24 iterations took 1.3 h in total (Supplementary Table ST18a).

To examine how the cost grows with the size of the lattice, the design iteration was timed on plates with 24, 51, 88 and 110 cells. These plates, with the proportions and the cut of the plate above, are clamped along the uncut long side and loaded in plane on the opposite face (Figure 16); the largest has 32.7 million DOFs in the cut finite element model and 1.62 million free retained DOFs. Each plate was run for its first four design iterations, with learned substructures beyond 4 GiB of GPU memory streamed from CPU memory. The time of a design iteration grew in proportion to the number of cells, at 10.1 to 10.9 s per cell, and reached 19.4 min for 110 cells; the conjugate gradients needed 138 to 173 iterations. The CPU memory, which holds the streamed state of the learned cells, grew by about 0.7 GiB per cell. With the learned cells on the GPU held within a fixed 4 GiB, the GPU memory grew with the assembled retained system, whose stiffness matrix $\mathbb{K}_{PP}$ the preconditioner factorises directly (Supplementary Table ST20). Exact checks of the 24- and 51-cell plates at their first design iteration give the compliance and gradient errors listed in Table 3, and every gradient component has the exact sign.

Table 6: Thickness optimisation cases. Compliance under a unit traction load on a face. Start: initial design, whose volume exceeds V^* by 25% in every run. Final: last design iteration. Compliances are NICE values unless marked otherwise. Exact verification: exact discrete compliance of the designs, with the amount by which the NICE compliance lies below it in parentheses. Iterations: number of design iterations. Time: mean time of a design iteration, including the geometry generation of every cell. Unlike the analyses of Table 5, a design iteration regenerates the geometry of every cell and solves for one load (Supplementary Note S6.1). Hom: homogenised macroscale model with 4,464 trilinear elements. Supplementary Note S6 and Supplementary Tables ST16–ST21 give the details.

Case	Cells (cut)	Support and load	Iterations	Start C	Final C	Exact verification	Time (s)
A, NICE	8 (4)	Clamped face; normal traction on the opposite face	24	24.5908	24.1038	24.5936 (0.011%), 26.0063 (0.018%), 24.1105 (0.028%) at iterations 0, 12, 23	74
A, exact condensation	8 (4)	As A	23	24.5936[a]	24.1108[a]	Exact throughout; final design 0.0013% above the NICE design	469
Plate, NICE	24 (8)	Clamped cut-plane elements; in-plane traction on the end face	24	75.591	78.264	75.599 (0.011%), 78.287 (0.030%) at iterations 0, 23	200
Plate, Hom	Macroscale	As plate	23	55.19[b] (NICE 75.59)	55.27[b] (NICE 80.17)	Final design: 80.198 (0.032%)	4[b]
Scale: plates of 24, 51, 88, 110 cells	24–110 (8–18)	Clamped long side; in-plane traction on the opposite face	4 timed	94.34, 89.12, 86.75, 85.38	—	24 and 51 cells: 94.353 (0.013%), 89.129 (0.011%) at iteration 0	242; 547; 962; 1,164

[a] Exact compliance. [b] Homogenised macroscale model, with the time measured on the CPU; the value in parentheses is the NICE compliance of the same design on the cut geometry.

6. Discussion

6.1. Retained degrees of freedom, network and correction

Using the complete set of retained DOFs, without reducing the interface, separates two choices that a reduced substructure model combines: the displacement patterns that neighbouring cells can exchange, and the accuracy with which the interior responds to each of them. The network and the correction affect only the interior response, whereas the ablation of Section 5.7 isolates the effect of the exchanged displacement patterns. Even with exact interior displacements, restricting the cell-face displacements changes the assembled response, and a polynomial degree that gives an accurate compliance can still leave an appreciable error in the sensitivity. In the two-cell assemblies of the cut cells examined, restricting the cell-face displacements produced larger errors in the local sensitivity than the interior approximation of NICE (Sections 5.6 and 5.7). This comparison was made at a fixed cell size and without the refinement of the partition or the enrichment that methods with reduced boundary DOFs use. In addition, the retained DOFs of the cut-plane elements carry the support of the plate of Section 5.10 directly, whereas the homogenised model needs a separate constraint on the plane of the cut for this support. Where the cut surfaces carry no loads or supports, however, these DOFs enlarge the assembled system (Section 6.3), whose solution takes most of the time of a NICE analysis (Supplementary Table ST12d).

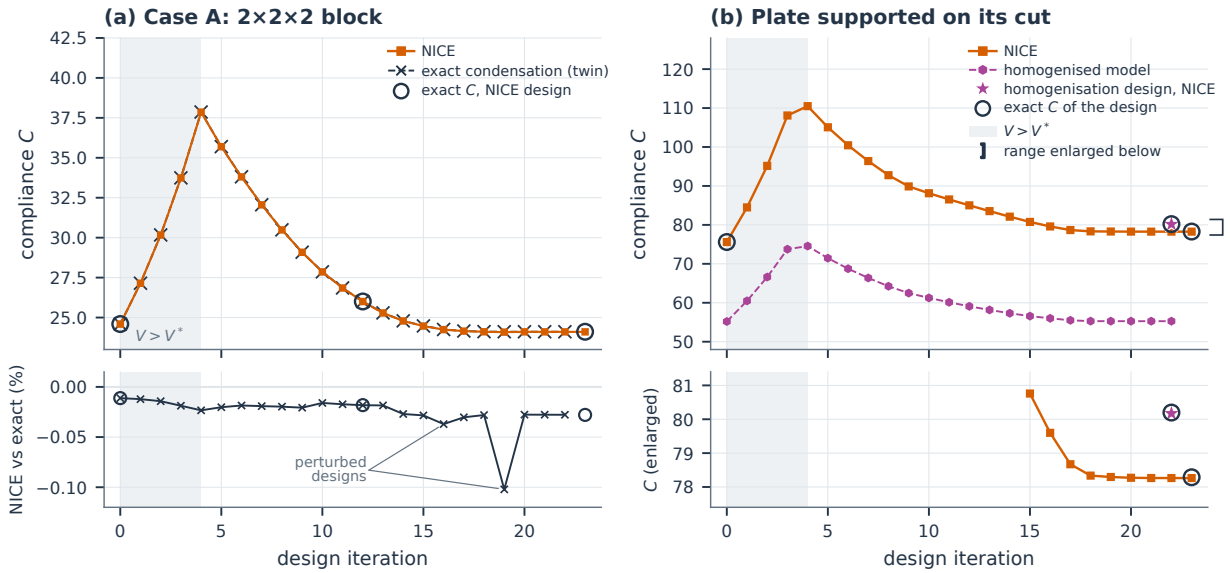


Figure 14: **Thickness optimisation with NICE.** (a) Case A: compliance histories of the NICE optimisation and of the twin optimisation with exact static condensation. Open circles give the exact compliance of the NICE designs at iterations 0, 12 and 23. The compliance first rises while the volume is reduced from $1.25V^*$ to V^* (shading, iterations 0–3). Lower strip: NICE compliance relative to the twin run at the same design iteration (line) and relative to the exact compliance of the same design (circles: -0.011%, -0.018%, -0.028%). The designs of the two runs coincide at iterations 0–4 and differ afterwards. At iterations 16 and 19, labelled 'perturbed designs', the NICE design had been perturbed by the fallback of the geometry generator (Supplementary Table ST19a). (b) Plate supported on its cut. The curves show the NICE compliance from the uniform start and the compliance of the homogenised macroscale model along its own optimisation (dashed line). The star marks the homogenisation design analysed with NICE on the cut geometry, placed at the last macroscale iteration. Open circles give the exact compliance of the uniform start, the final NICE design and the homogenisation design. Shading: design iterations with $V > V^*$. Lower strip: the range marked by the bracket, enlarged, from iteration 15.

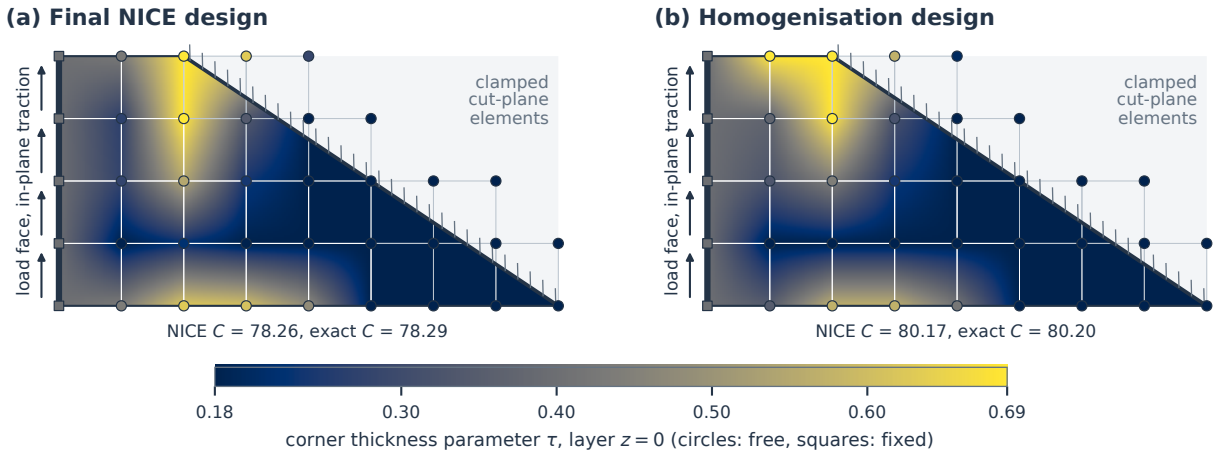


Figure 15: **Designs of the plate supported on its cut.** (a,b) Corner thickness parameters of the final NICE design and of the homogenisation design in the layer $z = 0$, drawn with the long side horizontal. Circles denote design variables, and squares denote the fixed vertices in the plane of the loaded face. Vertices drawn in the removed region are corners of cut cells. The thick hatched line marks the clamped cut-plane elements, and the arrows mark the loaded face and the direction of the in-plane traction. The compliances in (a,b) are the NICE and exact values of the same design (Supplementary Table ST21).

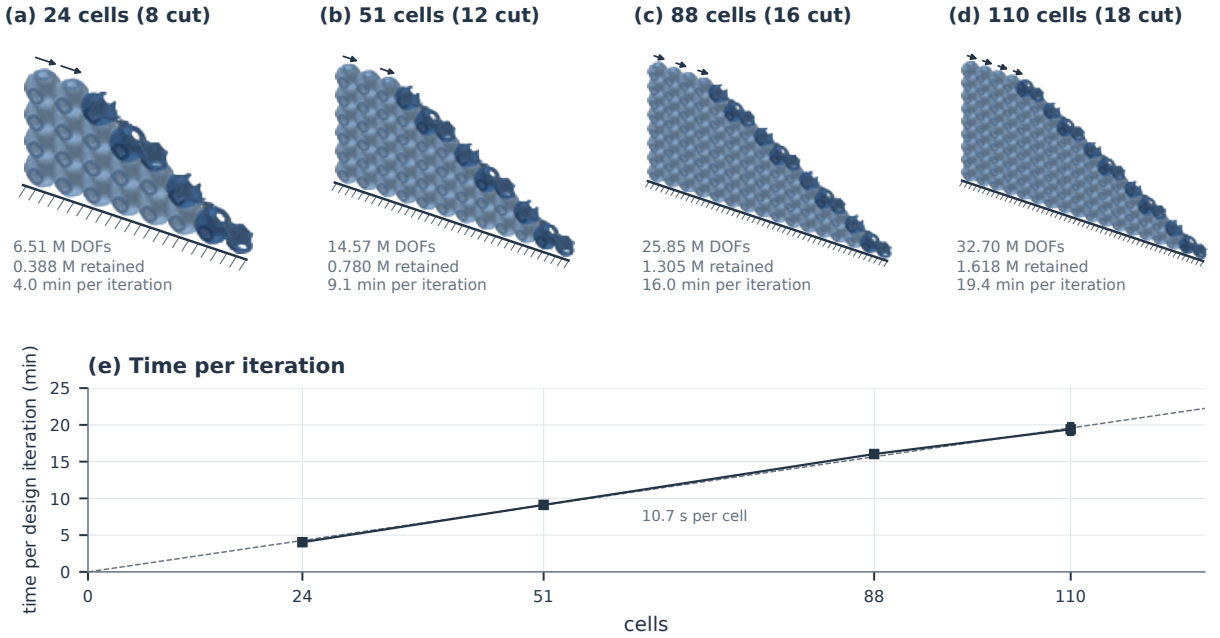


Figure 16: **Plates of the scale study.** (a–d) Plates of 24, 51, 88 and 110 cells with the proportions and cut of the plate in Figure 15, at the uniform start $\tau = 0.40$, drawn to the same scale. Each plate stands on its clamped long side (hatched), and the opposite face carries an in-plane traction along the long side (arrows). Light: uncut cells; dark: cut cells. Each panel gives the DOFs of the cut finite-element model, the free retained DOFs and the mean time per design iteration. (e) Time per design iteration against the number of cells (squares: mean; bars: range over the timed iterations). The dashed line is proportional to the number of cells, with the mean time of 10.7 s per cell. Memory use is given in Supplementary Table ST20.

The structure of the network makes it possible to learn the displacement recovery on the complete set of retained DOFs (Section 3.2). Its weights act on local node positions, feature channels and grid levels, so one network of about 6×10^5 parameters serves every set of retained DOFs. The same network, unchanged, was used for the 80 validation geometries, the sixteen cells of the lattices and the cut cells of the plate supported on its cut (Sections 5.3, 5.8 and 5.10). Because the network output is exactly linear in the retained displacements, the relations of Sections 3.1 and 4 apply to the network itself (Section 3.2), and because the correction is linear and fixed for each geometry, the network can be trained with the correction applied (Section 3.5). The condensed stiffness is applied without being formed, so each cell stores the quantities determined by its geometry rather than a dense matrix on its retained DOFs (Section 3.6).

Within the interior, the network prediction and the correction are complementary. By the energy formulation of the condensed stiffness, if an interior correction does not increase the A-energy error for any retained displacements, the corrected condensed stiffness lies between the uncorrected and the exact condensed stiffness, whatever the network (Section 3.4). An interior displacement error of about 1% gives an energy error of 1.8 to 35% because the error is stiffer than the exact solution (Section 5.4). Chebyshev smoothing removes the error components with large Rayleigh quotients of the Jacobi-scaled interior stiffness matrix, and the coarse-grid correction removes the smooth components that smoothing reduces slowly. Under the same correction, the network prediction leaves lower energy errors than zero interior displacements and the graph-harmonic field (Table 4); even with eight times the smoothing work, the graph-harmonic field leaves a larger error than the eight-step result of the base network in five of six cells (Section 5.5). Conversely, the correction reduces the error components that the network represents poorly. On the moderately cut cell M1, it lowers the energy error of the base network from 13.5% to 0.19% without any change of the network parameters, and in the two-cell configurations Base network + correction already stays below the 3% lines. Training with the correction applied adds less than the correction itself: it reduces the mean energy error by a further factor of 1.31 (Section 5.3).

6.2. Computational value

A number of correction steps that suffices for the compliance can leave the local sensitivity inaccurate, so the relevant cost is the total work needed to reach the prescribed accuracy in both quantities. On the lattices of Section 5.9, NICE is faster than the whole-lattice direct solution and conventional exact static condensation and needs less memory than these two direct routes; on the two eight-cell lattices on which iterative solvers were compared, it is also faster than whole-lattice conjugate gradients with algebraic multigrid. The advantage comes from the way the interior is resolved. The exact routes, direct or iterative, solve the fine-scale problem of every cell or of the whole lattice, whereas NICE obtains the interior displacements of each cell from one network evaluation and a fixed number of correction steps and solves the assembled problem on the retained DOFs only (Section 5.9). Inexact BDDC and FETI-DP methods Li & Widlund (2007), Klawonn & Rheinbach (2007), Hirschler et al. (2024) use approximate subdomain and coarse solves only inside the preconditioner and therefore converge to the exact discrete solution. The lattice solve of NICE, in contrast, iterates on the assembled condensed stiffness matrices $\widehat{S}$ of the learned substructures, so the

approximation lies in the operator itself. Fewer smoothing steps would lower the cost of applying the condensed stiffness, at the loss of accuracy quantified in Section 5.5.

6.3. Limitations

The trained network applies to the setting in which it was trained and tested: Schwarz-P-type cells of Eq. (1) with eight corner parameters in [0.175, 0.699] and with corner span and gradient norm of at most 0.47 (Supplementary Table ST02). Each cell has at most one planar cut, whose normal is the image of $(\cos\vartheta, \sin\vartheta, 0)$ under a symmetry of the cube. The network is also tied to one discretisation, with $n = 32$, $E_Y = 1$, $\nu = 0.3$ and $\gamma = 10^{-4}$, to which its grid hierarchy, with 65, 33, 17 and 9 positions per axis, is matched. Other TPMS families need further training. On gyroid sheet cells, the network trained on Schwarz-P cells gives mean energy errors under loads and supports that are 56 to 234 times those on the corresponding Schwarz-P cells. A further 8,000 training steps with eighteen gyroid cells lower these errors six- to tenfold, but they remain 6 to 23 times the Schwarz-P level (Supplementary Note S7). Other families therefore require new training data and network parameters, as do cells with several cuts or curved boundaries, other resolutions and other materials; training on several families and testing on a family held out would show how far the network itself transfers, which is left to future work. The network is not equivariant under the symmetries of the cube: for a rotated or reflected cell, its prediction is in general not the correspondingly rotated or reflected prediction, a property that symmetry-decoupled shape functions for voxel substructures have Jiang, C. et al. (2026). NICE was evaluated mainly in the identity orientation. In a second orientation, obtained by a symmetry of the cube and also used in checkpoint selection, its mean energy error on the 80 validation geometries over the seven load classes evaluated in both orientations is 4% higher in relative terms, and under traction loads 12% higher (Supplementary Table ST03c).

The properties of the condensed stiffness shown in Section 3.1 hold for any input to the network, but the accuracy does not, and no certified error bound is provided (Appendix G.4). Because $\widehat{S} \succeq S$, the learned model is stiffer than the reference model and underestimates the compliance (Eq. (15)), so its compliance is not a conservative estimate. The interior residual of Eq. (7) is available at run time, but the energy error that it determines requires one interior solve per test displacement. The lower bound of Appendix I needs no reference solution and can detect a large error for a test displacement, but it cannot confirm a small one. Accuracy is measured against the discrete reference model (Section 5.2). At $n = 32$, this model deviates from its refinement to $n = 64$ by at most 0.28% in the compliance and 0.44% in the sensitivity, except in the heavily cut cell H2, where the deviations reach about 1% and 3.6% (Supplementary Table ST10).

The evidence on assembled models consists of two-cell assemblies in fourteen configurations of cells used for checkpoint selection and eighteen configurations of cells outside checkpoint selection (Section 5.6), and of two eight-cell lattices and the two four-cell layers of the block (Sections 5.8 and 5.9). The designs checked against exact static condensation are three designs of the block in case A, five designs of the cut 24-cell plate and the first designs of the 24- and 51-cell plates of the scale study (Section 5.10). All comparisons between trained variants use a single random seed, and the effect of the sensitivity term in the loss of Eq. (13) was not isolated. The reported sensitivities are $\widetilde{s}_c = -\widehat{u}^T K_{,c} \widehat{u}$; they omit the design dependence of the recovery operator F, which enters the derivative of Eq. (18). A

central-difference check of the approximate compliance cannot decide whether $\widehat{C}_{,c}$ or $\widetilde{s}_c$ is the more accurate gradient. At least one of the discrete choices listed in Appendix B.1 switches in 21 to 48 of the 49 rebuilds of each lattice cell, and the difference quotient grows as the step decreases (Supplementary Note S4.3). The sensitivity $\widetilde{s}_c$ needs no difference quotient; for the two eight-cell lattices and the two four-cell layers, it gives the gradient with respect to the shared parameters to within 0.02–0.13% under the face loads and 0.22–0.35% under the random loads (Supplementary Table ST14). The optimisations of Section 5.10 take steps without a line search, because a line search would compare the approximate compliance with $\widetilde{s}_c$, which is not its derivative. Where the cut surfaces carry no loads or supports, the retained DOFs of the cut-plane elements enlarge the assembled system. In the lattices of Sections 5.8 and 5.9, the DOFs of the cut-plane elements that do not lie on the cell faces make up 37 to 50% of the free retained DOFs. A reduced representation of these DOFs where the cut carries no loads or supports is a natural next step. On one GPU, the size of the lattice that can be analysed is at present limited by the direct factorisation of $\mathbb{K}_{PP}$ in the preconditioner. The memory of this factorisation grows with the assembled retained system, whereas the memory of the learned cells stays within a fixed budget; a further level in the preconditioner or an iterative coarse solve is the natural way to remove this limit.

Fine-scale iterative solvers of the whole lattice were compared on the two eight-cell lattices only, using algebraic multigrid Vaněk et al. (1996), Henson & Yang (2002), Falgout & Yang (2002) in standard settings and the local factorisation cost of BDDC with exact local solvers Dohrmann (2003) (Section 5.9). FETI-DP Farhat et al. (2001) and tuned multigrid were not compared. An optimiser must keep the designs inside the parameter domain stated above, as the bounds and limits of Section 5.10 do, and the optimisations of that section are local.

7. Conclusions

NICE is a learned component model that keeps the properties of static condensation on the complete set of retained DOFs of a cut cell, a set that changes with the cell geometry. A network predicts the interior displacements, a fixed two-grid correction reduces the interior force imbalance that the network leaves, and the condensed stiffness is computed from the strain energy of the corrected field and applied with the transpose of the complete recovery operator. The condensed stiffness is therefore symmetric positive semidefinite, only the rigid-body modes have zero energy, and it is an upper bound on the exact condensed stiffness, with an error of second order in the interior displacement error (Proposition 1). These properties do not depend on how well the network is trained. A correction that satisfies the conditions of Proposition 2 cannot increase the error of the condensed stiffness, and in every example of Supplementary Table ST07, a larger number of smoothing steps reduced the error further. The network and the correction complement each other. Under the same correction, a graph-harmonic initial field leaves 5.4 to 265 times the error of NICE; applied to the base network without retraining, the correction alone lowers the mean energy error under traction loads on the 80 validation geometries from 6.89% to 0.0965%.

The error relations of Section 4 determine the quantities on which such a model must be checked: the interior error enters the assembled compliance weighted by the energy fraction of each cell (Proposition 3) and the thickness sensitivity through a term of first order in that error (Proposition 4). In cut cells, the displacement errors of the base network are about

1%, but they are stiffer than the exact solution and give energy errors of 1.8 to 35% (Section 5.4). These relations hold for any approximate recovery operator, so an approximate static condensation used in design must be checked on its sensitivities as well as on its compliance; NICE is trained and checked on both.

For cut thin-walled Schwarz-P cells with tens of thousands of retained DOFs, the mean energy error of NICE over the 80 validation geometries is 0.074%, and in every two-cell configuration its compliance and sensitivity errors are at most 0.27% and 1.49%. An analysis of an eight-cell lattice with sensitivities takes 79–105 s on one GPU, whereas on 16 CPU threads or processes the whole-lattice direct solution takes 868–1,042 s and whole-lattice conjugate gradients with algebraic multigrid take 637–752 s. Thickness optimisation with NICE reproduces the design that exact static condensation gives for an eight-cell lattice, to within 0.0051 in every corner parameter. For a 24-cell plate clamped at its cut, a homogenised model underestimates the compliance by 27%. At the five checked designs of this plate, the compliance and the gradient from NICE differ from those of exact static condensation by at most 0.032% and 0.32%, and the optimised design has a 32% lower compliance than the uniform design of the same volume. On the plates of the scale study, exact checks up to 51 cells give compliance errors below 0.014% and gradient errors below 0.05%, and a design iteration of a 110-cell plate with 32.7 million DOFs takes 19 min.

The periodic boundary conditions assumed by homogenisation fail without scale separation or when cells are cut, and a fine-scale model that discretises every cell must be reassembled and solved at every design iteration. Between these two approaches, NICE provides a component model that keeps the properties of static condensation without factorising the interior stiffness matrix of each cell. Including the coarse-grid correction in the network, as a coarse level that solves the coarse-grid problem with the matrix $V^T A V$, would leave the smoothing as the only fixed stage. The properties of the condensed stiffness follow from its computation through the strain energy and do not depend on the network. The error relations and the correction therefore apply to any family of components with a symmetric positive semidefinite stiffness matrix and a positive definite interior block; for a new family of cells, the network has to be retrained, whereas the error relations and the correction carry over unchanged.

Code and data availability

Upon acceptance, the code and data listed below will be archived as a cleaned, frozen snapshot in a public Zenodo repository. The code comprises the implementation of the network recovery operator, the two-grid correction and its transpose, the cut-cell preprocessing and the reference static condensation. It also comprises the scripts for assembly, benchmarks and figures. The data comprise the trained network parameters of the variants of Table 2 and Supplementary Note S8 and the parameter files of all validation and lattice geometries. They also comprise the result records from which every table and figure is generated. Training data are available from the corresponding author on reasonable request.

CRediT authorship contribution statement

[Author contributions (CRediT taxonomy).]

Declaration of competing interest

[Competing-interest statement.]

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[To be written by the author according to the journal's current policy (tool or service, purpose, and the authors' responsibility for the content).]

Appendix A. Discrete construction and metric definitions

Appendix A.1. Active degrees of freedom and element integration

A background element is active if its intersection with $\Omega(\eta)$ has positive measure. This is decided with closed-form interval enclosures of ϕ and τ over the element clipped by the cut plane. The element is excluded if an enclosure shows that a constraint is violated everywhere, accepted if both constraints hold strictly at its centre or a vertex, and subdivided otherwise. The active background elements use tensor-product Q_2 displacements, with 27 nodes and 81 displacement degrees of freedom per element. The retained cell-face node set is the union of the nine face nodes of every certified positive-area material patch. The cut node set is the union of all 27 nodes of each active element carrying a positive-area cut-surface patch. Their union is deduplicated in active-node order. The DOFs of the cut-plane elements that lie off the cut plane are also retained, because their basis functions determine the displacement and the virtual work on the cut plane.

The discrete stiffness is assembled as

$$K(\eta) = \sum_e L_e^T K_e(\eta) L_e + \gamma \sum_{f \in \mathcal{F}_g} L_f^T G_f^T G_f L_f, \qquad K_e(\eta) = \sum_{\alpha \in \{0,\dots,4\}^3} M_{e\alpha}(\eta) T_\alpha. \tag{A.1}$$

Here L_e, L_f extract element and ghost-face DOFs. The fixed elasticity templates $T_\alpha \in \mathbb{R}^{81\times 81}$ use the physical basis derivatives and isotropic Lamé constants $\lambda_L = E_Y\nu/[(1+\nu)(1-2\nu)]$ and $\mu_L = E_Y/[2(1+\nu)]$. The 125 moments $M_{e\alpha}$ integrate local monomials with coordinatewise exponents from zero to four over the material part of an element. The ghost-face set, its factors and coefficient specify the stabilisation contribution. The sum defines the stabilised discrete energy, including its ghost-penalty contribution.

The ghost-face set $\mathcal{F}_g$ consists of interior background faces whose two neighbouring elements are active and are not both certified as completely filled with material. These faces lie within each substructure. For background-element width $h = 1/n$, the unit-coefficient stabilisation bilinear form is

$$g_h(v,u) = (\lambda_{\mathrm{ref}} + 2\mu_{\mathrm{ref}}) \sum_{f \in \mathcal{F}_g} \sum_{j=1}^{2} h^{2j-1} \int_f [\partial_{n_f}^j v] \cdot [\partial_{n_f}^j u] \, \mathrm{d}A.$$

The derivative uses a common positive coordinate normal on both neighbouring elements, and the jump is the value on the first element minus that on the second. Integration covers the complete background face. The stabilisation form g_h, and hence G_f, uses fixed reference values $E_{\mathrm{ref}} = 1$, $\nu_{\mathrm{ref}} = 0.3$ and their Lamé constants $\lambda_{\mathrm{ref}}, \mu_{\mathrm{ref}}$; these equal the material values E_Y, ν of the 80 validation geometries. Each face integral uses a tensor-product three-point Gauss rule in its two tangential coordinates. Stacking the square-root-weighted jump evaluations for both derivative orders and all three displacement components gives $G_f \in \mathbb{R}^{54\times 135}$, acting on the 45 distinct nodes of the two-element patch. Thus the second term in Eq. (A.1) is the matrix representation of γg_h.

The moments are integrated on 4^3 initial subcells per active background element. Partially occupied subcells are refined once into 2^3 children, so that the finest subcell width near the material boundary is $h/8$. Refinement is local to partially occupied subcells and does not subdivide every active element uniformly. Fully occupied subcells contribute analytic tensor-product monomial moments. At the finest partial level, each subcell is divided into six Kuhn tetrahedra and clipped successively by the three interpolated inequalities defining the implicit band and the cut plane. The clipped tetrahedra are integrated with a conical Gauss–Jacobi product rule of four points per coordinate direction (64 points per tetrahedron). The accumulated moments are multiplied by the physical Jacobian $(2n)^{-3}$. Every quadrature weight is positive. The integrand that the templates expand in monomials is $B^T \mathsf{C} B$, where B is the strain–displacement matrix and C the elasticity tensor. Each numerically integrated K_e is therefore a positive combination of positive semidefinite matrices, and $K \succeq 0$ holds for the numerical moments as well.

Geometry generation accepts a cell only if its active background elements form a single face-connected set carrying retained cell-face DOFs. This establishes the connectivity of the stabilised discrete model, in which material connected only through partially filled elements is coupled through shared basis functions and the ghost penalty. Every training, validation, neighbour and lattice cell used in this work passes this check. This check yields the rigid-body kernel assumed in Section 2.2, provided the element integration is exact and the elasticity tensor is positive definite. Suppose $v^T K v = 0$. Both terms of Eq. (A.1) are positive semidefinite, so each vanishes. On every active element the material part has positive measure, so the Q_2 field v_h has zero symmetric gradient on a set of positive measure. Since $\nabla^{\mathrm{s}} v_h$ is polynomial, it then vanishes on the whole element, and v_h is rigid there. Two face-adjacent active elements share the nine nodes of their common face, which are not collinear, so their rigid motions coincide. Face-connectivity therefore makes v_h one global rigid motion. Conversely, a global rigid field is affine, so it has zero strain and zero jumps of its first and second normal derivatives, and the ghost term vanishes on it. Hence $\ker K = \operatorname{range} R$. With numerical moments, the same conclusion requires the discrete element matrices to keep this kernel. That is an assumption, supported by the rigid-body energy ratios of Table ST04.

For the population statistics of Section 5, the energy error of a geometry is the mean over the evaluated test displacements of one load class. Population means then give equal weight to each available geometry.

Appendix B. Variational identity, rigid-body kernel and energy-error measures

Appendix B.1. Assumptions

In Appendices B and C, $f \equiv f_g$ denotes the assembled retained load.

For each cell, $K = K^T \succeq 0$, the retained and interior DOF sets P, I are fixed, and $A = K_{II} \succ 0$. The exact recovery E is defined by $E_P = I_p$ and $E_I = -A^{-1}K_{IP}$. A linear recovery operator F has $F_P = I_p$. Set $H = F_I - E_I$, so $F = E + J_I^T H$, $S = E^T K E$, and $\widehat{S} = F^T K F$. All transposes are taken of the complete recovery operator, including the correction. Interior body loads are zero; equivalently, the load functional depends only on the retained DOFs. Boundary loads integrated consistently have this property when all DOFs with nonzero boundary virtual work are retained.

The cell interiors are disjoint, all intercell couplings act through the retained DOFs, and the assembly maps B_m include fixed homogeneous supports. The supported exact assembled stiffness $\mathbb{K} = \sum_m B_m^T S_m B_m$ is positive definite. The same local stabilised matrices define both the reference and the learned assemblies. The applied global load $f \neq 0$ is fixed.

Design statements additionally use a differentiable design interval on which the following discrete choices are fixed: the active elements; the ghost-face set $\mathcal{F}_g$; the retained and interior DOF sets P, I; the assembly maps B_m and the supports; the load; the network's binary node indicators and the sets of element and face stencils of weakly connected regions (Appendix G.1); and the coarse-column support screen and the coarse-factor shift ξ (Appendix F.1). A change in any of them is a switch of the discrete model, and derivatives are not taken across it. Stiffness derivatives include every varying term in the chosen discrete specification. In the sensitivities reported here, moment differences vary the body stiffness while the selected ghost matrix and its coefficient are fixed.

Appendix B.2. Variational identity, transpose of the recovery operator and rigid-body kernel (proof of Proposition 1)

Let the selectors satisfy $J_P^T J_P + J_I^T J_I = I_{n_a}$, and assume the symmetric stiffness in Eq. (2) has $A \succ 0$. For any field v with $J_P v = q$, there is a unique $w \in \mathbb{R}^i$ such that $v = Eq + J_I^T w$. Since $J_I K E = 0$,

$$v^T K v = q^T S q + w^T A w. \tag{B.1}$$

This proves the energy-minimising property and uniqueness of Eq. Applying the same expansion to an admissible linear recovery operator $F = E + J_I^T H$ yields Eq. (6). At fixed q, the stabilised discrete strain energy of Fq therefore exceeds that of Eq by one half of $q^T H^T A H q$. The reported energy error uses the quadratic energy without the factor one half. Moreover, $r_I = J_I K F q = A H q$, which proves both expressions in Eq. (7).

Suppose additionally that $\ker K = \operatorname{range} R$, R_P has rank six, and $F R_P = R$. The exact field associated with $R_P a$ is Ra, because it takes these retained displacements and satisfies interior equilibrium. Thus $E R_P = R$ and $H R_P = 0$. Rigid reproduction gives $\widehat{S} R_P = F^T K R = 0$. Conversely, if $q^T \widehat{S} q = 0$, positive semidefiniteness implies $Fq \in \ker K$. Write $Fq = Ra$; selecting the retained entries gives $q = R_P a$. Hence

$$\ker S = \ker \widehat{S} = \operatorname{range} R_P. \tag{B.2}$$

The inclusion $\ker \widehat{S} \subseteq \ker S$ holds without these assumptions: $\widehat{S} - S \succeq 0$ and $S \succeq 0$, so $q^T \widehat{S} q = 0$ implies $q^T S q = 0$ and hence $Sq = 0$. An approximate recovery operator therefore cannot add zero-energy modes to the condensed stiffness. A zero-energy mode other than the retained rigid-body modes could only belong to the discrete model itself. Appendix A.1 excludes such a mode under its connectivity condition and the assumption that the numerically integrated element matrices keep the rigid-body kernel. For the lattices of Table 5, the supported whole-lattice systems admitted a Cholesky factorisation, which excludes a zero-energy mode of the supported assembled system.

Rigid reproduction and symmetry give

$$\widehat{S} R_P = 0, \qquad R_P^T \widehat{S} = 0. \tag{B.3}$$

For the construction in Eq. (9), $C_R R_P = I_6$ and $\Pi_P R_P = 0$, and a linear raw displacement map sends zero to zero, so $\widehat{E} R_P = R$. Corrections driven by the interior residual leave this field unchanged, so $F R_P = R$ and Eq. (B.3) holds for the corrected recovery operator.

Let $Z \in \mathbb{R}^{p \times (p-6)}$ have orthonormal columns spanning the orthogonal complement of the retained rigid-body modes, and define $S_* = Z^T S Z \succ 0$. The largest energy error over this space is

$$\varepsilon_* = \sup_{q \perp R_P,\ q \neq 0} \varepsilon(q) = \| A^{1/2} H Z S_*^{-1/2} \|_2^2, \qquad \mu_* = 1 + \varepsilon_*. \tag{B.4}$$

Let $q = Za$, with $Z^T Z = I$ spanning the rigid complement. Its relative energy error is

$$\frac{a^T Z^T H^T A H Z a}{a^T S_* a} = \frac{\| A^{1/2} H Z S_*^{-1/2} b \|_2^2}{\|b\|_2^2}, \qquad b = S_*^{1/2} a. \tag{B.5}$$

Taking the supremum proves Eq. (B.4).

The retained block of the unbalanced discrete force, used alone, would be

$$(KF)_P = S + K_{PI} H.$$

The additional variational reaction is $F_I^T (KF)_I = (E_I + H)^T A H$. Since $E_I^T A = -K_{PI}$, its term $E_I^T A H$ cancels $K_{PI} H$. Thus

$$(KF)_P + F_I^T (KF)_I = S + H^T A H. \tag{B.6}$$

The transpose F^T thus ensures reciprocity and cancels the first-order stiffness error. A retained-force extraction without this term generally has an $O(H)$ error and need not be symmetric.

For any global vector, the quadratic error is the sum of nonnegative local errors in Eq. (C.1). Consequently $\widehat{\mathbb{K}} \succeq \mathbb{K} \succ 0$, which proves that the supported system is solvable even if some cells are only positive semidefinite.

Appendix B.3. Displacement magnitude and stiffness ratio

The energy error weights displacement errors of equal magnitude differently. To make this dependence explicit, remove the rigid-body part of the retained displacements by taking $q \perp R_P$, let $d = Fq - Eq$, and introduce a displacement norm induced by $\mathsf{W} \succ 0$:

$$\delta_{\mathsf{W}}(q) = \frac{\|d\|_{\mathsf{W}}}{\|Eq\|_{\mathsf{W}}}, \qquad \rho_{K,\mathsf{W}}(v) = \frac{v^T K v}{v^T \mathsf{W} v}, \qquad \kappa_{\mathsf{W}}(q,d) = \frac{\rho_{K,\mathsf{W}}(d)}{\rho_{K,\mathsf{W}}(Eq)}. \tag{B.7}$$

For nonzero field error and nonzero reference energy, cancellation of $d^T\mathsf{W}d$ and $u^T\mathsf{W}u$, with $u = Eq$, gives

$$\boxed{\varepsilon(q) = \delta_{\mathsf{W}}(q)^2 \kappa_{\mathsf{W}}(q,d).} \tag{B.8}$$

Section 5.4 and Table ST04 use $\mathsf{W} = \mathrm{diag}(0, D)$ with $D = \mathrm{diag}(K_{II})$, i.e. a Jacobi-weighted interior norm. This weighting is only positive semidefinite. The identity still holds, provided $d_I \neq 0$ and $u_I \neq 0$, because d vanishes on the retained DOFs; the condition $q \perp R_P$ is then not needed. The factor κ_{W} compares the stiffness content of the error with that of the equilibrium response; it is not a matrix condition number. An error concentrated in deformation stiffer than the loaded response has a large κ_{W}, so that even a small relative displacement error is mechanically significant.

The denominator of κ_{W} measures the physical response to the chosen test displacement. The smoothing acts on the spectrum of $Av = \lambda Dv$ with the retained displacements fixed, so the two notions of softness differ. For example, $A = \mathrm{diag}(\epsilon_0, 1)$ with $0 < \epsilon_0 \ll 1$ has a very soft unscaled mode, while $D^{-1/2}AD^{-1/2} = I$. The cumulative Jacobi-scaled low-mode error energy (Eq. (10)) therefore identifies components that the smoothing reduces slowly, but does not determine the measured κ.

Appendix C. Assembly and compliance ordering

For a global retained vector U, minimising the sum of substructure energies over their disjoint interior DOFs separates into the individual minimisations in Eq. (B.1). The minimised quadratic form is $\sum_m U^T B_m^T S_m B_m U$. Under the coupling assumptions of Section 2.3, local condensation followed by assembly is therefore equivalent to eliminating the interior DOFs from the assembled system of all substructures.

Let $\mathbb{D} = \widehat{\mathbb{K}} - \mathbb{K}$. From Eq. (6),

$$U^T \mathbb{D} U = \sum_m \|H_m B_m U\|^2_{A_m} \geq 0. \tag{C.1}$$

If homogeneous supports make $\mathbb{K} \succ 0$, then both global systems are invertible and matrix inversion reverses their order. For any fixed nonzero load,

$$\widehat{C} = f_g^T \widehat{\mathbb{K}}^{-1} f_g \leq C = f_g^T \mathbb{K}^{-1} f_g. \tag{C.2}$$

The order reversal follows by setting $Q = \mathbb{K}^{-1/2}\widehat{\mathbb{K}}\mathbb{K}^{-1/2} \succeq I$. Its eigenvalues are at least one, so $Q^{-1} \preceq I$; a congruence gives the inverse inequality.

Appendix C.1. Compliance error as the error energy of the recovered field (proof of Proposition 3)

Let $U = \mathbb{K}^{-1} f$, $\widehat{U} = \widehat{\mathbb{K}}^{-1} f$, $q_m = B_m U$, $\widehat{q}_m = B_m \widehat{U}$, $u_m = E_m q_m$, and $\widehat{u}_m = F_m \widehat{q}_m$. The matrix $\mathbb{D} = \widehat{\mathbb{K}} - \mathbb{K}$ of Eq. (C.1) equals $\sum_m B_m^T H_m^T A_m H_m B_m$.

For conforming cell fields, use the energy form $a(v,v)=\sum_m v_m^T K_m v_m$. Shared retained DOFs are counted through their separate cell energy contributions. Exact local equilibrium and the global equation give $a(u,v)=f^T Y$, where Y denotes the free assembled retained displacements of v. At the equilibrium $\widehat{\mathbb{K}}\widehat{U}=f$, $a(\widehat{u},\widehat{u})=f^T\widehat{U}=\widehat{C}$. Hence

$$\boxed{C-\widehat{C}=a(\widehat{u}-u,\widehat{u}-u)=\|\widehat{U}-U\|_{\mathbb{K}}^2+\sum_m \|H_m\widehat{q}_m\|_{A_m}^2.}$$

The second equality follows from $\widehat{u}_m-u_m=E_m(\widehat{q}_m-q_m)+J_{I,m}^T H_m\widehat{q}_m$. The two terms are K_m-orthogonal because $J_{I,m}K_mE_m=0$. The underestimation of the compliance thus measures the total energy error of the recovered field, and the error of the retained solution accounts for only one part of it.

Define

$$\beta=\frac{U^T\mathbb{D}U}{C}=\sum_{m:q_m^T S_m q_m>0} w_m\varepsilon_m(q_m),\qquad w_m=\frac{q_m^T S_m q_m}{C}.$$

For zero-energy cells with exact rigid reproduction, the corresponding numerator is also zero and its contribution is defined directly as zero. Insert $Y=\alpha U$ in the maximum principle

$$\widehat{C}=\max_Y\{2f^TY-Y^T\widehat{\mathbb{K}}Y\}.$$

Since $U^T\widehat{\mathbb{K}}U=(1+\beta)C$, optimising the scalar gives $\alpha=(1+\beta)^{-1}$ and $\widehat{C}\geq C/(1+\beta)$, i.e. $(C-\widehat{C})/C\leq\beta/(1+\beta)\leq\beta$. With Eq. (C.2), this proves Eq. (15). The statement is specific to the load, since w_m and ε_m are evaluated at the same exact assembled retained displacements.

Appendix C.2. Inexact assembled solves

Let $\bar{U}$ be an approximate solution of the same fixed variational operator, $\rho=f-\widehat{\mathbb{K}}\bar{U}$, $\bar{C}=f^T\bar{U}$, and $\bar{u}_m=F_mB_m\bar{U}$. The exact identity becomes

$$\boxed{C-\bar{C}=a(\bar{u}-u,\bar{u}-u)+\bar{U}^T\rho,\quad J(\bar{U})=\bar{C}+\bar{U}^T\rho\leq\widehat{C}\leq C.}$$

The error of the corrected functional is $\widehat{C}-J(\bar{U})=\rho^T\widehat{\mathbb{K}}^{-1}\rho$, and $\widehat{C}-\bar{C}=\widehat{U}^T\rho$. Thus the work $f^T\bar{U}$ can violate the variational compliance ordering if the algebraic residual is appreciable, while the concave work functional retains a quadratic residual error.

If an applied action $y(\bar{U})$ is not numerically identical to the variational energy, define $\omega=\bar{U}^Ty(\bar{U})-\sum_m\bar{u}_m^TK_m\bar{u}_m$ and $\rho=f-y(\bar{U})$. Then the first identity acquires an additional $+\omega$. This separates the residual of the solve from the inconsistency between the applied action and the energy. A recursive Krylov residual need not equal this applied-action residual. Evaluating the terms in Eq. (16) additionally requires the signed residual work and ω. For the two-cell configurations and the lattices, both are recorded together with the dual-norm bound $|\bar{U}^T\rho|\leq\sqrt{\bar{U}^T\mathbb{K}\bar{U}}\sqrt{\rho^T\mathbb{K}^{-1}\rho}\leq\sqrt{\bar{U}^Ty(\bar{U})-\omega}\,\sqrt{\rho^T\mathbb{K}^{-1}\rho}$; the second inequality holds because $\widehat{\mathbb{K}}\succeq\mathbb{K}$ (Supplementary Note S4.3). ## Polynomial smoothing and coarse projections

For $0<a<b$, the degree-k Chebyshev error polynomial and its energy bound are

$$p_k(t) = \frac{T_k((a+b-2t)/(b-a))}{T_k((a+b)/(b-a))}. \tag{D.1}$$

$$d_I^{\mathrm{sm}} = p_k(D^{-1}A)d_I, \qquad \|d_I^{\mathrm{sm}}\|_A^2 \le \psi_k^2 \|d_I\|_A^2, \qquad \psi_k = \max_{\lambda \in \sigma(D^{-1/2}AD^{-1/2})} |p_k(\lambda)|. \tag{D.2}$$

Here $D = \mathrm{diag}(A)$.

Write $\bar{\lambda} = (a+b)/2$, $\eta_s = (b-a)/2$, and $\zeta = \bar{\lambda}/\eta_s > 1$ for the centre and half-width of the smoothing interval and their ratio. Starting from u_0, keep its retained entries fixed and define the interior residual with the sign convention $r_j = (Ku_j)_I$. A recurrence producing the polynomial in Eq. (D.1) is

$$\begin{aligned} \varrho_0 &= 1/\zeta, & z_0 &= -D^{-1}r_0/\bar{\lambda}, \\ u_{j+1} &= u_j + J_I^T z_j, & \varrho_{j+1} &= (2\zeta - \varrho_j)^{-1}, \\ z_{j+1} &= \varrho_{j+1}\varrho_j z_j - \frac{2\varrho_{j+1}}{\eta_s} D^{-1} r_{j+1}. \end{aligned} \tag{D.3}$$

The last line is evaluated only when another step is needed. Zero steps return the initial field. To verify the polynomial, observe that $\varrho_j = T_j(\zeta)/T_{j+1}(\zeta)$, and apply $T_{j+2}(x) = 2xT_{j+1}(x) - T_j(x)$. After the first step the error is $d_1 = (I - D^{-1}A/\bar{\lambda})d_0$, and the same recurrence gives $d_k = p_k(D^{-1}A)d_0$.

The matrices $D^{-1}A$ and $D^{-1/2}AD^{-1/2}$ are similar, $p_k(D^{-1}A)$ is self-adjoint in the A-inner product, and

$$\|p_k(D^{-1}A)d\|_A^2 = \sum_j \lambda_j p_k(\lambda_j)^2 c_j^2, \qquad c_j = v_j^T D d. \tag{D.4}$$

This proves Eq. (D.2). If $a \le \lambda \le b$, the argument of T_k lies in $[-1, 1]$, where $|T_k| \le 1$. For $0 < \lambda < a$, it lies between 1 and ζ, where T_k increases from 1 to $T_k(\zeta)$. Hence $|p_k(\lambda)| \le 1$ throughout $(0, b]$. For $b < \lambda \le a + b$ the argument lies in $[-\zeta, -1)$, where $|T_k| \le T_k(\zeta)$, so $|p_k(\lambda)| \le 1$ on $(0, a+b]$, with equality at $a + b$. The polynomial therefore does not increase the error in the A-energy norm when the actual positive spectrum is contained in $(0, a+b]$, and in particular when it is contained in $(0, b]$. Modes below a can contract slowly, and eigenvalues above $a + b$ are amplified.

The operational upper endpoint is estimated using a random power vector, 40 iterations of $D^{-1}A$, Euclidean normalisation, and a factor of 1.05; the interval is computed once per geometry. The estimate sets the polynomial, while the result in Eq. (D.2) is conditional on the actual spectrum. A finite power estimate multiplied by a safety factor only estimates the endpoint and does not establish the containment used above. For the 80 validation geometries the operational endpoint was therefore compared with $\lambda_{\max}(D^{-1}A)$ computed by Lanczos iteration with full reorthogonalisation on $D^{-1/2}AD^{-1/2}$ (78–150 steps; relative Ritz residual at most 9.6×10^{-4}, median 2.2×10^{-6}). The operational endpoint exceeds the converged value by 2.1–5.0% (median 3.9%) on every geometry, which supports the containment required by Eq. (D.2) on all evaluated cells. A converged Ritz value is a numerical estimate, not a certified bound. The strict Gershgorin bound $\max_i \sum_j |A_{ij}|/A_{ii}$

is 4.7–19.4 times the operational endpoint (median 18.4); used as b, it would stretch the smoothing interval by that factor and slow the contraction of the upper spectrum accordingly.

For a full-column-rank coarse basis V, with $A_c = V^T A V \succ 0$, define $Q_V = V A_c^{-1} V^T A$. Direct multiplication gives $Q_V^2 = Q_V$ and $Q_V^T A = A Q_V$. Thus Q_V is an A-orthogonal projection and $C_V = I - Q_V$ is its complementary projection. With $b_r = V^T A d$,

$$\|d\|_A^2 = \|C_V d\|_A^2 + \|Q_V d\|_A^2, \qquad \|Q_V d\|_A^2 = b_r^T A_c^{-1} b_r. \tag{D.5}$$

This proves Eq. (11). Applying Eq. (D.2) before and after this projection yields $\|\Phi_k C_V \Phi_k d\|_A \leq \psi_k^2 \|d\|_A$. Because $\psi_k < 1$ whenever the positive spectrum lies in $(0, b]$, the cycle is an energy contraction, which gives the operator ordering stated in Section 3.4 (Appendix D.1). For the spectra of U1, M1 and M2 ($\lambda_1 \approx 4$–9×10^{-4} against $a \approx 0.17$; Supplementary Table ST05b), however, the energy contraction factor ψ_8^4 of the cycle is 0.97–0.99. The estimate therefore guarantees little more than that the cycle does not increase the energy error. It uses no approximation property of the coarse space, and the reductions by one to three orders of magnitude in Section 5.5 are empirical. A two-grid convergence estimate would require an approximation property of the Q_1 space for walls about one element thick Xu & Zikatanov (2002). This is the standard energy-projection mechanism of subspace correction Xu (1992).

Appendix C.3. Corrections, orderings and approximate coarse inverses (proof of Proposition 2)

A linear correction that is fixed for a given geometry acts on H through $H_{\text{new}} = \Theta H$. If $\Theta^T A \Theta \preceq A$, then

$$S \preceq \widehat{S}_{\text{new}} \preceq \widehat{S}_{\text{old}}.$$

Thus the exact assembled compliance increases monotonically towards the reference, and the total reconstructed energy error of Eq. (14) decreases; in particular, repeating a fixed cycle cannot increase the energy error. Changing the smoothing degree, or stopping the recurrence of Eq. (D.3) at an intermediate step, changes the polynomial, so the energy error need not decrease monotonically in k. The sensitivity error is not ordered by this matrix inequality. For example, take $K = I_3$, $P = \{1\}$, $u = (1, 0, 0)^T$ and the positive semidefinite derivative $K_{,c} = v v^T$, $v = (1, 1, -1)^T$, of the nested family $K(\tau) = I + \tau v v^T$. Projecting the second component out of the interior error $d = (0, t, t)^T$ halves its energy from $2t^2$ to t^2, whereas the sensitivity discrepancy of Eq. (17) changes from zero to $2t - t^2$ against a reference sensitivity of -1 (0.19 at $t = 0.1$).

For a symmetric approximate inverse G in place of A_c^{-1} in the coarse-grid correction $C_V = I - V A_c^{-1} V^T A$ of Eq. (D.5), the energy removed by the correction is

$$\|d\|_A^2 - \|d - V G V^T A d\|_A^2 = b_r^T (2G - G A_c G) b_r, \quad b_r = V^T A d. \tag{D.6}$$

Consequently, $2G - G A_c G \succeq 0$ suffices for the correction not to increase the energy error. A nonnegative shift $G = (A_c + \Lambda)^{-1}$, with $\Lambda \succeq 0$ and $A_c + \Lambda \succ 0$, satisfies this condition, since $2G - G A_c G = G(A_c + 2\Lambda)G$, although it is not the exact projection. Neither Eq. (D.6) nor this condition requires V to have full column rank. If A_c is singular because the

columns of V are dependent, a shift $\Lambda = \xi\,\mathrm{diag}(A_c)$ with $\xi > 0$ still gives $A_c + \Lambda \succ 0$, because every column kept in V carries positive energy. This statement concerns the coarse inverse while the prescribed fine-grid stiffness K remains fixed. If a probed matrix differs from A_c, contraction depends on the approximate inverse relative to the true A_c.

Appendix D. Transpose of the recovery operator

All transposes below use the Euclidean pairing of the stored displacement and nodal-force vectors. Write $M_I = J_I^T J_I$. Transposing Eq. (9) gives

$$\widehat{E}^T y = J_P y + C_R^T R^T M_I y + \Pi_P^T \mathcal{N}_\theta(\eta)^T M_I y. \tag{E.1}$$

The final overwrite of the retained values therefore keeps the direct retained contribution and masks the field sent through the network transpose. The geometry coefficients are held fixed during this displacement transpose.

Set $\varphi_k(t) = [1 - p_k(t)]/t$, with the polynomial continuation at zero. The full-field smoothing map $\mathcal{T}_k$ has the block representation

$$\mathcal{T}_k = \begin{bmatrix} I_p & 0 \\ -\varphi_k(D^{-1}A)D^{-1}K_{IP} & p_k(D^{-1}A) \end{bmatrix}, \qquad \mathcal{T}_k^T y = \begin{bmatrix} y_P - K_{PI}\varphi_k(D^{-1}A)D^{-1}y_I \\ Dp_k(D^{-1}A)D^{-1}y_I \end{bmatrix}. \tag{E.2}$$

The final interior field equals its equilibrium value plus $p_k(D^{-1}A)$ times the initial error. Replacing $(I - p_k(D^{-1}A))A^{-1}$ by $\varphi_k(D^{-1}A)D^{-1}$ yields the first expression. The second follows from $(D^{-1}A)^T D = A = D(D^{-1}A)$, which implies $p_k(D^{-1}A)^T = Dp_k(D^{-1}A)D^{-1}$ and the corresponding identity for φ_k. Although the forward map leaves the retained displacements unchanged, its transpose contributes a force on the retained DOFs through the upper block in Eq. (E.2).

For the full-field coarse-grid correction,

$$\mathcal{W}_c = I_{n_a} - J_I^T V A_c^{-1} V^T J_I K, \qquad \mathcal{W}_c^T = I_{n_a} - K J_I^T V A_c^{-1} V^T J_I. \tag{E.3}$$

A cycle $\mathcal{W} = \mathcal{T}_{\mathrm{post}} \mathcal{W}_c \mathcal{T}_{\mathrm{pre}}$ therefore uses the reverse sequence $\mathcal{W}^T = \mathcal{T}_{\mathrm{pre}}^T \mathcal{W}_c^T \mathcal{T}_{\mathrm{post}}^T$. The resulting reaction is

$$\widehat{S} q = \widehat{E}^T \mathcal{W}^T K \mathcal{W} \widehat{E} q. \tag{E.4}$$

The same formulas hold for the symmetric shifted inverse of Appendix F.1 when it replaces the coarse inverse consistently in both actions.

Appendix E. Coarse space, factorisation, and arithmetic

Appendix E.1. Coarse basis and factorisation

The principal coarse basis V consists of trilinear (Q_1) vector shape functions on a 17^3-vertex grid over the reference box, with three displacement DOFs per vertex. The shape functions are evaluated at the active background-node coordinates and restricted to the interior DOFs I; columns with an absolute interior-support sum at most 10^{-14} are removed.

Alternative coarse spaces (Q_1 on 9^3 and 33^3 vertices, Q_2 on 17^3 nodes and enriched partition-of-unity spaces) are compared in Supplementary Table ST06.

The coarse matrix V^TAV is recovered from coloured stiffness probes. With a stencil reach of four fine-grid node spacings and h_g fine-node spacings per coarse-vertex spacing, the probe radius is $R_g = 2 + \lceil 4/h_g \rceil$. Colours use vertex coordinates modulo $2R_g + 1$ and the displacement component, and the resulting colour separation resolves every interacting coarse pair. The recovered matrix is Jacobi scaled, with the corresponding column scaling of V, and factorised by Cholesky, trying diagonal shifts ξ in the order $0, 10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}$. The selected value, the coarse-factor shift, is stored with the geometry factorisation. With the correction in single precision, as in the optimisation runs of Section 5.10, a factor is accepted only if its smallest squared pivot is at least 10^{-7} of the unit diagonal. Otherwise the next shift is tried. This keeps the forward and transposed coarse solves consistent for cut cells near the lower thickness bound. A shift replaces the exact projection by the approximate inverse $G_s = (A_s + \xi I)^{-1}$ of the scaled Galerkin matrix A_s. In the unscaled basis this is the shift $\Lambda = \xi \operatorname{diag}(V^TAV)$ of Appendix D.1. The support screen removes only columns that are numerically zero, so V need not have full column rank. A positive shift then still gives $A_s + \xi I \succ 0$. Since $2G_s - G_sA_sG_s = G_s(A_s + 2\xi I)G_s \succeq 0$, Eq. (D.6), with the scaled basis in place of V, shows that the shifted coarse-grid correction still does not increase the A-energy error, with or without full column rank. This requires the forward and transposed actions to use the same solve.

Appendix E.2. Arithmetic

The network's interior displacements, the rigid-body reconstruction and the prescribed retained entries use single precision (float32). Quadratic energies and sparse stiffness products use double precision (float64), and convolutions use true float32 arithmetic, without TF32 or lower precisions. The correction, including the smoothing and the coarse algebra, runs in double precision in training and in the accuracy studies of Sections 5.2–5.8. It runs in single precision in the timed lattice route of Table 5 and in the optimisations of Section 5.10. The fixed-network correction studies of Section 5.5 (Table 4 and Supplementary Tables ST05a, ST06 and ST07) apply the same correction in double precision to the single-precision network field. For the base network they estimate the smoothing interval from their own power-iteration start, which is also the interval compared with the spectrum in Section 5.4 (Supplementary Table ST05b). Sensitivity contractions use float32 element products with float64 accumulation. These arithmetic choices approximate the real linear maps of the preceding derivations. ## Network coefficients and training test displacements

Appendix E.3. Geometry features and displacement maps

The architecture in Figure 3 separates the generation of geometry-dependent coefficients from the propagation of displacements. The displacement features carry 32 feature channels per node on the fine grid (level $k = 0$, 65 positions per axis) and on the coarse grids of levels $k = 1, 2, 3$ (33, 17 and 9 positions per axis). The network processes several test displacements simultaneously, each with its own displacement features. The geometry embeddings are the 64-component element and node feature vectors of Section 3.2 (Figure 3a); they carry no test-displacement dimension. Element, face, retained-node and grid-transfer incidences are constructed before the network is evaluated. The symbols a, b, k, p_i, r_i, σ and $\mathcal{R}$ are local

to this appendix and unrelated to the smoothing interval, polynomial and residual notation of the main text and Appendix D.

Geometry encoding. Each element has 126 input features: material volume fraction, its logarithm, and the remaining 124 moments normalised by material volume. The 11 node features contain indicators of retained, cell-face, cut-plane-element and weakly connected nodes, a normalised logarithmic stiffness diagonal, and six sine/cosine coordinate entries. The stiffness summary is the Frobenius norm of the node's diagonal 3×3 block. A node is designated weakly connected when this norm is less than 0.01 times its median over active nodes.

The element encoder maps 126 inputs to a 64-component embedding. The mean of the incident element embeddings, concatenated with the 11 node features, forms the 75 inputs of the node encoder. Two residual message-passing rounds (the two rounds of exchange in Section 3.2) then update elements from the mean embeddings of their 27 nodes and update nodes from their incident elements. Each update uses the current embedding and the aggregated neighbouring embedding as a 128-component input. A ghost-face embedding is formed from the two adjacent element embeddings and a three-component indicator of the face axis. All encoders and coefficient heads use two affine layers separated by GELU, with hidden width 64.

Local linear interactions. The three retained displacement components are lifted by a shared matrix $W_{\mathrm{in}} \in \mathbb{R}^{3\times 32}$, while interior features are initially zero. Denote this initial feature field by $X^0(q)$; it depends on q through q_d. For each element or face stencil t, local node position s and head h, the interaction first gathers and mixes channels:

$$Z_{th} = \left(\sum_{s=1}^{27} a_{tsh}(\eta) X_{i(t,s)} \right) W_{\ell h}, \qquad \Delta X_i = \sum_{(t,s):i(t,s)=i} \sum_{h=1}^{4} \omega_{ts}\, b_{tsh}(\eta) Z_{th}.$$

Here $W_{\ell h} \in \mathbb{R}^{32\times 32}$ is a shared learned channel map for layer ℓ and head h. The scalar gather and scatter coefficients a and b are generated separately from the element or face embedding, the incident node embedding and an eight-dimensional position embedding. The position embedding is a learned 27×8 table indexed by the local node position, with one table for element stencils and one for face stencils. The 136-component concatenation of the three embeddings identifies both the local material context and the position within the stencil. The four heads provide distinct gather--mix--scatter contributions on the same fixed incidence. Equation (8) adds their update to the incoming state and restores the prescribed retained features after every local interaction.

The element stencil contains all 27 Q_2 nodes. The learned ghost-face stencil contains 18 owner-side nodes and nine neighbour-side nodes, ordered consistently by face orientation. It defines a learned communication map on a face neighbourhood; the full mechanical ghost-penalty contribution remains in K.

The node scatter accounts for the number and relative stiffness of incident contributions. If k_{es} is the Frobenius norm of the element's diagonal 3×3 block at local node position s, the element weights are

$$\pi_{es} = \frac{k_{es}}{\sum_{(e',s')\mapsto i} k_{e's'}}, \qquad \omega_{es} = \frac{1-\lambda_{\mathrm{mix}}}{\deg(i)} + \lambda_{\mathrm{mix}} \pi_{es}, \qquad (e,s) \mapsto i. \tag{G.1}$$

The face weights use the corresponding ghost-contribution diagonal blocks and face incidences. When all incident stiffness norms at a node vanish, the stiffness fraction is defined by the inverse incidence count. Each layer has its own learned λ_{mix}, initialised at zero (plain incidence averaging). For the additional weak-region layers, a stencil is selected when it contains a weakly connected node; its weights keep the normalisation of the full element or face incidence set.

Multilevel propagation. The grid hierarchy enlarges the spatial range of communication, while additive skip connections keep the fine-grid field. Let t_{ia} be the fixed trilinear incidence weight between a fine node i and a coarse vertex a. Positive geometry coefficients r_i and p_i define restriction and prolongation componentwise:

$$(\mathcal{R}X)_a = \frac{\sum_i t_{ia} r_i X_i}{\sum_i t_{ia} r_i}, \qquad (\mathcal{I}X_c)_i = p_i \frac{\sum_a t_{ia} X_{c,a}}{\sum_a t_{ia}}.$$

Only coarse vertices reached by the stored trilinear incidences participate. The denominators depend on geometry alone, so both maps are linear in displacement. Restriction and prolongation are separately parameterised. The network transpose $\mathcal{N}_\theta(\eta)^T$ in Eq. (E.1), and hence the action of the condensed stiffness in Eq. (5), is therefore obtained by transposing the complete composition of operations; one transfer map is not identified with the transpose of the other.

On each coarse level, a residual convolution takes the form

$$X \leftarrow X + g_j^{(k)}(\eta) \odot \mathrm{Conv}_j^{(k)}(X).$$

Each convolution has a $3 \times 3 \times 3$ kernel, 32 input and output channels, unit padding, no bias and no activation on the displacement features. Participating coarse-grid nodes are inserted into the corresponding background grid, with unused positions set to zero; the convolution output is sampled back at those nodes. The geometry branch supplies one multiplicative coefficient per participating node, channel and convolution. Its coarse-grid embeddings are obtained by fixed trilinear weighted averaging.

The full displacement sequence is specified below. Grid sizes count background positions per axis, whereas the participating node counts depend on geometry. Every row carries 32 displacement channels until the final projection.

Stage	Operations in execution order
Fine input, 65 grid	Lift q_d on retained nodes, set interior features to zero; apply four element-then-face interaction pairs; save $Y^{(0)}$.
Downward, 33 grid	Restrict 65→33; apply two residual convolutions; save $Y^{(1)}$.
Downward, 17 grid	Restrict 33→17; apply two residual convolutions; save $Y^{(2)}$.
Downward, 9 grid	Restrict 17→9; apply two residual convolutions.
Upward, 9 grid	Apply two residual convolutions; prolong 9→17; add $Y^{(2)}$.
Upward, 17 grid	Apply two residual convolutions; prolong 17→33; add $Y^{(1)}$.

Stage	Operations in execution order
Upward, 33 grid	Apply two residual convolutions; prolong 33→65; add $Y^{(0)}$.
Fine output, 65 grid	Restore retained features; apply four element-then-face interaction pairs, then four weak-region element-then-face pairs; project 32→3.

At the upward step that returns to level $k \in \{0, 1, 2\}$, the combination is $X = Y^{(k)} + \sigma^{(k)}\mathcal{I}^{(k)}X_c$, where X_c is the field on level $k+1$ and $\sigma^{(k)}$ is a learned scalar. Thus convolution precedes prolongation, the skip connections are additive, and the 9-position grid receives two downward and two upward convolutions. There are twelve coarse convolution updates in total. The fine retained values are restored after the grid hierarchy returns to the active nodes.

Coefficient bounds and output. The local heads produce separate gather/scatter values for eight ordinary element layers, twelve face layers and four additional weak-region element layers, each with four heads and 27 local node positions. Raw head outputs are multiplied by 0.2. For a finite positive group bound $a_{\max}$, the coefficient map is

$$\mathcal{C}(z) = \begin{cases} z, & |z| \le k_b a_{\max}, \\ \operatorname{sgn}(z)\left[k_b a_{\max} + (1-k_b)a_{\max}\tanh\left(\dfrac{|z| - k_b a_{\max}}{(1-k_b)a_{\max}}\right)\right], & |z| > k_b a_{\max}. \end{cases}$$

All variants of Table 2 and Supplementary Table ST24 set $k_b = 0.5$. Each $a_{\max}$ is twice the largest magnitude of its coefficient group recorded in a calibration pass over training geometries, so the map is the identity up to that recorded maximum. Table ST15 (Supplementary Note S5) lists the groups, the initialisation and the parameter count of every block. Positive transfer coefficients apply the corresponding bound to softplus(z) and add 10^{-3}; convolution coefficients use $2\,\mathrm{sigmoid}(z)$.

A bias-free matrix $W_{\text{out}} \in \mathbb{R}^{32\times 3}$ returns the fine features to nodal displacement in the prescribed coordinate order. Reconstruction of the rigid-body motion and restoration of the retained values then give Eq. (9).

Frame consistency. Geometry augmentation in training samples the 48 cube symmetries. For an orthogonal cube transformation, the deterministic signed permutation maps satisfy

$$K' = T_a K T_a^T, \qquad E' = T_a E T_P^T, \qquad S' = T_P S T_P^T. \tag{G.2}$$

Here $T_a \in \mathbb{R}^{n_a \times n_a}$ and $T_P \in \mathbb{R}^{p\times p}$ are the signed permutation matrices that the cube transformation induces on the active and the retained DOFs of the cell, and primes denote quantities of the transformed cell. The augmented geometry and vector field are passed through the network and the predicted displacement is returned to the original frame for mechanical evaluation.

Appendix E.4. Test-displacement sets

The normalised test displacements are

$$q_j = \frac{\Pi_P q_j^{\mathrm{raw}}}{\sqrt{(\Pi_P q_j^{\mathrm{raw}})^T S (\Pi_P q_j^{\mathrm{raw}})}}, \tag{G.3}$$

where q_j^{raw} is the test displacement produced by its class before rigid-body removal and energy normalisation.

The load classes used in training have the following mechanical definitions.

Class	Construction before rigid removal and energy normalisation
`force`	Equilibrated nodal point loads from smooth plane waves or localised Gaussian patches, with a subset loading the cut boundary.
`force_c`	Consistently integrated, self-equilibrated traction loads on the material cell faces; a subset also loads the cut surface. Equilibrium is imposed in traction-quadrature space.
`face` / `face_c`	Equilibrated nodal point loads or consistently integrated traction loads on a single cell face.
`support`	Responses with soft spring support on one cell face and equilibrated loads on the other faces.
`support_k`	Consistently integrated loads with springs scaled by the local stiffness diagonal and a logarithmically sampled factor in $[0.3, 3]$; the cut surface remains unloaded.
`macro`	Equal sampling of uniform strain, quadratic, and cubic imposed displacement fields.
`grf`	Multiscale displacement fields spanning 0.5–24 spatial cycles across the reference box.
`glued`	Displacements imposed by a neighbouring cell that shares the interface degrees of freedom and is clamped or supported by springs on its far face; one quarter of the samples load only the neighbour.
`adv`	Test displacements selected by the generalised-energy search described below.

The class `glued` supplies test displacements shaped by the neighbouring cell, while the energy and sensitivity labels refer to the operator of the isolated test cell; each test displacement is normalised by Eq. (G.3).

Appendix E.5. Search for difficult test displacements

Test displacements with large energy ratios $q^T \widehat{S} q / q^T S q$ are found by a block search on the complement of the rigid-body modes in the space of retained displacements. Whenever

a geometry enters the training rotation, the search initialises eight candidate test displacements and performs four block iterations. Applying $S_\perp^{-1} = ZS_*^{-1}Z^T$, the inverse of S on this complement (with Z and S_* as in Eq. (B.4)), uses an equilibrated Neumann solve. Six deterministically selected displacement pins remove the rigid-body motion, and the result is projected onto the complement. The candidate energy ratios are observations for individual test displacements; maximising them does not give an upper bound over all test displacements. The training settings are listed in Supplementary Table ST01.

Appendix E.6. Coverage by the test displacements

With Z and $S_* = Z^T S Z \succ 0$ as in Eq. (B.4), let $T_* = S_*^{-1/2} Z^T H^T A H Z S_*^{-1/2}$. For an energy-normalised test displacement, $x_j = S_*^{1/2} Z^T q_j$ has unit Euclidean norm and, by Eq. (B.5), $\varepsilon_j = x_j^T T_* x_j$. For N_q sampled test displacements, define the coverage matrix $M_{N_q} = N_q^{-1} \sum_{j=1}^{N_q} x_j x_j^T$. Then

$$\bar{\varepsilon} = \operatorname{tr}(T_* M_{N_q}), \qquad M_{N_q} \succeq \alpha I, \alpha > 0 \implies \varepsilon_* \le \operatorname{tr}(T_*) \le \bar{\varepsilon}/\alpha. \tag{G.4}$$

If the test displacements do not span the complement, choose a unit v orthogonal to them and $T_* = \Theta\, vv^T$ with $\Theta > 0$. Every sampled error is then zero, whereas the worst error is Θ and can be arbitrarily large. Such a PSD perturbation can be realised as $H^T A H$ if there is at least one interior degree of freedom. Thus retaining all DOFs in P defines the approximation target, while adequate coverage by the test displacements controls how accurately it is learned. The coverage condition $M_{N_q} \succeq \alpha I$, $\alpha > 0$, on the $(p-6)$-dimensional complement requires $N_q \ge p - 6$, i.e. thousands to tens of thousands of test displacements per geometry here. The sampled sets of test displacements do not satisfy it, so Eq. (G.4) is not applied and no upper bound on ε_* is claimed.

Appendix F. Sensitivity identities and design intervals

The derivative of the exact condensed stiffness and the derivative of the energy at fixed displacement are

$$S_{,c} = E^T K_{,c} E, \qquad \frac{\partial}{\partial \tau_c}\left(\tfrac{1}{2} q^T S q\right) = \tfrac{1}{2}(Eq)^T K_{,c}(Eq). \tag{H.1}$$

On a differentiable interval with fixed active topology and fixed selectors J_P and J_I, differentiating $J_P E = I_p$ gives $J_P E_{,c} = 0$. Since $J_I K E = 0$, both terms $E_{,c}^T K E$ and $E^T K E_{,c}$ vanish. Differentiating $S = E^T K E$ therefore proves Eq. (H.1).

For a supported equilibrium, differentiating $\mathbb{K}U = f_g$ at a fixed load gives $U_{,c} = -\mathbb{K}^{-1}\mathbb{K}_{,c}U$ and hence $C_{,c} = -U^T \mathbb{K}_{,c} U$. With fixed gathers and Eq. (H.1), this becomes the sum of $-u_m^T K_{m,c} u_m$ over affected cells. The sensitivity labels hold the base-design nodal load fixed, including the nodal loads obtained by consistent integration of traction loads.

Expanding the quadratic sensitivity estimate at $u + d$ proves Eq. (17).

For the condensed stiffness $\widehat{S} = F^T K F$, direct differentiation gives

$$\widehat{S}_{,c} = F_{,c}^T K F + F^T K_{,c} F + F^T K F_{,c}. \tag{H.2}$$

Apply the compliance derivative at equilibrium to the assembly of learned substructures and use $J_P F_{,c} = 0$. With $F_{I,c} = J_I\, \partial F/\partial \tau_c$, so that $F_{,c} = J_I^T F_{I,c}$, the two terms that contain the derivative of the recovery operator become $2(F_{I,c}\widehat{q})^T r_I$, proving Eq. (18). If the load depends on design, the full compliance derivative also contains $2f_{g,c}^T \widehat{U}$, and correspondingly $2f_{g,c}^T U$ for the exact assembly. Changes of the DOF selectors, assembly maps or support maps contribute their own chain-rule terms, including the derivatives of $B_m^T S_m B_m$ when B_m varies.

In the reference computation, moment derivatives are approximated by

$$M_{e\alpha,c} \approx \frac{M_{e\alpha}(\boldsymbol{\tau} + h_c e_c) - M_{e\alpha}(\boldsymbol{\tau} - h_c e_c)}{2h_c}, \qquad h_c = 10^{-5}\tau_c. \tag{H.3}$$

The active set is fixed for this calculation, and the elasticity templates and ghost contribution are held constant. The moment quadrature is recomputed at each perturbed thickness, so its clipping and integration branches may change. The resulting matrices represent a numerical design derivative on that prescribed topology. The sensitivities of the NICE lattice analyses and optimisations (Sections 5.9 and 5.10) are instead obtained by reverse-mode differentiation of the moment integrals at the fixed recovered displacement field.

Appendix F.1. Sensitivity error bounds (proof of Proposition 4)

For a fixed retained displacement vector q, set $u = Eq$, $d = J_I^T Hq$, and $\mathcal{E} = d^T K d = (Hq)^T A (Hq)$. The derivatives $K_{,c}$, $c = 1, \dots, 8$, are symmetric. Define the reference vector $\boldsymbol{s}$ with components $s_c = -u^T K_{,c} u$, and assume $\|\boldsymbol{s}\|_2 > 0$. Eq. (17) then reads

$$\widetilde{s}_c - s_c = -2d^T K_{,c} u - d^T K_{,c} d.$$

Let $b_c = J_I K_{,c} u$, $K_{II,c} = J_I K_{,c} J_I^T$, and let the rows of $\mathsf{B}_s \in \mathbb{R}^{8\times i}$ be $b_c^T A^{-1/2}$. Define

$$L = \|\mathsf{B}_s\|_2, \qquad Q = \left(\sum_{c=1}^{8} \|A^{-1/2} K_{II,c} A^{-1/2}\|_2^2 \right)^{1/2}.$$

Cauchy–Schwarz and the quadratic-form operator bound give

$$\boxed{\|\widetilde{\boldsymbol{s}} - \boldsymbol{s}\|_2 \le 2L\sqrt{\mathcal{E}} + Q\mathcal{E}, \qquad e_s \le \frac{2L\sqrt{q^T S q}}{\|\boldsymbol{s}\|_2}\sqrt{\varepsilon} + \frac{Q q^T S q}{\|\boldsymbol{s}\|_2}\varepsilon.} \tag{H.4}$$

All eight components are included, and the denominator is the norm of the exact sensitivity vector for that test displacement. The constants separate the coupling through the stiffness derivative, the interior coercivity and the scale of the reference sensitivity. For fixed q and $H = tH_0$, the linear coefficient vanishes if and only if $(H_0 q)^T b_c = 0$ for every corner. It vanishes for all interior errors if $J_I K_{,c} E q = 0$ for every corner. A useful special case is $K_{,c} = \alpha_c K$: interior equilibrium removes the linear term, and the relative sensitivity error equals the energy error when the reference vector is nonzero.

Appendix F.2. The complete design derivative and the sign of the thickness derivative

With fixed selectors, differentiating $J_I K E = 0$ gives $A E_{I,c} = -J_I K_{,c} E$. With $K_{II,c}$ as in Appendix H.1, differentiating the variational stiffness error gives

$$\boxed{(\widehat{S} - S)_{,c} = H_{,c}^T A H + H^T K_{II,c} H + H^T A H_{,c}.} \tag{H.5}$$

Its norm is bounded by $2\|A\|\|H\|\|H_{,c}\| + \|K_{II,c}\|\|H\|^2$. Thus a value error $H = O(t)$ by itself does not imply a quadratic derivative error. For example, with $f = 1$, $K = I_2$, $E = (1, 0)^T$ and $F_t(\tau) = (1, t + \tau)^T$, the compliance gap at $\tau = 0$ is $C - \widehat{C}_t = t^2/(1 + t^2)$, whereas the derivative error is $\partial_\tau(\widehat{C}_t - C) = -2t/(1 + t^2)^2$. With $H_t(\tau) = t \sin(\tau/t^2 + \pi/4)$ the gap is $O(t^2)$, but the derivative error tends to -1. Neither family is uniformly C^1-small. A uniformly C^1-small recovery error, $H = tH_0(\tau)$ with bounded $H_0, H_{0,c}$, gives $(\widehat{S} - S)_{,c} = O(t^2)$. With uniformly bounded exact solutions and inverse stiffnesses, this also yields $\widehat{C}_{,c} - C_{,c} = O(t^2)$.

At the same retained displacement vector q, Eqs. (17) and (H.5) give

$$\widetilde{s}_c - s_c = 2(Hq)^T A E_{I,c} q - (Hq)^T K_{II,c} H q, \qquad -q^T \widehat{S}_{,c} q - s_c = -2(H_{,c} q)^T A\, Hq - (Hq)^T K_{II,c} H q. \tag{H.6}$$

The complete derivative of Eq. (18) evaluates $-q^T \widehat{S}_{,c} q$ at the assembled $\widehat{q}$, so its comparison with s_c also contains the change of this quantity between q and $\widehat{q}$. The residual term of Eq. (18) thus removes the term linear in the interior displacement error and replaces it by a term weighted by the design derivative of the recovery error. Which of the two gives the more accurate gradient depends on whether $\|H_{,c} q\|_A$ is small compared with $\|E_{I,c} q\|_A$.

The $O(t)$ term of the sensitivity estimate can cancel against the residual term. At the same retained displacement, $J_I K_{,c} E q = -A E_{I,c} q$. For $H = tH_0$, the linear error of the sensitivity estimate is $+2t(H_0 q)^T A E_{I,c} q$, while the linear contribution of the residual term is $-2t(E_{I,c} q)^T A H_0 q$. A sensitivity estimate with a first-order error and a complete derivative with a second-order error can therefore coexist, even with a positive semidefinite stiffness derivative. For $K(\tau) = \begin{bmatrix} 1 + \tau & \tau \\ \tau & 1 + \tau \end{bmatrix}$, $E = (1, -\tau/(1 + \tau))^T$, $F_t = E + (0, t)^T$ and $f = 1$ at $\tau = 0$, the exact derivative is -1, the sensitivity estimate $-(1+t)^2/(1+t^2)^2$ and the residual term $2t/(1 + t^2)^2$; their sum $-1/(1 + t^2)$ has error $t^2/(1 + t^2)$.

Under a fixed basis and exact integration, increasing one band parameter τ_c with nonnegative Q_1 shape functions enlarges the material domain. With fixed ghost stabilisation, this gives $K_{,c} \succeq 0$, $S_{,c} \succeq 0$, and $C_{,c} \leq 0$. The sensitivity estimate is then also nonpositive for any displacement field; the quadratic term in the signed sensitivity discrepancy of Eq. (17) is nonpositive, while the cross term can have either sign. That the variational condensed stiffness dominates the exact one at every design does not by itself make the approximate compliance monotone with respect to the design: the residual term can change the sign of its complete derivative. For a scalar design parameter $\gamma_1 > 0$, $K(\gamma_1) = \gamma_1 I_2$, $E = (1, 0)^T$ and $F = (1, g(\gamma_1))^T$, one has $\widehat{S} = \gamma_1(1 + g^2) \geq S = \gamma_1$; at $\gamma_1 = 1$, $g = 0.1$ and $g' = -10$, $\widehat{S}' = -0.99$, so $\widehat{C}' > 0$ for $f = 1$ although $C' = -1$, while the sensitivity estimate remains negative.

More explicitly, for $h > 0$, $N_c^{Q_1}(x) \geq 0$ implies $\Omega(\tau) \subseteq \Omega(\tau + he_c)$. For any discrete coefficient vector v, with interpolated displacement field v_h,

$$v^T[K(\tau + he_c) - K(\tau)]v = \int_{\Omega(\tau+he_c)\setminus\Omega(\tau)} \nabla^{\mathrm{s}} v_h : \mathsf{C} : \nabla^{\mathrm{s}} v_h \, dx \geq 0. \tag{H.7}$$

Here ∇^{s} is the symmetric gradient, the elasticity tensor C and the basis are fixed, and the ghost contribution cancels. Taking the differentiable limit proves that the thickness derivative is positive semidefinite. A centred difference of stiffness matrices assembled from exactly nested domains is also PSD. Changes of the adaptive integration subdivision, approximate moments or independently selected stabilisation can break this discrete nesting relation; a fixed active topology alone does not establish that it holds numerically. A step study of the central difference and an element-level check of the moment derivatives (Supplementary Note S1, Figure S01(c)) show that the numerical derivative is accurate on the fixed active set. They also show that the discrete moments preserve the nesting only approximately: some partially filled elements have indefinite derivative matrices. Eq. (H.7) therefore describes exact integration; the reference sensitivities of this study are derivatives of the discrete model and are compared as such.

Appendix G. A computable lower bound on the energy error

For any nonzero interior test vector z, Cauchy–Schwarz applied to $A^{1/2}z$ and $A^{-1/2}r_I$ gives

$$\Delta(q) := q^T(\widehat{S} - S)q = r_I^T A^{-1} r_I \geq L_z := \frac{(z^T r_I)^2}{z^T A z}. \tag{I.1}$$

A preconditioned residual, a coarse approximation, or a Krylov iterate can supply z. Evaluating the final one-vector quotient with the original A makes the inequality independent of how that vector was generated. Let $q \notin \operatorname{range} R_P$ and $e_h = \widehat{u}^T K \widehat{u} = q^T \widehat{S} q$. Then $e_h - \Delta(q) = q^T S q > 0$, so $e_h - L_z > 0$. Moreover $\varepsilon(q) = \Delta/(e_h - \Delta)$, and the map $x \mapsto x/(e_h - x)$ is increasing on $[0, e_h)$. Reproduction of rigid-body motions gives $HR_P = 0$ and $SR_P = 0$ (Appendix B.2), so $\varepsilon(q) = \varepsilon(\Pi_P q) \leq \varepsilon_*$. Therefore

$$\varepsilon(q) \geq \frac{L_z}{e_h - L_z}, \qquad \mu_* \geq 1 + \frac{L_z}{e_h - L_z}. \tag{I.2}$$

If the exact reference energy is known, $L_z/(q^T S q)$ gives a sharper lower estimate. A large value identifies a test displacement on which the condensed stiffness is inaccurate; a small value alone does not bound the error from above. These inequalities hold in real arithmetic; their numerical evaluation requires the original quadratic forms stated above.

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